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ACI-DAfStb Databases for Shear Tests on Slender Reinforced Concrete Beams with Stirrups

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An extensive database was created in 1999 that contained the shear test data of beams with shear reinforcement subjected to point loads. The data were not only collected but were also critically reviewed, and criteria were developed to achieve a reliable database for assessing shear design relationships. This database was considerably expanded until 2006 and then even more in the recent years by a joint ACI-DAfStb group.

Because many of the 886 collected tests do not fully conform to either ACI or European design code limits, two final databases are presented in this paper. The larger of the two released evaluation databases contains 157 tests that are appropriate for comparisons to ACI 318 shear provisions. The second small evaluation database is a subset of the large database, contains 87 tests, and is appropriate for comparisons to shear design provisions such as those of the EC2 or fib *and FIP standards.*

Keywords: beams with shear reinforcement; database; reinforced concrete; shear strength; shear tests.

INTRODUCTION

Code provisions for the shear capacity of structural concrete members with shear reinforcement have been, more or less, semi-empirically derived with only partial reference to a truss model. In the provisions of ACI 318-11 (ACI Committee 318 2011), the truss model is used to determine the stirrup contribution to shear capacity, and then an empirically derived concrete term is added. The shear design of most European codes is solely based on the truss model with strut angles flatter than 45 degrees. In both codes, the maximum shear capacity or the strength of the inclined struts is also based on tests.

DIBt (1999) and Reineck (1999) presented a database with shear tests on beams with shear reinforcement subjected to point loads that comprised 493 tests. This database was considerably extended to 818 tests in 2006 by merging it with the database collected by Kuchma, and published by Reineck et al. (2010, 2012). More recently, a joint ACI-DAfStb group was formed that consists of members of ACI Subcommittee 445-D, Shear Database, and a similar group in the German Committee for Structural Concrete, Deutscher Ausschuß für Stahlbeton (DAfStb). This group added new tests and checked the data so that the collection database now contains 886 tests, in which 511 are tests on slender beams with shear span-depth ratios $a/d > 2.4$.

The same conversion factors for the concrete strengths that were used by Reineck et al. (2003, 2010, 2012, 2013). Likewise, the sanctioned set of criteria defined in 2003 was kept to with only a few modifications, which were developed for accepting a test result into an evaluation-level database. Examples of these criteria include a minimum compressive strength, a minimum overall height and width, and checks against flexural and anchorage failures.

RESEARCH SIGNIFICANCE

Code provisions for the shear capacity of structural concrete members with shear reinforcement have been primarily derived from test data with respect to the required amount of shear reinforcement and the calculation of maximum shear capacity. Therefore, the selection of reliable tests is of utmost importance for calibrating the design methods. A database with shear tests of reinforced concrete (RC) beams with stirrups was jointly established by an ACI-DAfStb group based on several previous databases, and represents what the authors contend is the most significant international and representative database of shear test results of RC beams.

OVERVIEW OF DATABASE AND PROCEDURES

Overview of collected data

This section includes a brief summary on the databases for beams with vertical stirrups, and the principles are described concerning the collection and control of the data and the selection of the data for the evaluation database. The data were stored in a spreadsheet, and in the process, several files were generated. In this way, the collection of data was strictly separated from their subsequent control and of the selection of tests for the evaluation database. Each file was given a distinct name so that it was possible to recognize what kind of tests it contained and in which state of the process the database was in.

In control files of the database different control criteria koni were defined (the abbreviation "kon" is a German acronym for control criterion, and "*i*" is the running number) and were checked for:

koni = 0 not fulfilled; no transfer to evaluation file;

koni = 1 fulfilled; transfer.

The criteria were for removing those beams from the database that did not satisfy the relevant requirements; the remaining beams were transferred to the data evaluation database file. The individual criteria are defined in a following section and presented with the results of the checks.

The tests on beams with vertical stirrups were performed as simple span beams subjected to point loads, and were

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collected in the file vsw-RC-DS with 886 tests, from which 646 had a rectangular cross section. From the collected tests, 125 had to be removed from the database by the selection criterion konx (Fig. 1) because essential information was missing, such as the yield strength of the longitudinal reinforcement or the attained ultimate shear force. The criterion kon61 for the slenderness was then applied, and the data separated into two files, as presented in Fig. 1. The file vsw-DK-sl contains the tests on slender beams with shear span ratios of $a/d > 2.4$, as defined in the following section. The remaining tests with smaller a/d were transferred to the database vsw-DK-24. Only the databases with slender beams are further presented and used in this paper.

Definition and notation

The notation for beams with point loads is given in Fig. 2. For the 240 T-beams, many details for the shape were collected as described by Reineck et al. (2012). A typical test setup is shown in Fig. 2(a), where two point loads are applied each a distance a from the support axis. The distance a is the shear span from the center of the support to the center of the applied load that causes that shear; thus, care is required when considering cases of a single point load, as shown in Fig. 2(b). In this case, the total load is shown to be divided into two forces F, each equal to the shear force, and thus the distance a is defined in a consistent way for single or double point loads.

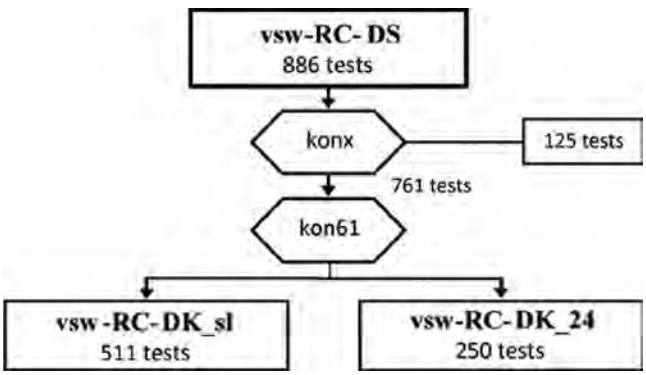

Fig. 1—Sorting and selecting tests according to primary selection criteria: (a) beam with point loads; and (b) definition of shear span a *in case of single point load.*

The values for the material strengths for the concrete and steel were determined as described in detail by Reineck et al. (2003, 2010, 2012). For the concrete compressive strength, the values determined on different control specimens were converted by different conversion factors into the uniaxial compressive strength f_{1c}, which corresponds to the compressive strength of a slender prism, and is equal to

$$f_{1c} = 0.95 \cdot f_{c,cyl} \tag{1}$$

where $f_{c,cyl}$ is the strength of a standard cylinder Ø150 mm; and h = 300 mm (6 x 12 in.).

Many test reports do not provide data for consideration of self-weight of the beams, so that the self-weight was determined in the way described by Reineck et al. (2003, 2010, 2012, 2013)

Determination of stirrup stress at failure

During the data collection, it was found that important data was often not reported or not measured in some of the tests. This observation particularly applied to the stirrup stress, because for only approximately half of the tests, either measurements of stirrup strains were reported or clear statements could be found in the reports that yielding of stirrups occurred at or before failure. From measurements of the stirrup strain, the stirrup stress at failure was determined by extrapolating the measured value from the load levels before failure, and in the database, this measured value was then entered under the value $\sigma_{sw,meas}$. The authors thereby acknowledge that strain measurements in stirrups can vary depending on the location of the gauge in relation to the crack.

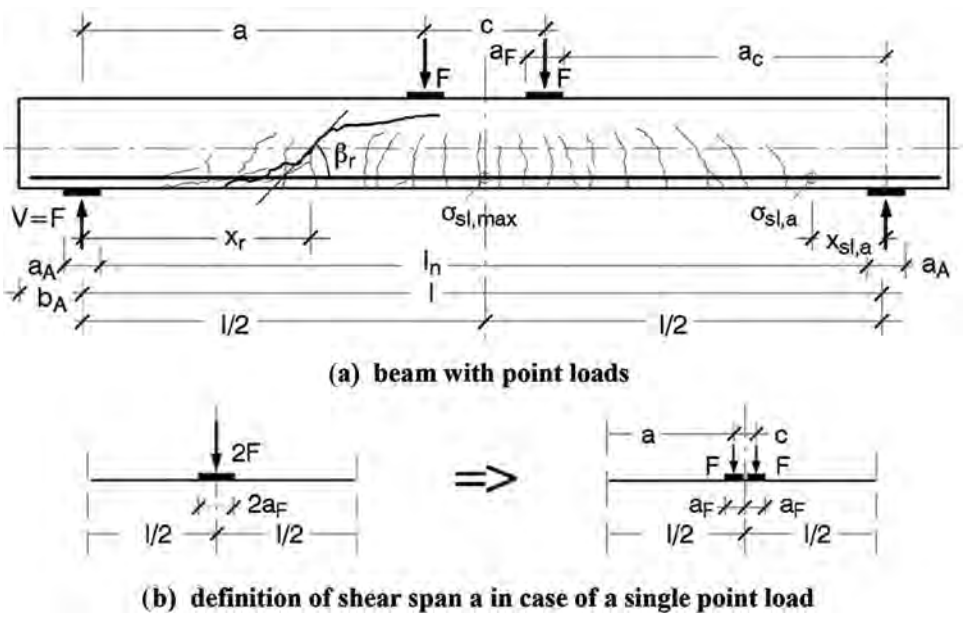

Fig. 2—Notation for simple span beams with point loads and definition of shear span.

If no measured value for the stirrup strain was reported, then in the report, a clear statement about stirrup yielding was searched for, and if such a statement was found, then the value $\sigma_{sw,ass} = f_{yw}$ was entered into a column of the data bank. If neither of these two cases applied, then $\sigma_{sw} = 0$ was entered.

SURVEY OF TESTS ON BEAMS WITH POINT LOADS

Control of data

The database vuct-RC-DK_sl for slender beams subjected to point loads comprises 511 tests (Fig. 1). Beam selection control criteria were used to either filter the tests into different datasets or to remove those beams from the database that did not satisfy specific requirements, as explained previously. The individual criteria are defined in Table 1, and further explained as follows along with the results of the data control. Many of these criteria followed the rules of the FIP Recommendations (1999), which is based on the internationally accepted CEB-FIP MC90 (1993). Thus, it is hoped that the resulting evaluation database is also internationally acceptable. If the often very different criteria of the many available national codes would be applied, it would result in many different evaluation databases, and could lead to some confusion when comparing design proposals.

It is noteworthy that criterion kon1 filtered out 9% of the tests that had a uniaxial compressive strength of only f_{1c} < 12 MPa (1740 psi). This limit corresponds to the minimum concrete classes required in many codes. The criteria kon2, kon3, kon4, and kon7 (for the height x of the flexural compression zone) are fulfilled by almost all tests. If, however, instead of kon3 the limit for the lowest allowable width is increased from b_w = 50 to 100 mm (2 to 4 in.) using criterion kon31, then 41 further tests would be eliminated. The similar increase for the height from h = 70 to 150 mm (2.76 to 6 in.) in the criterion kon41 had no effect.

The criterion kon8 (β_{flex} < 1.00) for the flexural check is only met by 65% of the tests. By accounting for some of the conservativeness of the standard flexural strength calculations, the criterion kon81 shows that an additional 10% allowance on the flexural check allows a significant increase by 108 tests (21%) that pass the flexural failure check.

The criterion kon9 confirms that the compressive stress in the diagonal struts of the web did not exceed the uniaxial compressive strength f_{1c}. The calculation of this stress depends on knowing the stirrup stress at failure, but this was only available for 247 tests, as checked by the criterion kon12. Of these 247 tests, only (247 – 232) = 15 beams failed to meet kon9.

The criterion kon101 showed that almost all beams (92%) had ribbed longitudinal reinforcing bars. According to the criterion kon102, however, only 54% of the beams had ribbed stirrups. Both conditions are fulfilled by only 263 tests (52%) according to the criterion kon10.

The criterion kon11 ($\beta_{lb} = l_{b,req}/l_{b,prov}$ < 1.0) for the check of the anchorage length l_b at the end support is performed according to the rules of the FIP Recommendations (1999), which is based on the internationally accepted CEB-FIP MC90 (1993). This criterion was not met by 65 beams (13%), and these were marked in the database as AF for "anchorage failure," meaning that the anchorage check was not fulfilled.

Table 1—Results of evaluation of tests in respect of individual criteria koni

Criterion	Condition for criterion	No. satisfying	Percent of 511	No. violating
kon1	f_{1c} > 12 MPa (1740 psi)	463	90.8	47
kon2	f_{1c} < 100 MPa (14,500 psi)	506	99.2	4
kon3	$b_w \geq 50$ mm (2 in.)	510	100	0
kon31	$50 \leq b_w < 100$ mm (4 in.)	41	8.0	—
kon4	h > 70 mm (2.76 in.)	510	100	0
kon41	70 < h < 150 mm (5.9 in.)	0	0	—
kon5	a/d > 2.89	378	74.1	132
kon6	$2.4 \leq a/d \leq 2.89$	123	24.1	387
kon7	$\xi_{test} = x/d \leq 0.5$	490	96.1	20
kon8	$\beta_{flex} = \mu_u/\mu_{flex}$ < 1.00	330	64.7	180
kon81	1.00 < β_{flex} < 1.10	108	21.2	—
kon9	$\nu_u = \sigma_{cw}/f_{1c}$ 1.0	232	45.5	278
kon101	$f_r = r$: ribbed	471	92.4	39
kon102	$f_{rw} = r$: ribbed	274	53.7	237
kon10	= kon101 · kon102	263	51.6	248
kon11	$\beta_{lb} = l_{b,req}/l_{b,prov}$ < 1.0	445	87.3	65
kon12	$\sigma_{sw,rep}$	247	48.4	263
kon131	$\rho_w > 0.06228 \cdot f_c'^{0.5}/f_{yw}$	468	91.8	42
kon132	$\rho_w > 0.08 \cdot f_{ck}^{0.5}/f_{yw}$	418	82.0	82
kon133	$\rho_w > 0.16 \cdot f_{1ct,cal}/f_{yw}$	406	79.6	94
kon134	$\rho_{w,min_ACI} < \rho_w < \rho_{w,min_DIN}$	62	12.2	—
kon141	$s_w \leq 0.7h$	369	72.4	141
kon142	$s_w \leq 300$ mm (12 in.)	462	90.6	48
kon14a	= kon141 · kon142	333	65.3	177
kon143	$s_w \leq 0.55d$	354	69.4	156
kon144	$s_w \leq 610$ mm (24 in.)	506	99.2	4
kon14b	= kon143 · kon144	354	69.4	156
kon15	No other failure type (oft)	415	81.4	95
kon161	$f_{yw} \leq 414$ MPa (6000 psi)	264	51.8	246
kon162	$f_{yw} \leq 552$ MPa (8000 psi)	405	79.4	105

The criterion kon12 $\sigma_{sw,rep}$ for the reported stirrups stress is to check whether the stirrups yielded at the shear failure or not. This fact was either reported or the stirrups strains were measured (refer to previous section) for only 247 beams (48%), which is a fairly low number.

The criteria kon131 to kon133 checked whether the tests exceeded the required minimum ratio of the web reinforcement defined in three different codes. The criterion kon131 (in SI units) is that of ACI 318-11, which is $\rho_{v,min} = 0.75 \cdot \sqrt{f_c'}/f_{yt}$ (in U.S. units), and it was fulfilled by 92% of the tests, that is, 8% of the 510 tests had reinforcing ratios lower than this minimum. Criterion kon132 is the minimum value for the *fib*

Table 2—Subsequent application of individual selection criteria for KONA0

Selection criterion	Combination of individual criteria	Added criterion	Remaining of 510	Difference
KONA0a	kon1·kon3·kon4	—	463	47
KONA0b	KONA0a·kon7	$\xi_{test} = x/d \leq 0.5$	444	19
KONA0c	KONA0b·kon15	No other failure type (oft)	356	88
KONA0d	KONA0c·kon101	$f_r = r$: ribbed	322	34
KONA0	KONA0d·kon143	$s_w \leq 0.55 \cdot d$	207	115

Table 3—Result of sorting tests with respect to slenderness and flexural check

Selection criterion	Combination of criteria	Fulfilled number	Percent of 207 dataset A0
KONA21a	KONA0·kon5·kon8	95	45.9
KONA22a	KONA0·kon5·kon81	32	15.5
KONA31a	KONA0·kon6·kon8	41	19.8
KONA32a	KONA0·kon6·kon81	12	5.8
Dataset (A2a+A3a)	Sum of aforementioned datasets	180	87.0

MC (2010), and was fulfilled by 82% of the tests. Criterion kon133 is that of DIN 1045-1 (2008), and was fulfilled by 80% of the tests. As the difference between the latter two is only marginal, the final database is based on the more restrictive kon133. The criterion kon134 defines the tests that fall between ACI and DIN limits. For comparisons with the *fib* MC (2010), care should be taken to include the appropriate number of tests that fall within the limits of kon134.

The maximum spacing s_w of the stirrups was checked for the rules of two codes. The first was the rules of EC2 respectively DIN 1045-1 checked by the criteria kon141and kon142, and the combined criterion kon14a was fulfilled by 65% of all tests. The second was the rules of ACI 318-11 checked by the criteria kon143 and kon144, and the combined criterion kon14b was fulfilled by 69% of all tests. This more restrictive criterion was selected in the following for the large dataset.

Criterion kon15 is a check against failure modes not reported as shear failures and these were marked in the database as "oft" for "other failure type" This check that no oft occurred was fulfilled by 81% of the collected tests.

Selection of tests for evaluation

To be transferred to the evaluation file, several criteria had to be fulfilled simultaneously for a beam—that is

$$\textbf{KONAi} = \text{kon1} \cdot \text{kon2} \cdot \text{kon3} \cdot \ldots \cdot \text{koni} \qquad (2)$$

As decribed for the individual criteria, it was checked
KONAi = 0 no transfer to evaluation file;
KONAi = 1 transfer

Table 4—Result of subsequent application of further criteria for 180 tests on reinforced concrete beams of dataset (A2a+A3a)

(A2a+A3a)					180
Large dataset (A2b+A3b) = (A2a+A3a)·kon131	A21b	A22b	A31b	A32b	157
	76	31	38	12	
	107		50		
(A2c+A2c) = (A2b+A3b)·kon12	A21c	A22c	A31c	A32c	100
	51	23	17	9	
	74		26		
(A2d+A3d) = (A2c+A3c)·kon9	A21d	A22d	A31d	A32d	96
	48	22	17	9	
	70		26		
Small dataset (A2+A3) = (A2d+A3d)·kon11	A21	A22	A31	A32	87
	44	17	17	9	
	61		26		

All of the evaluated tests had to fulfill the following set of criteria:

KONA0 = kon1·kon3·kon4·kon7·kon15·kon101·kon143

The results of the subsequent application of these criteria are listed in Table 2. In the resulting dataset A0, 207 tests of the 510 remained—that is, 41%.

Different alternatives were selected in the evaluation file for sorting the tests into different datasets with respect to flexural failures (kon8 and kon81) and slenderness (kon5 and kon6), and the results for the four datasets are listed in Table 3. From the 207 tests in the dataset A0, 180 tests (87%) remain in the end, so that altogether 27 tests were eliminated as flexural failures with $\beta_{flex} > 1.10$.

Subsequently, different criteria were applied to the 180 tests of the dataset (A2a+A3a), and the results are listed in Table 4. The criterion kon131 was selected because it eliminated the lowest number of 23 tests with very low web reinforcement ratios below the required minimum of ACI 318-11, so that 157 tests remained in the dataset (A2b+A3b). This large dataset was used for comparisons with some code relationships, as explained in a following section.

The criterion kon12 eliminated 57 tests where the stirrups stress at failure is not known, and this led to the dataset (A2c+A3c) with 100 tests. The role of this criterion for comparisons with code relationships is discussed in the following section. The next criterion, kon9, applied to a further four tests so that 96 remained in the dataset (A2d+A3d). From these, nine tests did not comply with the anchorage check so that for a strict selection, 87 tests remained in the final small dataset (A2+A3). This small dataset only comprises indisputable tests for the evaluations and for comparisons with code relationships.

If instead of kon3 the limit for the lowest allowable width was increased from b_w = 50 to 100 mm (2 to 4 in.) using the criterion kon31, however, then seven further tests would be eliminated.

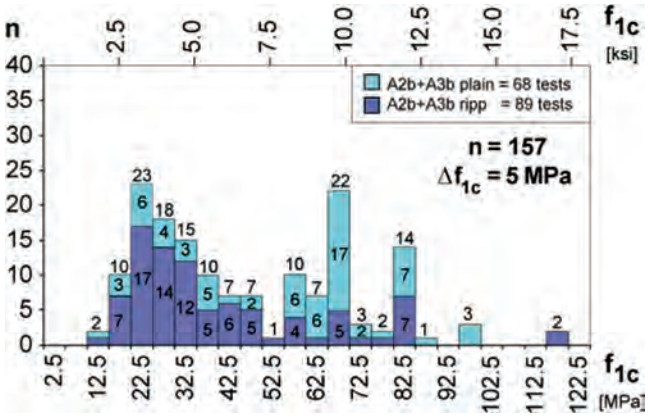

Fig. 3—Number of beams versus concrete compressive strength for large dataset A2b+A3b.

PRESENTATION OF LARGE DATASET

In this section, the number of the beams of the evaluation database vsw-RC_A2b+A3b with the large dataset (A2b+A3b) with 157 tests is plotted versus important test parameters to obtain an overview of the distribution of the test parameters. The parameters are thereby subdivided into class intervals, and the number of tests in each interval is plotted. In the diagrams, the 68 tests on beams with plain stirrups can be discerned from the 89 tests on beams with ribbed stirrups. Previous comparisons by Reineck et al. (2012) between the results of separate datasets for plain and ribbed stirrups showed that both yielded results in the same ranges, so that the following evaluations are performed for the whole dataset.

In Fig. 3, the number of beams n is plotted versus the uniaxial concrete compressive strength f_{1c} subdivided into class intervals of Δf = 5 MPa (725 psi). The predominant number of test beams had a uniaxial concrete compressive strength of f_{1c} < 40 MPa (5800 psi), that is, 78 tests (50%). A total of 54 beams (34%) had a uniaxial concrete compressive strength of f_{1c} > 60 MPa (8700 psi), and can thus be classified as high-strength concrete beams. The two tests with a compressive strength of f_{1c} >115 MPa (16,675 psi) were performed by Roller und Russel (1990).

In Fig. 4, the number of beams is plotted versus the yield strength of the stirrups f_{wy} of the beams subdivided in class intervals of Δf_{sy} = 50 MPa (7.25 ksi). Most beams were in the range of f_{yw} < 450 MPa (65.25 ksi), that is, 75 tests (48%). Only 59 beams (38%) had yield strengths of f_{yw} > 550 MPa (79.7 ksi).

In Fig. 5, the number of beams is plotted versus the mechanical reinforcement ratio $\omega_l = \rho_l \cdot f_{sy}/f_{1c}$ (with $\rho_l = A_s/(b \cdot d)$) of the longitudinal reinforcement subdivided into class intervals of $\Delta\omega_l$ = 0.05. Most tests of the evaluation database had mechanical reinforcement ratios of ω_l < 0.30. The majority of the beams with 113 tests (62%) were in the range of 0.10 < ω_l < 0.25. Only five beams (3%) had a low mechanical reinforcement ratio of ω_l < 0.10, and four of these had plain stirrups.

In Fig. 6, the number of beams is plotted versus the slenderness $\kappa = a/d$ subdivided in class intervals of $\Delta\kappa$ = 0.4. The majority of tests were carried out on beams with a slenderness of 2.4 < κ < 3.6—that is, 129 of the 157 beams, which is 82%.

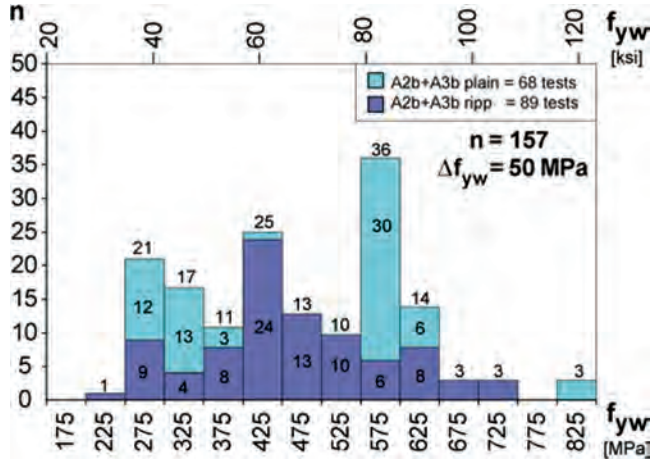

Fig. 4—Number of beams plotted versus yield strength of stirrups for large dataset A2b+A3b.

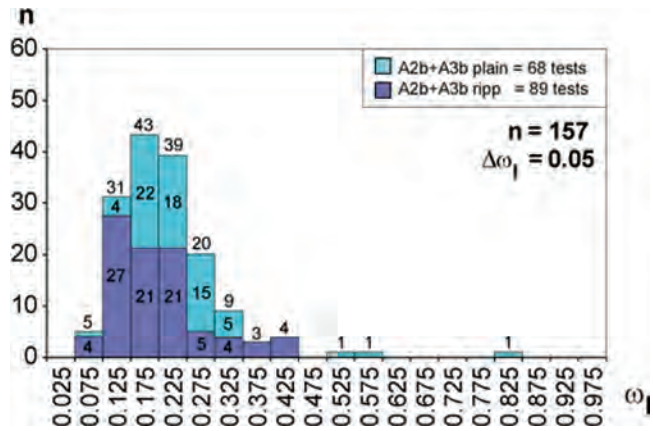

Fig. 5—Number of beams plotted versus mechanical reinforcement ratio of longitudinal reinforcement for large dataset A2b+A3b.

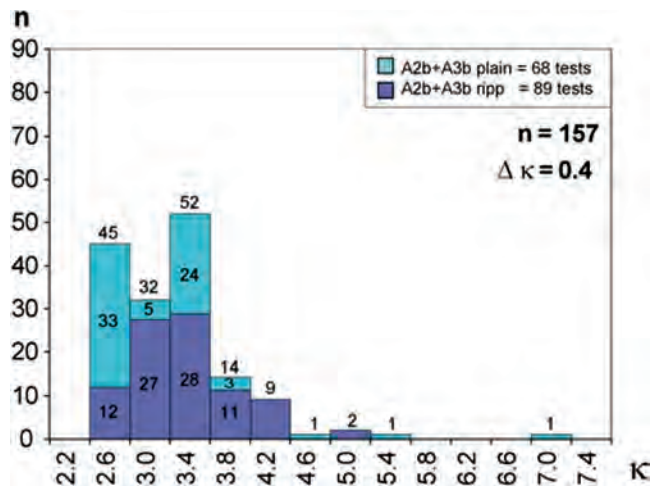

Fig. 6—Number of beams plotted versus slenderness $\kappa = a/d$ for large dataset A2b+A3b.

In Fig. 7, the number of the beams is plotted versus the effective depth d subdivided into class intervals of Δd = 100 mm (4 in.). Most beams had an effective depth 200 < d <

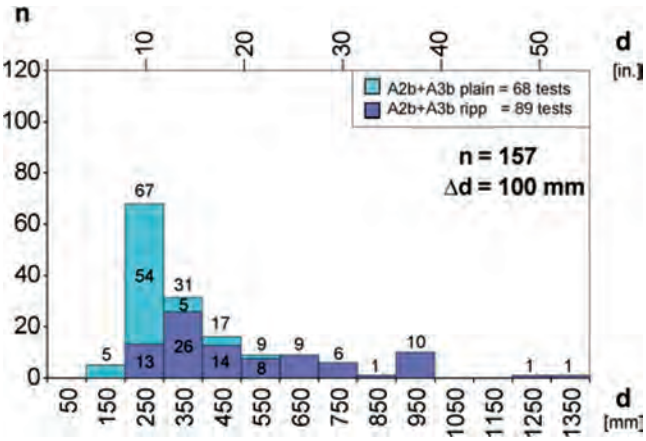

Fig. 7—Number of beams plotted versus effective depth d for large dataset A2b+A3b.

400 mm (8 in. < d < 16 in.), that is, 98 beams, respectively 62%. Altogether, 28 beams (12%) had an effective depth of d > 600 mm (23.6 in.). The two beams with d > 1200 mm (47.2 in.) were out of the series tested by Bhal (1968) and Aparicio et al. (1997).

COMPARISON OF TEST WITH DESIGN EQ. (11-3) IN ACI 318-11

Shear design in ACI 318-11

In Chapter 11 of ACI 318-11, the shear strength is defined by Eq. (11-2) for members with shear reinforcement

$$V_n = V_c + V_s \tag{3}$$

where V_n is the nominal shear force; V_c is the nominal shear force provided by the concrete; and V_s is the the nominal shear force provided by shear reinforcement.

The shear force V_c is defined in ACI 318-11, Section 11.2, and for non-prestressed members Eq. (11-3) applies

$$V_c = 2\sqrt{f_c'} \cdot b_w \cdot d \tag{4a}$$

where V_c, lb; f_c', psi; and b_w and d, in.

Converting into SI units gives

$$V_c = 0.166\sqrt{f_c'} \cdot b_w \cdot d \tag{4b}$$

where V_c, N; f_c', MPa; and b_w and d, mm.

It should also be noted that the value of $\sqrt{f_c'}$ is to be limited to 8.30 MPa (100 psi), so that the influence of concrete strengths f_c' above 69 MPa (10,000 psi) is neglected.

The shear force V_s carried by vertical shear reinforcement is defined by Eq. (11-15) of ACI 318-11

$$V_s = (A_{sw}/s_w) \cdot f_{yw} \cdot d \tag{5}$$

where A_{sw} is the area of shear reinforcement (A_v in ACI 318-11); s_w is the spacing of shear reinforcement (s in ACI 318-11); f_{yw} is the yield strength of stirrups (f_{yt} in ACI 318-11); and d is the effective depth.

From Eq. (3), (4b), and (5) follows (in SI units)

$$V_{u,\mathrm{ACI}} = (A_{sw}/s_w) \cdot f_{yw} \cdot d + 0.166\sqrt{f_c'} \cdot b_w \cdot d \tag{6}$$

The shear force should not exceed a maximum value. According to Section 11.4.7.9 of ACI 318-11, it is for V_s

$$V_{s,max} \le 8\sqrt{f_c'} \cdot b_w \cdot d \text{ in U.S. units} \tag{7a}$$

$$V_{s,max} \le 0.664\sqrt{f_c'}\ b_w \cdot d \text{ in SI units} \tag{7b}$$

Therefore, according to Eq. (3), the maximum shear force is

$$V_{n,max} = V_c + V_{s,max} = (2 + 8)\sqrt{f_c'} \cdot b_w \cdot d = 10\sqrt{f_c'} \cdot b_w \cdot d \text{ in U.S. units} \tag{8a}$$

$$V_{n,max} = (0.166 + 0.664)\sqrt{f_c'} \cdot b_w \cdot d = 0.830\sqrt{f_c'} \cdot b_w \cdot d \text{ in SI units} \tag{8b}$$

Because only the uniaxial compressive strength f_{1c} had been transferred to the evaluation database, the cylinder strength f_c' has to be calculated as $f_{c,cyl} = f_{1c}/0.95$ (Eq. (1)). From this average value, the characteristic cylinder strength was calculated assuming a scatter of Δf = 4 MPa (580 psi)

$$f_{ck} = f_{c,cyl} - 4 \quad \text{(MPa)} \tag{9}$$

This characteristic cylinder strength f_{ck} is used in European codes (EC 2, DIN 1045-1, CEB-FIP MC 90 1993, *fib* MC 2010; FIP Recommendations 1999), and it is a 5% fractile value. Because f_c' represents a 9% fractile, the following relationship can be derived

$$f_c' = f_{ck} + 1.60 \quad \text{(MPa)} \tag{10}$$

Further explanations are given by Reineck et al. (2003, 2010, 2012).

Comparison with tests of large dataset

The comparison of tests with the aforementioned relationships for the shear force is done by calculating the model safety factor

$$\gamma_{mod} = V_{u,test}/V_{u,cal} \tag{11}$$

where $V_{u,cal} = V_{u,\mathrm{ACI}}$ is either given by Eq. (6), or by Eq. (8), $V_{u,cal} = V_{n,max}$.

In Fig. 8, the model safety factor γ_{mod} for the shear strength calculated with ACI 318-11 is plotted versus the mechanical web reinforcement ratio ω_{wy} calculated for the 157 tests of the large dataset (A2b+A3b). It can clearly be seen that the design relationships of ACI 318-11 are very conservative with model safety factors up to 3; the average value is γ_{mod} = 1.59, and the coefficient of variation is v = 0.252. There are only six tests (4% of all) with values γ_{mod} < 1.0, which is less than the 5% fractile, but two values for beams

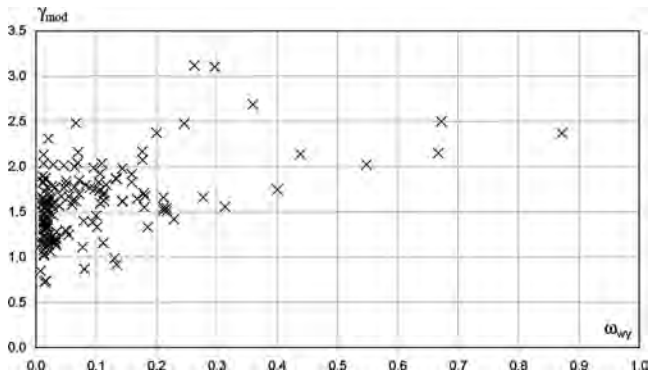

Fig. 8—Model safety factor for shear design of ACI 318-11 calculated for 157 tests of large dataset (A2b+A3b) plotted versus mechanical web reinforcement ratio ω_{wy}.

with very low web reinforcement ratios are very low with γ_{mod} = 0.72 and 0.73.

COMPARISON WITH SMALL DATASET

Design relationships in non-dimensional diagram

The comparison of different design relationships with tests is very instructive in a non-dimensional ω_{wu}-υ_u diagram as shown in Fig. 9, where the non-dimensional shear force υ_u is plotted versus the mechanical reinforcement ratio ω_{wu} for the stirrups, as explained by Reineck (2002). In a design the shear force is given, and it is plotted as an independent variable on the x-axis, and the required amount of stirrups is plotted on the y-axis. The terms ω_{wu} and υ_u are defined as follows

$$\upsilon_u = \frac{V_u}{b_w \cdot z \cdot f_{cwu}} \tag{12}$$

$$\omega_{wu} = \omega_{wy} \cdot \frac{\sigma_{sw}}{f_{yw}} \tag{13}$$

where σ_{sw} is stirrup stress at failure measured from stirrup strains.

For the truss model with struts inclined at the angle θ, the ultimate shear force V_u corresponding to yielding of the stirrups at failure ($\sigma_{sw} = f_{yw}$) is

$$V_u = \frac{A_{sw}}{s_w} \cdot f_{yw} \cdot z \cdot \cot\theta \tag{14}$$

where A_{sw}/s_w is the area of stirrups at spacing s_w; f_{yw} is yield strength of stirrups; z is the inner lever arm between tension and compression chord; and θ is the angle of inclined struts in the web.

The inner lever arm z between the tension and compression chords is calculated for the test beams, and is thus given. For the design relationships of RC beams, normally a value of $z = 0.9 \cdot d$ is assumed.

Dividing Eq. (14) by ($b_w \cdot z \cdot f_{cwu}$) leads to the non-dimensional shear force

$$\upsilon_u = \frac{A_{sw}}{b_w \cdot s_w} \cdot \frac{f_{yw}}{f_{cwu}} \cdot \cot\theta = \rho_w \cdot \frac{f_{yw}}{f_{cwu}} \cdot \cot\theta = \omega_{wy} \cdot \cot\theta \tag{15}$$

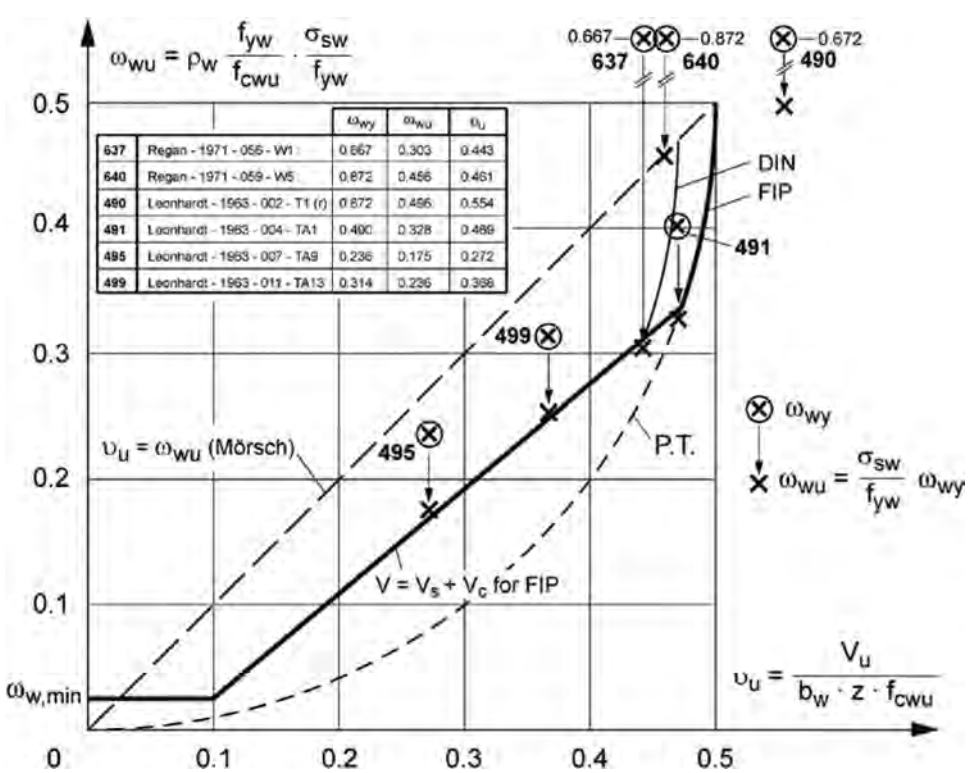

Fig. 9—Location of some tests in ω_{wu}-υ_u-dimensioning diagram in case of false assumption (⊗) of stirrup yielding and correct location (×) in case of measured stirrup stresses.

where

$$\omega_{wy} = \rho_w \cdot f_{yw}/f_{cwu} \tag{16}$$

$$\rho_w = A_{sw}/(b_w \cdot s_w) = \text{reinforcement ratio for stirrups} \tag{17}$$

where b_w is the width of web; f_{cwu} is $0.80f_{1c}$ = strength of inclined struts; and f_{1c} is the uniaxial compressive strength $= 0.95f_c'$.

For the truss model of Mörsch (1912), the struts are inclined at $\theta = 45$ degrees, and thus, in Eq. (15), the value cot45 degrees = 1. The strength calculated for this model is shown with a dashed line ($\upsilon_u = \omega_{wy}$) in Fig. 9. The dashed circle corresponds to the plasticity theory (PT) for which the stirrups yield, and likewise, the struts attain their strength $f_{cwu} = 0.8f_{1c}$ (Thürlimann 1978). The second inner circle applies to the German code DIN 1045-1 (2008) for the slightly lower strut strength of $f_{cwu} = 0.75f_{1c}$. In Fig. 9, as an example, the design relationship $V = V_s + V_c$ of the FIP Recommendations (1999) is also plotted, which is comparable to that of DIN 1045-1 for low concrete strengths.

The non-dimensional format for ACI 318-11 is attained by dividing Eq. (6) and (8) by ($b_w \cdot z \cdot f_{cwu}$), and this gives (in SI units)

$$\upsilon_{u,ACI} = \frac{A_{sw}}{b_w} \cdot \frac{f_{yw}}{f_{cwu}} \cdot \frac{d}{z} + 0.166 \cdot \frac{\sqrt{f_c'}}{f_{cwu}} \cdot \frac{d}{z} \tag{18a}$$

and with the definitions for ρ_w and ω_{wy}, this gives

$$\upsilon_{u,ACI} = \left[\omega_{wy} + 0.166 \cdot \frac{\sqrt{f_c'}}{f_u}\right] \cdot \frac{d}{z} \tag{18b}$$

From Eq. (8b) follows the non-dimensional maximum shear force (in SI units)

$$\upsilon_{u,max,ACI} = 0.830 \cdot \frac{\sqrt{f_c'}}{f_{cwu}} \cdot \frac{d}{z} \tag{19}$$

where f_c' and f_{cwu} need to be inserted in MPa.

Location of tests in diagram for design relationships

The reason for applying kon12 and using the small dataset (Table 4) is that in all design approaches for members with shear reinforcement using the relationship $V = V_s + V_c$, the required amount of stirrups is determined from V_s assuming that the yield stress is reached in the stirrups. Therefore, it is logical that this condition should also be fulfilled for a test used for comparison with this design approach, so that at first, only those tests can be considered where yielding of the stirrups occurred. In cases where the stirrups did not yield but the stirrup strains were measured, the tests could also be considered with the measured stirrup stress as explained previously.

From a very strict experimental perspective, if tests were considered without knowing the stirrup stresses but assuming that always yielding occurred then the validity and safety of a design equation $V = V_s + V_c$ cannot be assessed, and this is explained in Fig. 9. In many cases of beams with a large quantity of stirrups, the ultimate load predicted by a design equation is not reached, which means that in Fig. 9 such a test point lies well above the line representing the design equation because stirrup yielding is assumed, although this was not the case. If, instead, the stirrup stress actually attained is considered, however, then the test points lie far lower in the diagram of Fig. 9, such as demonstrated for tests No. 637 and 640 by Regan (1971) and tests No. 490, 491, 495, and 499 by Leonhardt and Walther (1963). Some of these tests are now on or near the safe side compared with the design relationship $V = V_s + V_c$ of the FIP Recommendations (1999). Test No. 640 ends on the Mörsch line slightly to the left of the plasticity circle for the German code DIN 1045-1 (2008).

For the comparison with Eq. (19) of ACI 318-11 or of any code using the relationship $V = V_s + V_c$, it is therefore justified to eliminate the tests where the stirrup stress is not known by means of the criterion kon12, and this leads to the dataset (A2c+A3c) with 142 tests with known stirrup stresses (Table 4).

This dataset still contains nine tests with AF, that is, the tests where the anchorage check at the end support was not fulfilled. Likewise, this dataset contains the tests with excessive high shear forces (kon9), which exhibit stresses in the inclined struts of the truss model exceeding the uniaxial compressive strength, and are located outside the outer circle in the diagram. These tests are disputable, and should not be used as a proof for the safety of any design relationship. Therefore, it appears to be justified to not use the tests without AF (kon11) and the tests complying with kon9 for comparisons with design relationships, and this leads to the indisputable small dataset (A2+A3) with 87 tests (Table 4). Only for comparisons with other codes than ACI 318-11, the limit kon131 for the minimum reinforcing ratio needs to be reconsidered, and leads to further elimination of some tests in the very low shear range by using criteria kon132 or kon133 (Table 1).

The result is shown in the non-dimensional Fig. 10, where ω_{wu} is plotted versus the ultimate shear force υ_u as defined by Eq. (12) and (13). This diagram now clearly shows that the shear strength calculated with ACI 318-11 is very conservative in the whole range of data with respect to $V_{u,ACI} = V_c + V_s$ of Eq. (18b), as well as with respect to the limit for maximum shear of Eq. (19). The FIP Recommendations (1999) appear to better fit to the results.

SUMMARY AND CONCLUSIONS

A database for shear tests on beams with shear reinforcement was established by a joint ACI-DAfStb group, which is an extended version of the database published by Reineck et al. (2010, 2012). From the originally collected 886 tests, 125 tests were filtered out due to missing important data; 511 tests on slender beams remained. The sanctioned set of criteria were kept, which had been defined in previous

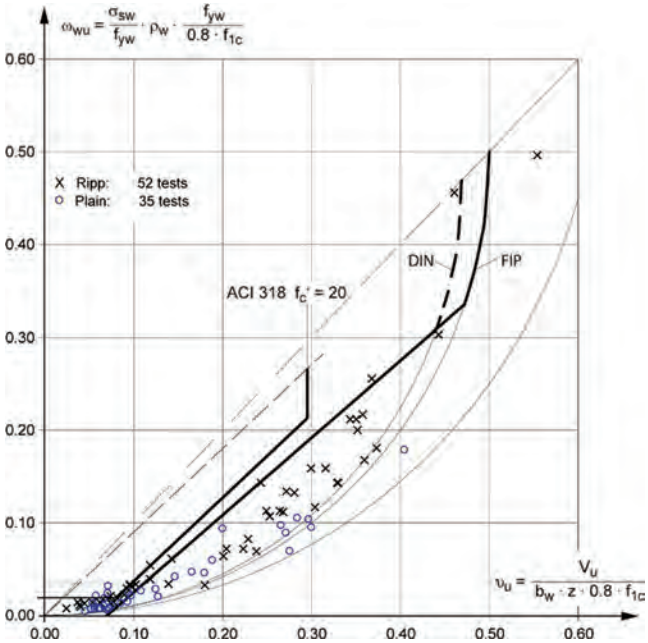

Fig. 10—142 tests of small dataset (A2+A3) compared with shear design of ACI 318-11 (ACI Committee 318 2011), DIN 1045-1 (2008), and FIP Recommendations (1999).

databases since 1999 for accepting a test result into an "evaluation-level" database. After applying basic filtering criteria including a flexural failure check, a subset of 180 tests remained. Of those, 23 had reinforcement ratios below the minimum required by ACI 318-11, resulting in an evaluation data set with 157 tests. The comparisons with these tests showed that the shear strengths of ACI 318-11 are very conservative (Fig. 8), and this observation particularly applies to the maximum limit for shear. If comparisons are performed with design relationships assuming yielding of the stirrups such as in the commonly used relationship $V = V_c + V_s$, however, then the small dataset with 87 tests should be used, because only for these either stirrup stresses were reported or clear statements were found that the stirrups yielded. Likewise, four tests were eliminated because they had excessive failure loads (kon9), and nine tests were eliminated because they did not comply with the anchorage check at the end support. Thus, these 87 tests are indisputable tests that can be used for the comparison of the test results with shear design relationships assuming yielding of the stirrups such as in DIN 1045-1 and FIP Recommendations (1999), which fit well with the test results.

Finally, it should be noted that the MS Excel version of the evaluation database is provided to the research community with this paper. The original collection database and the control database are passed on to interested researchers as PDF and spreadsheet versions under the conditions described in Appendix A,* and can be obtained by contacting the first or the fourth author of this paper. The authors acknowledge that the datasets might still contain errors despite several rounds of checking, and therefore would appreciate if these are reported to the ACI-DAfStb group so that an updated version of the database can be provided.

*The Appendix is available at **www.concrete.org/publications** in PDF format, appended to the online version of the published paper. It is also available in hard copy from ACI headquarters for a fee equal to the cost of reproduction plus handling at the time of the request.

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	ACI-DAfStb Large Evalution Database vsw-RC_A2b-A3b_sl																							
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
					Querschnittswerte - Section properties											Laststellung und Geometrie - loading and geometry							
	Nr.	Lit.	Bez.	Einheiten	b	b_w	h	h_f	$h_{h,top}$	h_w	h_h	$h_{h,bot}$	x	A_c	z_{c2}	a_A	a_F	b_A	L	c	a	a/d	c_c
				E	b	bw	h	hf	hh,top	hw	hft	hh,bot	bft	Ac	zc2	aa	af	ba	L	c	a	kap	cc
	-	-	-	SI / A	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm²	mm	mm	mm	mm	mm	mm	mm	-	mm
Ahmad_1996_003_NNW-3	19	Ahmad;	NNW-3	A	127,0	127,0	254,0							32258	127,0	50,8	31,8	127,0	1219,2	31,8	593,7	2,92	28,6
Ahmad_1996_006_NHW-3	22	Xie;	NHW-3	A	127,0	127,0	254,0							32258	127,0	50,8	31,8	127,0	1188,7	31,8	578,5	2,92	28,6
Ahmad_1996_007_NHW-3a	23	Yu (1996)	NHW-3a	A	127,0	127,0	254,0							32258	127,0	50,8	31,8	127,0	1188,7	31,8	578,5	2,92	28,6
Ahmad_1996_008_NHW-3b	24		NHW-3b	A	127,0	127,0	254,0							32258	127,0	50,8	31,8	127,0	1188,7	31,8	578,5	2,92	28,6
Ahmad_1996_009_NHW-4	25		NHW-4	A	127,0	127,0	254,0							32258	127,0	50,8	31,8	127,0	1585,0	31,8	776,6	3,92	28,6
Angelakos_1999_001_DB0.530M	40	Angelakos; Bentz;	DB0.530M	SI	300,0	300,0	1000,0							300000	500,0	152,0	76,0	308,8	5400,0	76,0	2662,0	2,88	
Angelakos_1999_004_DB140M	42	Collins P. (1999)	DB140M	SI	300,0	300,0	1000,0							300000	500,0	152,0	76,0	308,8	5400,0	76,0	2662,0	2,88	
Aparicio_1997_001_VHA	45	Aparicio; Calavera	VHA	SI	585,0	75,0	800,0	140,0	87,1	285,8	200,0	87,1	585,0	277821	420,5			600,0	12000,0	0,0	2400,0	3,41	
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Bhal_1968_001_B1S	62	Bhal (1968)	B1S	SI	240,0	240,0	350,0							84000	175,0	180,0	180,0	300,0	1800,0	180,0	810,0	2,70	
Bhal_1968_002_B2S	63		B2S	SI	240,0	240,0	650,0							156000	325,0	180,0	240,0	300,0	3600,0	240,0	1680,0	2,80	
Bhal_1968_003_B3S	64		B3S	SI	240,0	240,0	950,0							228000	475,0	210,0	270,0	300,0	5400,0	270,0	2565,0	2,85	
Bhal_1968_004_B4S	65		B4S	SI	240,0	240,0	1250,0							300000	625,0	210,0	300,0	300,0	7200,0	300,0	3450,0	2,88	
Bresler_1963_001_A-1	111	Bresler;	A-1	A	307,3	307,3	561,3							172522	280,7	127,0	152,4	0,0	3657,6	152,4	1752,6	3,76	42,9
Bresler_1963_007_C-1	117	Scordelis (1963)	C-1	A	154,9	154,9	558,8							86580	279,4	127,0	152,4	0,0	3657,6	152,4	1752,6	3,78	42,9
Cladera_2002_010_H 100/2	132	Cladera; Mari (2002)	H 100/2	SI	200,0	200,0	400,0							80000	200,0	150,0	75,0	220,0	2160,0	75,0	1042,5	2,95	25,0
Cederwall_1974_003_734-46	137	Cederwall; Hedman; Losberg (1974)	734-46	SI	139,0	139,0	260,0							36140	130,0			50,0	3000,0	1800,0	600,0	2,56	14,0
Guidi_1963_001_V	213	Guidi; Radogna (1963)	V	SI	250,0	80,0	400,0	60,0	40,0	200,0	60,0	40,0	250,0	59200	200,0	100,0		150,0			1500,0	3,95	
Guralnik_1960_004_IB-2R	217	Guralnik (1960)	IB-2R	A	584,2	177,8	387,4	101,6		285,8				110161	140,1	152,4	76,2	304,8	2463,8	635,0	914,4	2,97	
Guralnik_1960_005_IC-1R	218		IC-1R	A	584,2	177,8	387,4	101,6		285,8				110161	140,1	152,4	76,2	304,8	2438,4	609,6	914,4	3,06	
Guralnik_1960_006_IC-2R	219		IC-2R	A	584,2	177,8	387,4	101,6		285,8				110161	140,1	152,4	76,2	304,8	2489,2	660,4	914,4	2,95	
Guralnik_1960_008_ID-2R	221		ID-2R	A	584,2	177,8	387,4	101,6		285,8				110161	140,1	152,4	76,2	304,8	2514,6	685,8	914,4	2,99	
Hamadi_1976_002_GT-2	243	Hamadi (1976)	GT-2	SI	500,0	120,0	400,0	100,0		300,0				86000	133,7	150,0	75,0	250,0	2500,0	75,0	1212,5	3,46	22,0
Hamadi_1976_004_GT-4	245	Hamadi, Regan (1980)	GT-4	SI	500,0	120,0	400,0	100,0		300,0				86000	133,7	150,0	75,0	250,0	2500,0	75,0	1212,5	3,46	22,0
Hamadi_1976_005_GT-5	246		GT-5	SI	500,0	120,0	400,0	100,0		300,0				86000	133,7	150,0	75,0	250,0	2500,0	75,0	1212,5	3,46	22,0
Hegger_2001_001_SVB 3b	247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	SI	220,0	180,0	500,0							100000	241,7			200,0	3000,0		1500,0	3,33	
Hegger_2003_001_NSC 3L	248	Hegger, J; Görtz, S. (2003)	NSC 3L	SI	400,0	80,0	500,0	90,0	20,0	280,0	90,0	20,0	400,0	104000	250,0	100,0	100,0	450,0	4500,0	1500,0	1500,0	3,19	
Hegger_2003_002_NSC 3R	249		NSC 3R	SI	400,0	80,0	500,0	90,0	20,0	280,0	90,0	20,0	400,0	104000	250,0	100,0	100,0	450,0	3000,0	100,0	1450,0	3,09	
Johnson_1989_001_1	260	Johnson;	1	A	304,8	304,8	609,6							185806	304,8	101,6	228,6	235,0	4254,5	914,4	1670,1	3,10	25,4
Johnson_1989_002_2	261	Ramirez (1989)	2	A	304,8	304,8	609,6							185806	304,8	101,6	228,6	235,0	4254,5	914,4	1670,1	3,10	25,4
Johnson_1989_005_5	264		5	A	304,8	304,8	609,6							185806	304,8	101,6	228,6	235,0	4254,5	914,4	1670,1	3,10	25,4

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2	3	4	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
			Längsbewehrung (Zugbewehrung) - longitudinal tensile reinforcement														compr. Reinforcement					Bügelbewehrung - Stirrups										
Nr.	Lit.	Bez.	d_s	d	ns	Φ_{st}	f_{sr}	A_s	α_s	ρ_s	ρ_{sw}	f_{sy}	ε_{sy}	f_{sk}	f_{sk}/f_{sy}	ε_{euk}	d_{s2}	n_{s2}	Φ_{st2}	A_{s2}	f_{sy2}	Φ_w	n_{sw}	A_{sw}	f_{rw}	s_w	s_w / h	sw/d	z_{w1}	z_{w2}	ρ_w	f_{yw}
			ds	d	ns	dst	fR	As	alphaa	rhos	rhosw	fsy	esy	ftk	ftk/fsy	euk	ds2	ns2	dst2	As2	fsy2	dw	nsw	Asw	frw	sw	sw/h		zw1	zw2	rhow	fyw
-	-	-	mm	mm	-	mm	-	mm²	-	%	%	MPa	‰	MPa	-	‰	mm	-	mm	mm²	MPa	mm	-	mm²	-	mm	-	-	mm	mm	%	MPa
19	Ahmad;	NNW-3	203,2	203,2	2	19,1	r	823,4	1,00	3,1907	3,1907	420,7	2,10		0,00		38,1	2	12,7	253,4	421	6,4	2	63,3	0	101,6	0,40	0,50	171	146	0,4909	324,1
22	Xie;	NHW-3	198,1	198,1	4	19,1	r	1140,1	1,00	4,5311	4,5311	420,7	2,10		0,00		38,1	2	12,7	253,4	421	6,4	2	63,3	0	99,1	0,39	0,50	166	141	0,5035	324,1
23	Yu (1996)	NHW-3a	198,1	198,1	4	19,1	r	1140,1	1,00	4,5311	4,5311	420,7	2,10		0,00		38,1	2	12,7	253,4	421	6,4	2	63,3	0	76,2	0,30	0,38	166	141	0,6545	324,1
24		NHW-3b	198,1	198,1	4	19,1	r	1140,1	1,00	4,5311	4,5311	420,7	2,10		0,00		38,1	2	12,7	253,4	421	6,4	2	63,3	0	63,5	0,25	0,32	166	141	0,7854	324,1
25		NHW-4	198,1	198,1	4	19,1	r	1140,1	1,00	4,5311	4,5311	420,7	2,10		0,00		38,1	2	12,7	253,4	421	6,4	2	63,3	0	99,1	0,39	0,50	166	141	0,5035	324,1
40	Angelakos; Bentz;	DB0.530M	925,0	925,0	2	29,9	r	1400,0	1,00	0,5045	0,5045	550,0	2,75	710,00	1,29		75,0	2		400,0	483	9,5	1	71,3	r	300,0	0,30	0,32	877	0	0,0792	508,0
42	Collins P. (1999)	DB140M	925,0	925,0	4	29,9	r	2800,0	1,00	1,0090	1,0090	550,0	2,75	710,00	1,29		75,0	2		400,0	483	9,5	1	71,3	r	300,0	0,30	0,32	877	0	0,0792	508,0
45	Aparicio; Calavera	VHA	704,0	704,0	24	20,0	r	7640,4	1,00	1,8552	14,4704	500,3	2,50		0,00			4	12,0	653,5	500	12,0	2	226,2	r	170,0	0,21	0,24	644	0	1,7741	500,3
46	del Pozo (1997)	HA-45	1369,0	1369,0	24	32,0	r	19301,9	1,00	0,9724	9,3995	520,2	2,60	636,77	1,22			8	10,0	1030,4	500	12,0	2	226,2	r	90,0	0,06	0,07	665	0	1,6755	582,0
62	Bhal (1968)	B1S	300,0	300,0	2	24,0	r	904,8	1,00	1,2566	1,2566	425,8	2,13	513,06	1,21	15	25,0	2	10,0	157,1	443	6,0	2	56,5	r	160,0	0,46	0,53	270	0	0,1473	441,5
63		B2S	600,0	600,0	4	24,0	r	1809,6	1,00	1,2566	1,2566	425,8	2,13	513,06	1,21	15	25,0	2	10,0	157,1	443	6,0	2	56,5	r	160,0	0,25	0,27	570	0	0,1473	441,5
64		B3S	900,0	900,0	6	24,0	r	2714,3	1,00	1,2566	1,2566	425,8	2,13	513,06	1,21	15	25,0	2	10,0	157,1	443	6,0	2	56,5	r	160,0	0,17	0,18	870	0	0,1473	441,5
65		B4S	1200,0	1200,0	8	24,0	r	3619,1	1,00	1,2566	1,2566	425,8	2,13	513,06	1,21	15	25,0	2	10,0	157,1	443	6,0	2	56,5	r	160,0	0,13	0,13	1170	0	0,1473	441,5
111	Bresler;	A-1	466,1	466,1	4	28,7	r	2578,9	0,01	1,8003	1,8003	555,2	2,78	957,93	1,73		50,8	2	12,7	253,4	346	6,4	2	63,3	r	209,6	0,37	0,45	434	380	0,0983	325,5
117	Scordelis (1963)	C-1	463,6	463,6	2	28,7	r	1289,5	0,01	1,7953	1,7953	555,2	2,78	957,93	1,73		50,8	2	12,7	253,4	346	6,4	2	63,3	r	209,6	0,38	0,45	432	378	0,1951	325,5
132	Cladera; Mari (2002)	H 100/2	353,0	353,0	2	32,0	r	1608,5	1,00	2,2783	2,2783	500,0	2,50		0,00		35,0	2	8,0	100,5	540	6,0	2	56,5	r	165,0	0,41	0,47	323	303	0,1714	530,0
137	Cederwall; Hedman; Losberg (1974)	734-46	234,0	234,0	3	12,0	r	339,3		1,0431	1,0431	818,5	4,09	848,24	1,04			2	8,0	100,5		6,0	2	56,5	r	100,0	0,38	0,43	204	206	0,4068	436,5
213	Guidi; Radogna (1963)	V	380,0	380,0	8	12,0	r	1017,9	1,00	1,0714	3,3483	490,5	2,45	755,37	1,54		20,0	4	6,0	113,1	491	6,0	2	56,5	r	100,0	0,25	0,26	350	0	0,7069	490,5
217	Guralnik (1960)	IB-2R	307,8	307,8	2	22,2	r	775,9	1,00	0,4314	1,4175	583,5	2,92	1011,72	1,73			2	22,2	775,9	584	6,4	2	63,3	r	146,1	0,38	0,47	276	0	0,2439	529,0
218		IC-1R	299,2	299,2	2	32,3	r	2410,4	1,00	1,3790	4,5309	576,9	2,88	967,76	1,68			2	32,3	2410,4	577	12,7	2	253,4	r	88,9	0,23	0,30	236	0	1,6029	340,7
219		IC-2R	309,6	309,6	2	32,3	r	2410,4	1,00	1,3326	4,3785	576,9	2,88	967,76	1,68			2	32,3	2410,4	577	6,4	2	63,3	r	146,1	0,38	0,47	278	0	0,2439	529,0
221		ID-2R	306,1	306,1	2	22,2	r	1345,9	1,00	0,7527	2,4733	601,6	3,01	1020,49	1,70			2	22,2	1345,9	602	6,4	2	63,3	r	146,1	0,38	0,48	274	0	0,2439	529,0
243	Hamadi (1976)	GT-2	350,0	350,0	4	20,0	r	1256,6	1,00	0,7181	2,9920	400,0	2,00	565,00	1,41			4	12,0	452,4	350	6,0	2	56,5	r	125,0	0,31	0,36	320	306	0,3770	320,0
245	Hamadi, Regan (1980)	GT-4	350,0	350,0	4	20,0	r	1256,6	1,00	0,7181	2,9920	655,0	3,28	820,00	1,25			4	12,0	452,4	350	8,0	2	100,5	r	160,0	0,40	0,46	310	306	0,5236	480,0
246		GT-5	350,0	350,0	4	20,0	r	1256,6	1,00	0,7181	2,9920	655,0	3,28	820,00	1,25			4	12,0	452,4	350	8,0	2	100,5	r	125,0	0,31	0,36	310	306	0,6702	480,0
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	450,0	450,0	6	25,0	r	2945,2	1,00	2,9750	3,6361	532,0	2,66	673,00	1,27							8,0	2	100,5	r	100,0	0,20	0,22	410	0	0,5585	514,0
248	Hegger, J; Görtz, S. (2003)	NSC 3L	470,0	470,0	5	20,0	r	3581,4	1,00	1,9050	9,5250	500,0	2,50		0,00							6,0	2	56,5	r	100,0	0,20	0,21	440	0	0,7069	550,0
249		NSC 3R	470,0	470,0	5	20,0	r	3581,4	1,00	1,9050	9,5250	500,0	2,50		0,00							6,0	2	56,5	r	100,0	0,20	0,21	440	0	0,7069	550,0
260	Johnson;	1	538,7	538,7	5	32,3	r	4086,3	0,70	2,4885	2,4885	524,8	2,62		0,00		46,0	2	28,7	1289,5	540	6,4	2	63,3	r	133,4	0,22	0,25	507	488	0,1558	479,3
261	Ramirez (1989)	2	538,7	538,7	5	32,3	r	4086,3	0,70	2,4885	2,4885	524,8	2,62		0,00		46,0	2	28,7	1289,5	540	6,4	2	63,3	r	266,7	0,44	0,50	507	488	0,0779	479,3
264		5	538,7	538,7	5	32,3	r	4086,3	0,70	2,4885	2,4885	524,8	2,62		0,00		46,0	2	28,7	1289,5	540	6,4	2	63,3	r	133,4	0,22	0,25	507	488	0,1558	479,3

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2	3	4	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76
							compressive strength									mech. reinforcement ratios								
Nr.	Lit.	Bez.	$f_{yw} \cdot \rho_w/100$	f_{wtk}	f_{wtk}/f_{yw}	ϵ_{wuk}	Φba	f_{1c}		f_{cwu}	$f_{1ct,test}$		$\beta_{ct,test}$	$f_{1ct,m,cal}$	$\beta_{ct,cal}$	ω_s	ω_b	ω_{wy}	g	V_g	F	$V_{u,g+F,Rep}$	$V_{u,Rep}$	$V_{u,g+F}$
			fyw_rhow_100	fwtk	fwtk/fyw	ewuk	diaa	f1c	Method	fcwu	f1cttest	Method	betacttest	f1ctmcal	betactcal	oms	oml	omwy	g	Vg	F	Vu_F+g,rep	Vu_Rep	Vu_gF
-	-	-	MPa	MPa	-	‰	mm	MPa	-	MPa	MPa	-	-	MPa	-	-	-	-	kN/m	kN	kN	kN	kN	kN
19	Ahmad;	NNW-3	1,59		0,00		19,1	39,46	cyl	31,57	0,00		0,0000	3,39	0,0858	0,3401	0,3401	0,0504	0,77	0,24	87,03		87,27	87,27
22	Xie;	NHW-3	1,63		0,00		19,1	95,17	cyl	76,14	0,00		0,0000	5,11	0,0537	0,2003	0,2003	0,0214	0,77	0,24	102,37		102,61	102,61
23	Yu (1996)	NHW-3a	2,12		0,00		19,1	87,18	cyl	69,74	0,00		0,0000	4,96	0,0568	0,2187	0,2187	0,0304	0,77	0,24	108,22		108,46	108,46
24		NHW-3b	2,55		0,00		19,1	99,99	cyl	80,00	0,00		0,0000	5,20	0,0520	0,1906	0,1906	0,0318	0,77	0,24	122,54		122,78	122,78
25		NHW-4	1,63		0,00		19,1	95,74	cyl	76,59	0,00		0,0000	5,12	0,0535	0,1991	0,1991	0,0213	0,77	0,31	93,72		94,03	94,03
40	Angelakos; Bentz;	DB0.530M	0,40	778,00	1,53		10,0	30,40	cyl	24,32	0,00		0,0000	2,78	0,0916	0,0913	0,0913	0,0165	7,20	9,86	256,00		265,86	265,86
42	Collins P. (1999)	DB140M	0,40	778,00	1,53		10,0	36,10	cyl	28,88	0,00		0,0000	3,17	0,0878	0,1537	0,1537	0,0139	7,20	9,50	270,00		279,50	279,50
45	Aparicio; Calavera	VHA	8,88		0,00			42,25	cyl	33,80	0,00		0,0000	3,56	0,0843	0,2197	0,2197	0,2626	6,67	8,00			886,82	894,83
46	del Pozo (1997)	HA-45	9,75	669,63	0,00			41,14	cyl	32,91	4,21	sp	0,1024	3,49	0,0849	0,1183	0,1183	0,2963	19,92	44,32			3384,45	3428,77
62	Bhal (1968)	B1S	0,65	567,51	1,29		30,0	25,88	cu	20,70	2,26	fl	0,0873	2,46	0,0950	0,2067	0,2067	0,0314	2,02	1,17	127,53		128,70	128,70
63		B2S	0,65	567,51	1,29		30,0	24,18	cu	19,34	2,39	fl	0,0988	2,33	0,0964	0,2213	0,2213	0,0336	3,74	1,87	247,70		249,57	249,57
64		B3S	0,65	567,51	1,29		30,0	25,34	cu	20,27	2,67	fl	0,1055	2,42	0,0955	0,2111	0,2111	0,0321	5,47	3,97	365,42		369,39	369,39
65		B4S	0,65	567,51	1,29		30,0	24,49	cu	19,59	2,14	fl	0,0875	2,36	0,0962	0,2185	0,2185	0,0332	7,20	13,50	459,11		472,61	472,61
111	Bresler;	A-1	0,32	429,66	1,32		19,1	22,87	cyl	18,29	2,57	fl	0,1123	2,23	0,0975	0,4371	0,4371	0,0175	4,14	3,94	233,52		237,46	237,46
117	Scordelis (1963)	C-1	0,64	429,66	1,32		19,1	28,11	cyl	22,49	2,81	fl	0,1000	2,62	0,0933	0,3546	0,3546	0,0282	2,08	1,98	155,68		157,66	157,66
132	Cladera; Mari (2002)	H 100/2	0,91	680,00	1,28		12,0	82,65	cyl	66,12	0,00		0,0000	4,86	0,0588	0,1378	0,1378	0,0137	1,92	1,07			225,55	226,62
137	Cederwall; Hedman; Losberg (1974)	734-46	1,78		0,00			28,69	cu	22,96	0,00		0,0000	2,66	0,0929	0,2975	0,2975	0,0774	0,87	1,04			95,65	96,69
213	Guidi; Radogna (1963)	V	3,47	755,37	1,54		15,0	32,37	cu	25,90	0,00		0,0000	2,92	0,0902	0,1623	0,1623	0,1339	1,42	1,07			122,63	123,69
217	Guralnik (1960)	IB-2R	1,29	591,03	1,12	1	15,9	15,99	cyl	12,79	1,42	fl	0,0891	1,65	0,1035	0,1575	0,1575	0,1009	2,64	2,05			140,11	142,16
218		IC-1R	5,46	595,66	1,75	2	15,9	32,30	cyl	25,84	2,16	fl	0,0668	2,92	0,0903	0,2463	0,2463	0,2113	2,64	2,01			411,88	413,90
219		IC-2R	1,29	591,03	1,12	1	15,9	32,30	cyl	25,84	2,16	fl	0,0668	2,92	0,0903	0,2380	0,2380	0,0499	2,64	2,08			225,07	227,15
221		ID-2R	1,29	591,03	1,12	1	15,9	32,30	cyl	25,84	2,16	fl	0,0668	2,92	0,0903	0,1402	0,1402	0,0499	2,64	2,12			217,51	219,62
243	Hamadi (1976)	GT-2	1,21	430,00	1,34		20,0	23,63	cu	18,90	0,00		0,0000	2,29	0,0969	0,1216	0,1216	0,0638	2,06	1,33			148,50	149,83
245	Hamadi, Regan (1980)	GT-4	2,51	720,00	1,50		20,0	29,43	cu	23,54	0,00		0,0000	2,72	0,0923	0,1598	0,1598	0,1067	2,06	1,33			227,50	228,83
246		GT-5	3,22	720,00	1,50		20,0	18,56	cu	14,85	0,00		0,0000	1,88	0,1013	0,2534	0,2534	0,2166	2,06	1,33			222,50	223,83
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	2,87		0,00			66,50	cyl	53,20	0,00		0,0000	4,51	0,0678	0,2380	0,2380	0,0540	2,40	1,80			427,00	428,80
248	Hegger, J; Görtz, S. (2003)	NSC 3L	3,89		0,00			27,55	cyl	22,04	0,00		0,0000	2,58	0,0937	0,3457	0,3457	0,1764	2,50	3,74			334,00	337,74
249		NSC 3R	3,89		0,00			27,55	cyl	22,04	0,00		0,0000	2,58	0,0937	0,3457	0,3457	0,1764	2,50	1,93			349,00	350,93
260	Johnson;	1	0,75		0,00		19,1	34,59	cyl	27,67	0,00		0,0000	3,07	0,0888	0,3775	0,3775	0,0270	4,46	5,76	338,50		344,26	344,26
261	Ramirez (1989)	2	0,37		0,00		19,1	34,59	cyl	27,67	0,00		0,0000	3,07	0,0888	0,3775	0,3775	0,0135	4,46	5,76	222,40		228,16	228,16
264		5	0,75		0,00		19,1	53,07	cyl	42,46	0,00		0,0000	4,16	0,0784	0,2461	0,2461	0,0176	4,46	5,76	382,53		388,29	388,29

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2	3	4	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99
			Versuch - Test results								Bruchschnittgrößen			Kontrolle Biegung											
Nr.	Lit.	Bez.	β_r	x_r	$\sigma_{sl,a}$	$\sigma_{sw,meas}$	$\sigma_{sw,ass}$	σ_{sw}	ω_{wu}	andere Bruchart	M_u	μ_u	$\upsilon_{u,test}$	κ_c	σ_{sl}	ξ_{test}	ζ_{test}	z_{test}	ζ_c	z	μ_{flex}	$\mu_{u,flex}$	$\upsilon_{u,flex}$	β_{flex}	Bbem
			betar_meas	xr_meas	xr	sigswmeas	sigswass	sigsw	omwu	andbr	Mu	muu	vutest	kapc	sigsl	xsitest	zetatest	z_test	zeta	z_	muflex	Mu_flex	Vu_flex	betaflex	BB
-	-	-	[°]	mm	mm	MPa	MPa	MPa	-	-	kNm	-	-	-	MPa	-	-	mm	-	mm	-	-	kN	-	-
19	Ahmad;	NNW-3			297	324		324	0,0504		51,81	0,2504	0,1309	0,842	378	0,363	0,818	166	0,798	162	0,2714	56,17	94,61	0,92	
22	Xie;	NHW-3			289			0	0,0000		59,36	0,1251	0,0605	0,619	297	0,228	0,886	176	0,838	166	0,1679	79,66	137,70	0,75	
23	Yu (1996)	NHW-3a			289			0	0,0000		62,74	0,1444	0,0708	0,651	318	0,254	0,873	173	0,832	165	0,1820	79,07	136,69	0,79	
24		NHW-3b			289	324		324	0,0318		71,03	0,1425	0,0707	0,600	365	0,275	0,862	171	0,841	167	0,1603	79,93	138,17	0,89	
25		NHW-4			388	324		324	0,0213		73,03	0,1530	0,0571	0,617	378	0,290	0,855	169	0,839	166	0,1670	79,69	102,62	0,92	
40	Angelakos; Bentz;	DB0.530M			1331	508		508	0,0165		707,71	0,0907	0,0416	0,878	550	0,104	0,948	877	0,948	877	0,0865	675,24	253,66	1,05	BB
42	Collins P. (1999)	DB140M		1380	1380	508		508	0,0139		744,04	0,0803	0,0367	0,856	302	0,099	0,951	879	0,910	842	0,1399	1296,52	487,05	0,57	
45	Aparicio; Calavera	VHA			1200	297		297	0,1557		2128,38	0,1738	0,5638	0,831	449	0,237	0,881	620	0,868	611	0,1907	2335,34	973,06	0,91	
46	del Pozo (1997)	HA-45			2108	357		357	0,1818		14180,23	0,1269	0,5390	0,835	520	0,142	0,929	1272	0,929	1272	0,1099	12284,65	2932,03	1,15	BB
62	Bhal (1968)	B1S		321	321	438		438	0,0312		104,25	0,1865	0,0976	0,896	426	0,231	0,885	265	0,885	265	0,1829	102,24	126,22	1,02	BB
63		B2S		1300	1300	441		441	0,0336		419,29	0,2007	0,1021	0,903	426	0,245	0,878	527	0,878	527	0,1942	405,64	241,45	1,03	BB
64		B3S		1975	1975	441		441	0,0321		947,48	0,1923	0,0956	0,899	426	0,235	0,883	794	0,883	794	0,1863	917,89	357,85	1,03	BB
65		B4S			1725	441		441	0,0332		1630,50	0,1926	0,0953	0,902	426	0,242	0,879	1055	0,879	1055	0,1920	1625,11	471,05	1,00	BB
111	Bresler;	A-1			876			0	0,0000		416,18	0,2726	0,1110	0,909	424	0,368	0,816	380	0,759	354	0,3320	506,79	289,16	0,82	
117	Scordelis (1963)	C-1			876			0	0,0000		276,31	0,2953	0,1220	0,888	555	0,400	0,800	371	0,800	371	0,2838	265,55	151,52	1,04	BB
132	Cladera; Mari (2002)	H 100/2			521	530		530	0,0137		235,14	0,1142	0,0533	0,669	457	0,188	0,906	320	0,897	317	0,1236	254,67	244,29	0,92	
137	Cederwall; Hedman; Losberg (1974)	734-46			300			0	0,0000		57,39	0,2628	0,1540	0,885	818	0,336	0,832	195	0,832	195	0,2475	54,06	90,10	1,06	BB
213	Guidi; Radogna (1963)	V			750			0	0,0000		183,94	0,1574	0,1718	0,871	491	0,186	0,907	345	0,907	345	0,1472	172,03	114,69	1,07	BB
217	Guralnik (1960)	IB-2R			457			0	0,0000		128,12	0,1448	0,2185	0,936	584	0,168	0,916	282	0,916	282	0,1442	127,65	139,60	1,00	BB
218		IC-1R			457			0	0,0000		376,63	0,2229	0,3490	0,871	577	0,283	0,859	257	0,859	257	0,2115	357,25	390,69	1,05	BB
219		IC-2R			457			0	0,0000		205,80	0,1138	0,1702	0,871	297	0,141	0,930	288	0,863	267	0,2055	371,73	406,53	0,55	
221		ID-2R			457			0	0,0000		198,89	0,1125	0,1662	0,871	519	0,139	0,931	285	0,919	281	0,1289	227,88	249,21	0,87	
243	Hamadi (1976)	GT-2			606		320	320	0,0638		180,06	0,1244	0,2005	0,906	400	0,134	0,933	327	0,933	327	0,1134	164,12	135,36	1,10	BB
245	Hamadi, Regan (1980)	GT-4			606		480	480	0,1067		275,84	0,1530	0,2530	0,882	655	0,181	0,909	318	0,909	318	0,1453	261,99	216,08	1,05	BB
246		GT-5			606		480	480	0,2166		269,78	0,2373	0,4133	0,926	655	0,274	0,863	302	0,863	302	0,2187	248,66	205,08	1,08	BB
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b			750		514	514	0,0540		640,50	0,2162	0,1183	0,734	532	0,324	0,838	377	0,838	377	0,1994	590,78	393,85	1,08	BB
248	Hegger, J; Görtz, S. (2003)	NSC 3L			750	0		0	0,0000		501,00	0,2058	0,4651	0,890	343	0,267	0,867	407	0,806	379	0,2786	678,12	452,08	0,74	
249		NSC 3R			725	0		0	0,0000		506,05	0,2079	0,4869	0,890	348	0,270	0,865	407	0,806	379	0,2786	678,12	467,67	0,75	
260	Johnson;	1			835			0	0,0000		574,93	0,1879	0,0865	0,862	298	0,249	0,875	472	0,781	421	0,2948	902,25	540,25	0,64	
261	Ramirez (1989)	2			835			0	0,0000		381,04	0,1245	0,0545	0,862	188	0,157	0,922	496	0,781	421	0,2948	902,25	540,25	0,42	
264		5			835			0	0,0000		648,46	0,1381	0,0617	0,788	326	0,194	0,903	486	0,844	455	0,2077	974,90	583,75	0,67	

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2	3	4	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126
			Querkraft - PT (mit γ=0,80)				Querkraft - PT im Versuch				anchorage																		
Nr.	Lit.	Bez.	$\sin^2\theta_p$	θ_p	$\cot\theta_p$	υ_{up}	γ_{wp}	$\cot\theta_u$	θ_u	υ_u	β_{ib}									f_{1ck}	$f_{cm,cyl}$	f_{ck}	$f^{\prime}c$						
			sin2thp	thp	cotthp	vup	gamwp	cotthu	thu	nueu	betalb	VB	A2+A3	kon34	KONA4	KONA5	A4+A5	Differenz	AF	f1ck	fcm_cyl	fck	fc_prime	Vct_ACI	Vs_ACI	Vcal_sACI	Vmax_ACI	rat_smax	VACI_cal
-	-	-	-	[°]	-	-	-	-	[°]	-	-	-	-	-	-	-	-	-		MPa	MPa	MPa	MPa	kN	kN	kN	kN		kN
19	Ahmad;	NNW-3	0,05	12,97	4,34	0,22	0,60	2,60	21,06	0,31	1,02	VB	0	1	0	0	0	0	AF	35,7	41,5	37,5	39,1	26,82	41,06	67,88	134,08	0,506	67,88
22	Xie;	NHW-3	0,02	8,42	6,76	0,14	0,42	0,00	0,00	0,00	0,62		0	1	0	0	0	0		91,4	100,2	96,2	97,8	41,32	41,06	82,39	206,62	0,399	82,39
23	Yu (1996)	NHW-3a	0,03	10,04	5,65	0,17	0,41	0,00	0,00	0,00	0,68		0	1	0	0	0	0		83,4	91,8	87,8	89,4	39,51	53,38	92,89	197,53	0,470	92,89
24		NHW-3b	0,03	10,28	5,52	0,18	0,40	2,22	24,22	0,15	0,73		1	1	1	0	1	0		96,2	105,3	101,3	102,9	42,38	64,05	106,44	211,92	0,502	106,44
25		NHW-4	0,02	8,39	6,78	0,14	0,40	2,68	20,47	0,14	0,57		1	1	1	0	1	0		91,9	100,8	96,8	98,4	41,45	41,06	82,51	207,26	0,398	82,51
40	Angelakos; Bentz;	DB0.530M	0,02	7,39	7,71	0,13	0,33	2,51	21,70	0,10	0,80		1	1	0	1	1	0		26,6	32,0	28,0	29,6	250,76	111,61	362,37	1253,79	0,289	362,37
42	Collins P. (1999)	DB140M	0,01	6,78	8,41	0,12	0,31	2,63	20,79	0,09	0,37		1	1	0	1	1	0		32,3	38,0	34,0	35,6	275,00	111,61	386,61	1375,00	0,281	386,61
45	Aparicio; Calavera	VHA	0,26	30,83	1,68	0,44	1,28	3,62	15,44	1,76	0,15		0	0	0	0	0	0		38,4	44,5	40,5	42,1	56,88	468,65	525,53	284,40	1,848	284,40
46	del Pozo (1997)	HA-45	0,30	32,98	1,54	0,46	1,18	2,97	18,64	1,42	0,36		0	1	0	0	0	0		37,3	43,3	39,3	40,9	218,12	2002,57	2220,69	1090,62	2,036	1090,62
62	Bhal (1968)	B1S	0,03	10,21	5,55	0,17	0,56	3,13	17,71	0,27	0,69		1	1	0	1	1	0		22,1	27,2	23,2	24,8	59,60	46,81	106,41	298,02	0,357	106,41
63		B2S	0,03	10,56	5,36	0,18	0,57	3,04	18,22	0,28	0,56		1	1	0	1	1	0		20,4	25,5	21,5	23,1	114,83	93,61	208,45	574,17	0,363	208,45
64		B3S	0,03	10,32	5,49	0,18	0,54	2,98	18,55	0,25	0,47		1	1	0	1	1	0		21,5	26,7	22,7	24,3	176,75	140,42	317,17	883,75	0,359	317,17
65		B4S	0,03	10,50	5,40	0,18	0,53	2,87	19,20	0,25	0,44		1	1	0	1	1	0		20,7	25,8	21,8	23,4	231,28	187,23	418,51	1156,42	0,362	418,51
111	Bresler;	A-1	0,02	7,60	7,49	0,13	0,85	0,00	0,00	0,00	0,01		0	1	0	0	0	0		19,1	24,1	20,1	21,7	110,75	45,86	156,61	553,76	0,283	156,61
117	Scordelis (1963)	C-1	0,03	9,67	5,87	0,17	0,74	0,00	0,00	0,00	0,01		0	1	0	0	0	0		24,3	29,6	25,6	27,2	62,20	45,61	107,81	310,99	0,347	107,81
132	Cladera; Mari (2002)	H 100/2	0,01	6,73	8,47	0,12	0,46	3,88	14,44	0,18	0,56		1	1	1	0	1	0		78,9	87,0	83,0	84,6	107,85	64,12	171,97	539,27	0,319	171,97
137	Cederwall; Hedman; Losberg (1974)	734-46	0,08	16,15	3,45	0,27	0,58	0,00	0,00	0,00	3,43	VB	0	1	0	0	0	0		24,9	30,2	26,2	27,8	28,49	57,77	86,25	142,43	0,606	86,25
213	Guidi; Radogna (1963)	V	0,13	21,46	2,54	0,34	0,50	0,00	0,00	0,00	0,34		0	0	0	0	0	0		28,6	34,1	30,1	31,7	28,42	105,40	133,82	142,09	0,942	133,82
217	Guralnik (1960)	IB-2R	0,10	18,52	2,99	0,30	0,73	0,00	0,00	0,00	1,45	VB	0	1	0	0	0	0		12,2	16,8	12,8	14,4	34,53	70,62	105,15	172,66	0,609	105,15
218		IC-1R	0,21	27,37	1,93	0,41	0,85	0,00	0,00	0,00	1,26	VB	0	1	0	0	0	0		28,5	34,0	30,0	31,6	49,67	290,51	340,18	248,35	1,370	248,35
219		IC-2R	0,05	12,91	4,36	0,22	0,78	0,00	0,00	0,00	0,63		0	1	0	0	0	0		28,5	34,0	30,0	31,6	51,40	71,03	122,43	257,00	0,476	122,43
221		ID-2R	0,05	12,91	4,36	0,22	0,76	0,00	0,00	0,00	0,75		0	1	0	0	0	0		28,5	34,0	30,0	31,6	50,81	70,21	121,02	254,05	0,476	121,02
243	Hamadi (1976)	GT-2	0,06	14,63	3,83	0,24	0,82	3,14	17,66	0,56	0,57		1	1	1	0	1	0		19,8	24,9	20,9	22,5	33,07	50,67	83,73	165,33	0,506	83,73
245	Hamadi, Regan (1980)	GT-4	0,11	19,07	2,89	0,31	0,82	2,37	22,88	0,57	0,74		1	1	1	0	1	0		25,6	31,0	27,0	28,6	37,29	105,56	142,85	186,46	0,766	142,85
246		GT-5	0,22	27,74	1,90	0,41	1,00	1,91	27,66	0,80	1,06	VB	0	1	0	0	0	0	AF	14,8	19,5	15,5	17,1	28,88	135,11	163,99	144,40	1,136	144,40
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	0,05	13,43	4,19	0,23	0,52	2,19	24,53	0,25	0,54		1	1	1	0	1	0		62,7	70,0	66,0	67,6	110,61	232,53	343,14	553,06	0,620	343,14
248	Hegger, J; Görtz, S. (2003)	NSC 3L	0,18	24,83	2,16	0,38	1,22	0,00	0,00	0,00	0,19		0	0	0	0	0	0		23,8	29,0	25,0	26,6	32,21	146,18	178,39	161,04	1,108	161,04
249		NSC 3R	0,18	24,83	2,16	0,38	1,28	0,00	0,00	0,00	0,20		0	0	0	0	0	0		23,8	29,0	25,0	26,6	32,21	146,18	178,39	161,04	1,108	161,04
260	Johnson;	1	0,03	9,46	6,00	0,16	0,53	0,00	0,00	0,00	0,41		0	1	0	0	0	0		30,8	36,4	32,4	34,0	159,06	122,65	281,71	795,30	0,354	281,71
261	Ramirez (1989)	2	0,01	6,67	8,55	0,12	0,47	0,00	0,00	0,00	0,27		0	1	0	0	0	0		30,8	36,4	32,4	34,0	159,06	61,32	220,39	795,30	0,277	220,39
264		5	0,02	7,62	7,47	0,13	0,47	0,00	0,00	0,00	0,33		0	1	0	0	0	0		49,3	55,9	51,9	53,5	199,42	122,65	322,06	997,08	0,323	322,06

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2	3	4	127
Nr.	Lit.	Bez.	gmod_ACI
-	-	-	
19	Ahmad;	NNW-3	1,286
22	Xie;	NHW-3	1,245
23	Yu (1996)	NHW-3a	1,168
24		NHW-3b	1,154
25		NHW-4	1,140
40	Angelakos; Bentz;	DB0.530M	0,734
42	Collins P. (1999)	DB140M	0,723
45	Aparicio; Calavera	VHA	3,118
46	del Pozo (1997)	HA-45	3,103
62	Bhal (1968)	B1S	1,209
63		B2S	1,197
64		B3S	1,165
65		B4S	1,129
111	Bresler;	A-1	1,516
117	Scordelis (1963)	C-1	1,462
132	Cladera; Mari (2002)	H 100/2	1,312
137	Cederwall; Hedman; Losberg (1974)	734-46	1,109
213	Guidi; Radogna (1963)	V	0,916
217	Guralnik (1960)	IB-2R	1,332
218		IC-1R	1,658
219		IC-2R	1,838
221		ID-2R	1,797
243	Hamadi (1976)	GT-2	1,773
245	Hamadi, Regan (1980)	GT-4	1,593
246		GT-5	1,541
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	1,244
248	Hegger, J; Görtz, S. (2003)	NSC 3L	2,074
249		NSC 3R	2,167
260	Johnson;	1	1,222
261	Ramirez (1989)	2	1,035
264		5	1,206

				E	b	bw	h	hf	hh,top	hw	hft	hh,bot	bft	Ac	zc2	aa	af	ba	L	c	a	kap	cc
	-	-	-	SI / A	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm²	mm	mm	mm	mm	mm	mm	mm	-	mm
Kong_1997_001_S1-1	359	Kong; Rangan (1997)	S1-1	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_002_S1-2	360		S1-2	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_003_S1-3	361		S1-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_004_S1-4	362		S1-4	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_005_S1-5	363		S1-5	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_006_S1-6	364		S1-6	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_007_S2-1	365		S2-1	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_008_S2-2	366		S2-2	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_009_S2-3	367		S2-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_010_S2-4	368		S2-4	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_011_S2-5	369		S2-5	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_013_S3-1	371		S3-1	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	360,0	1980,0	500,0	740,0	2,49	35,0
Kong_1997_014_S3-2	372		S3-2	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	360,0	1980,0	500,0	740,0	2,49	35,0
Kong_1997_015_S3-3	373		S3-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,49	35,0
Kong_1997_016_S3-4	374		S3-4	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,49	35,0
Kong_1997_017_S3-5	375		S3-5	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	380,0	1940,0	500,0	720,0	2,51	35,0
Kong_1997_018_S3-6	376		S3-6	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	380,0	1940,0	500,0	720,0	2,51	35,0
Kong_1997_019_S4-1	377		S4-1	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	350,0	3100,0	500,0	1300,0	2,48	35,0
Kong_1997_021_S4-3	379		S4-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	2160,0	500,0	830,0	2,50	35,0
Kong_1997_022_S4-4	380		S4-4	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	470,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_024_S4-6	382		S4-6	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	450,0	1500,0	500,0	500,0	2,53	35,0
Kong_1997_025_S5-1	383		S5-1	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	320,0	2260,0	500,0	880,0	3,01	35,0
Kong_1997_026_S5-2	384		S5-2	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	400,0	2100,0	500,0	800,0	2,74	35,0
Kong_1997_027_S5-3	385		S5-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_031_S7-1	389		S7-1	SI	250,0	250,0	350,0							87500	175,0	100,0	50,0	380,0	1940,0	50,0	945,0	3,40	35,0
Kong_1997_032_S7-2	390		S7-2	SI	250,0	250,0	350,0							87500	175,0	100,0	50,0	380,0	1940,0	50,0	945,0	3,40	35,0
Kong_1997_033_S7-3	391		S7-3	SI	250,0	250,0	350,0							87500	175,0	100,0	50,0	380,0	1940,0	50,0	945,0	3,40	35,0
Kong_1997_034_S7-4	392		S7-4	SI	250,0	250,0	350,0							87500	175,0	100,0	50,0	380,0	1940,0	50,0	945,0	3,40	35,0
Kong_1997_035_S7-5	393		S7-5	SI	250,0	250,0	350,0							87500	175,0	100,0	50,0	380,0	1940,0	50,0	945,0	3,40	35,0
Kong_1997_036_S7-6	394		S7-6	SI	250,0	250,0	350,0							87500	175,0	100,0	50,0	380,0	1940,0	50,0	945,0	3,40	35,0
Kong_1997_037_S8-1	395		S8-1	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_038_S8-2	396		S8-2	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_039_S8-3	397		S8-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_040_S8-4	398		S8-4	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_041_S8-5	399		S8-5	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_042_S8-6	400		S8-6	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Krefeld_1966_010_26-1	420	Krefeld; Thurston (1966)	26-1	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_012_29b-1	422		29b-1	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_015_29a-2	425		29a-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_016_29b-2	426		29b-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_017_29c-2	427		29c-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_018_29d-2	428		29d-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_020_29f-2	430		29f-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_021_29g-2	431		29g-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_024_24.5-3	434		24.5-3	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_034_39-3	444		39-3	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	

			ds	d	ns	dst	fR	As	alphaa	rhos	rhosw	fsy	esy	ftk	ftk/fsy	euk	ds2	ns2	dst2	As2	fsy2	dw	nsw	Asw	frw	sw	sw/h		zw1	zw2	rhow	fyw
-	-	-	mm	mm	-	mm	-	mm²	-	%	%	MPa	‰	MPa	-	‰	mm	-	mm	mm²	MPa	mm	-	mm²	-	mm	-	-	mm	mm	%	MPa
359	Kong;	S1-1	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
360	Rangan (1997)	S1-2	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
361		S1-3	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
362		S1-4	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
363		S1-5	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
364		S1-6	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
365		S2-1	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	150,0	0,43	0,51	267	222	0,1022	569,0
366		S2-2	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	125,0	0,36	0,43	267	222	0,1227	569,0
367		S2-3	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
368		S2-4	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
369		S2-5	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	75,0	0,21	0,26	267	222	0,2044	569,0
371		S3-1	297,0	297,0	2	28,0	r	1232,0	0,70	1,6593	1,6593	450,0	2,25	562,00	1,25		41,0	2	12,0	224,0	462	4,0	2	24,6	0	100,0	0,29	0,34	277	227	0,0985	632,0
372		S3-2	297,0	297,0	2	28,0	r	1232,0	0,70	1,6593	1,6593	450,0	2,25	562,00	1,25		41,0	2	12,0	224,0	462	4,0	2	24,6	0	100,0	0,29	0,34	277	227	0,0985	632,0
373		S3-3	293,0	293,0	2	36,0	r	2046,0	0,70	2,7932	2,7932	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,0	2	24,6	0	100,0	0,29	0,34	273	223	0,0985	632,0
374		S3-4	293,0	293,0	2	36,0	r	2046,0	0,70	2,7932	2,7932	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,0	2	24,6	0	100,0	0,29	0,34	273	223	0,0985	632,0
375		S3-5	287,0	287,0	6	24,0	r	2760,0	0,70	3,8467	3,8467	442,0	2,21	544,00	1,23		41,0	2	12,0	224,0	462	4,0	2	24,6	0	100,0	0,29	0,35	267	217	0,0985	632,0
376		S3-6	287,0	287,0	6	24,0	r	2760,0	0,70	3,8467	3,8467	442,0	2,21	544,00	1,23		41,0	2	12,0	224,0	462	4,0	2	24,6	0	100,0	0,29	0,35	267	217	0,0985	632,0
377		S4-1	524,0	524,0	4	36,0	r	4092,0	0,70	3,1237	3,1237	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,19	499	454	0,1533	569,0
379		S4-3	332,0	332,0	4	28,0	r	2464,0	0,70	2,9687	2,9687	450,0	2,25	562,00	1,25		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,30	307	262	0,1533	569,0
380		S4-4	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
382		S4-6	198,0	198,0	3	24,0	r	1380,0	0,70	2,7879	2,7879	442,0	2,21	544,00	1,23		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,51	173	128	0,1533	569,0
383		S5-1	292,0	292,0	2	36,0	r.	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	452	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
384		S5-2	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	452	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
385		S5-3	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
389		S7-1	278,0	278,0	4	32,0	r	3284,0	0,70	4,7252	4,7252	433,0	2,17	541,00	1,25		47,0	2	24,0	920,0	442	4,9	2	38,3	0	150,0	0,43	0,54	253	208	0,1022	569,0
390		S7-2	278,0	278,0	4	32,0	r	3284,0	0,70	4,7252	4,7252	433,0	2,17	541,00	1,25		47,0	2	24,0	920,0	442	4,9	2	38,3	0	125,0	0,36	0,45	253	208	0,1227	569,0
391		S7-3	278,0	278,0	4	32,0	r	3284,0	0,70	4,7252	4,7252	433,0	2,17	541,00	1,25		47,0	2	24,0	920,0	442	4,9	2	38,3	0	100,0	0,29	0,36	253	208	0,1533	569,0
392		S7-4	278,0	278,0	4	32,0	r	3284,0	0,70	4,7252	4,7252	433,0	2,17	541,00	1,25		47,0	2	24,0	920,0	442	4,9	2	38,3	0	80,0	0,23	0,29	253	208	0,1917	569,0
393		S7-5	278,0	278,0	4	32,0	r	3284,0	0,70	4,7252	4,7252	433,0	2,17	541,00	1,25		47,0	2	24,0	920,0	442	4,9	2	38,3	0	70,0	0,20	0,25	253	208	0,2190	569,0
394		S7-6	278,0	278,0	4	32,0	r	3284,0	0,70	4,7252	4,7252	433,0	2,17	541,00	1,25		47,0	2	24,0	920,0	442	4,9	2	38,3	0	60,0	0,17	0,22	253	208	0,2556	569,0
395		S8-1	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	150,0	0,43	0,51	267	222	0,1022	569,0
396		S8-2	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	125,0	0,36	0,43	267	222	0,1227	569,0
397		S8-3	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
398		S8-4	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0
399		S8-5	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	80,0	0,23	0,27	267	222	0,1917	569,0
400		S8-6	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462	4,9	2	38,3	0	70,0	0,20	0,24	267	222	0,2190	569,0
420	Krefeld; Thurston (1966)	26-1	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	352,4	1,76		0,00							6,4	2	63,3	0	152,4	0,30	0,33	424	0	0,1636	341,4
422		29b-1	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	338,6	1,69		0,00							6,4	2	63,3	0	228,6	0,45	0,50	424	0	0,1091	341,4
425		29a-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	383,4	1,92		0,00							6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4
426		29b-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	364,1	1,82		0,00							6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4
427		29c-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	271,0	1,36		0,00							6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4
428		29d-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	290,3	1,45		0,00							6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4
430		29f-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	406,2	2,03		0,00							6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4
431		29g-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	276,6	1,38		0,00							6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4
434		24.5-3	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	406,2	2,03		0,00							6,4	2	63,3	r	114,3	0,23	0,25	424	0	0,2182	237,2
444		39-3	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	406,2	2,03		0,00							9,5	2	141,9	r	228,6	0,45	0,50	408	0	0,2444	275,9

			fyw_rhow_100	fwtk	fwtk/fyw	ewuk	diaa	f1c	Method	fcwu	f1cttest	Method	betacttest	f1ctmcal	betactcal	oms	oml	omwy	g	Vg	F	Vu_F+g,rep	Vu_Rep	Vu_gF
-	-	-	MPa	MPa	-	‰	mm	MPa	-	MPa	MPa	-	-	MPa	-	-	-	-	kN/m	kN	kN	kN	kN	kN
359	Kong;	S1-1	0,87	614,00	1,08		7,0	58,51	cyl	46,81	0,00		0,0000	4,31	0,0736	0,2165	0,2165	0,0186	2,10	1,31			228,30	229,61
360	Rangan (1997)	S1-2	0,87	614,00	1,08		7,0	58,51	cyl	46,81	0,00		0,0000	4,31	0,0736	0,2165	0,2165	0,0186	2,10	1,36			208,30	209,66
361		S1-3	0,87	614,00	1,08		7,0	58,51	cyl	46,81	0,00		0,0000	4,31	0,0736	0,2165	0,2165	0,0186	2,10	1,31			206,10	207,41
362		S1-4	0,87	614,00	1,08		7,0	58,51	cyl	46,81	0,00		0,0000	4,31	0,0736	0,2165	0,2165	0,0186	2,10	1,41			277,90	279,31
363		S1-5	0,87	614,00	1,08		7,0	58,51	cyl	46,81	0,00		0,0000	4,31	0,0736	0,2165	0,2165	0,0186	2,10	1,29			253,30	254,59
364		S1-6	0,87	614,00	1,08		7,0	58,51	cyl	46,81	0,00		0,0000	4,31	0,0736	0,2165	0,2165	0,0186	2,10	1,18			224,10	225,28
365		S2-1	0,58	614,00	1,08		7,0	66,70	cyl	53,36	3,25	sp	0,0487	4,51	0,0676	0,1899	0,1899	0,0109	2,10	1,08			260,30	261,38
366		S2-2	0,70	614,00	1,08		7,0	66,70	cyl	53,36	3,25	sp	0,0487	4,51	0,0676	0,1899	0,1899	0,0131	2,10	1,34			232,50	233,84
367		S2-3	0,87	614,00	1,08		7,0	66,70	cyl	53,36	3,25	sp	0,0487	4,51	0,0676	0,1899	0,1899	0,0164	2,10	1,31			253,30	254,61
368		S2-4	0,87	614,00	1,08		7,0	66,70	cyl	53,36	3,25	sp	0,0487	4,51	0,0676	0,1899	0,1899	0,0164	2,10	1,20			219,40	220,60
369		S2-5	1,16	614,00	1,08		7,0	66,70	cyl	53,36	3,25	sp	0,0487	4,51	0,0676	0,1899	0,1899	0,0218	2,10	1,41			282,10	283,51
371		S3-1	0,62	687,00	1,09		7,0	62,01	cyl	49,61	3,28	sp	0,0528	4,40	0,0709	0,1204	0,1204	0,0126	2,10	1,33			209,20	210,53
372		S3-2	0,62	687,00	1,09		7,0	62,01	cyl	49,61	3,28	sp	0,0528	4,40	0,0709	0,1204	0,1204	0,0126	2,10	1,28			178,00	179,28
373		S3-3	0,62	687,00	1,09		7,0	62,01	cyl	49,61	3,28	sp	0,0528	4,40	0,0709	0,2036	0,2036	0,0126	2,10	1,18			228,60	229,78
374		S3-4	0,62	687,00	1,09		7,0	62,01	cyl	49,61	3,28	sp	0,0528	4,40	0,0709	0,2036	0,2036	0,0126	2,10	1,22			174,90	176,12
375		S3-5	0,62	687,00	1,09		7,0	62,01	cyl	49,61	3,28	sp	0,0528	4,40	0,0709	0,2742	0,2742	0,0126	2,10	1,28			296,60	297,88
376		S3-6	0,62	687,00	1,09		7,0	62,01	cyl	49,61	3,28	sp	0,0528	4,40	0,0709	0,2742	0,2742	0,0126	2,10	1,28			282,90	284,18
377		S4-1	0,87	614,00	1,08		7,0	80,32	cyl	64,25	3,72	sp	0,0463	4,82	0,0600	0,1758	0,1758	0,0136	2,10	2,03			354,00	356,03
379		S4-3	0,87	614,00	1,08		7,0	80,32	cyl	64,25	3,72	sp	0,0463	4,82	0,0600	0,1663	0,1663	0,0136	2,10	1,40			243,40	244,80
380		S4-4	0,87	614,00	1,08		7,0	80,32	cyl	64,25	3,72	sp	0,0463	4,82	0,0600	0,1577	0,1577	0,0136	2,10	1,26			258,10	259,36
382		S4-6	0,87	614,00	1,08		7,0	80,32	cyl	64,25	3,72	sp	0,0463	4,82	0,0600	0,1534	0,1534	0,0136	2,10	1,05			202,90	203,95
383		S5-1	0,87	614,00	1,08		7,0	82,25	cyl	65,80	4,80	sp	0,0583	4,86	0,0590	0,1540	0,1540	0,0133	2,10	1,37			241,70	243,07
384		S5-2	0,87	614,00	1,08		7,0	82,25	cyl	65,80	4,80	sp	0,0583	4,86	0,0590	0,1540	0,1540	0,0133	2,10	1,37			259,90	261,27
385		S5-3	0,87	614,00	1,08		7,0	82,25	cyl	65,80	4,80	sp	0,0583	4,86	0,0590	0,1540	0,1540	0,0133	2,10	1,34			243,80	245,14
389		S7-1	0,58	614,00	1,08		7,0	68,82	cyl	55,05	3,68	sp	0,0535	4,56	0,0663	0,2973	0,2973	0,0106	2,10	1,04			217,20	218,24
390		S7-2	0,70	614,00	1,08		7,0	68,82	cyl	55,05	3,68	sp	0,0535	4,56	0,0663	0,2973	0,2973	0,0127	2,10	1,04			205,40	206,44
391		S7-3	0,87	614,00	1,08		7,0	68,82	cyl	55,05	3,68	sp	0,0535	4,56	0,0663	0,2973	0,2973	0,0158	2,10	1,04			246,50	247,54
392		S7-4	1,09	614,00	1,08		7,0	68,82	cyl	55,05	3,68	sp	0,0535	4,56	0,0663	0,2973	0,2973	0,0198	2,10	1,04			273,60	274,64
393		S7-5	1,25	614,00	1,08		7,0	68,82	cyl	55,05	3,68	sp	0,0535	4,56	0,0663	0,2973	0,2973	0,0226	2,10	1,04			304,40	305,44
394		S7-6	1,45	614,00	1,08		7,0	68,82	cyl	55,05	3,68	sp	0,0535	4,56	0,0663	0,2973	0,2973	0,0264	2,10	1,22			310,60	311,82
395		S8-1	0,58	614,00	1,08		7,0	68,63	cyl	54,91	3,38	sp	0,0493	4,56	0,0664	0,1846	0,1846	0,0106	2,10	1,27			272,10	273,37
396		S8-2	0,70	614,00	1,08		7,0	68,63	cyl	54,91	3,38	sp	0,0493	4,56	0,0664	0,1846	0,1846	0,0127	2,10	1,31			251,00	252,31
397		S8-3	0,87	614,00	1,08		7,0	68,63	cyl	54,91	3,38	sp	0,0493	4,56	0,0664	0,1846	0,1846	0,0159	2,10	1,22			309,60	310,82
398		S8-4	0,87	614,00	1,08		7,0	68,63	cyl	54,91	3,38	sp	0,0493	4,56	0,0664	0,1846	0,1846	0,0159	2,10	1,29			265,80	267,09
399		S8-5	1,09	614,00	1,08		7,0	68,63	cyl	54,91	3,38	sp	0,0493	4,56	0,0664	0,1846	0,1846	0,0199	2,10	1,29			289,20	290,49
400		S8-6	1,25	614,00	1,08		7,0	68,63	cyl	54,91	3,38	sp	0,0493	4,56	0,0664	0,1846	0,1846	0,0227	2,10	1,13			283,90	285,03
420	Krefeld; Thurston (1966)	26-1	0,56		0,00			38,13	cyl	30,50	0,00		0,0000	3,30	0,0866	0,2059	0,2059	0,0183	3,10	2,93			206,83	209,76
422		29b-1	0,37		0,00			35,77	cyl	28,62	0,00		0,0000	3,15	0,0880	0,2109	0,2109	0,0130	3,10	2,93			160,13	163,06
425		29a-2	0,41		0,00			35,31	cyl	28,25	0,00		0,0000	3,12	0,0883	0,2419	0,2419	0,0144	3,10	2,93			216,62	219,55
426		29b-2	0,41		0,00			39,31	cyl	31,45	0,00		0,0000	3,38	0,0859	0,2064	0,2064	0,0129	3,10	2,93			202,38	205,31
427		29c-2	0,41		0,00			22,93	cyl	18,34	0,00		0,0000	2,24	0,0975	0,2634	0,2634	0,0221	3,10	2,93			161,46	164,39
428		29d-2	0,41		0,00			28,89	cyl	23,11	0,00		0,0000	2,68	0,0927	0,2239	0,2239	0,0176	3,10	2,93			165,02	167,95
430		29f-2	0,41		0,00			39,70	cyl	31,76	0,00		0,0000	3,40	0,0857	0,2280	0,2280	0,0128	3,10	2,93			234,41	237,34
431		29g-2	0,41		0,00			14,94	cyl	11,95	0,00		0,0000	1,56	0,1043	0,4125	0,4125	0,0340	3,10	2,93			149,90	152,83
434		24.5-3	0,52		0,00			33,68	cyl	26,94	0,00		0,0000	3,01	0,0893	0,2688	0,2688	0,0192	3,10	2,93			232,63	235,56
444		39-3	0,67		0,00			40,56	cyl	32,44	0,00		0,0000	3,45	0,0852	0,2232	0,2232	0,0208	3,10	2,93			239,75	242,68

			betar_meas	xr_meas	xr	sigswmeas	sigswass	sigsw	omwu	andbr	Mu	muu	vutest	kapc	sigsl	xsitest	zetatest	z_test	zeta	z_	muflex	Mu_flex	Vu_flex	betaflex	BB
-	-	-	[°]	mm	mm	MPa	MPa	MPa	-	-	kNm	-	-	-	MPa	-	-	mm	-	mm	-	-	kN	-	-
359	Kong;	S1-1	50	355	355			0	0,0000		166,66	0,1336	0,0740	0,766	309	0,193	0,903	264	0,859	251	0,1859	231,87	317,64	0,72	
360	Rangan (1997)	S1-2	55	332	332			0	0,0000		152,06	0,1219	0,0668	0,766	279	0,174	0,913	267	0,859	251	0,1859	231,87	317,64	0,66	
361		S1-3	41	354	354	314		314	0,0103		150,45	0,1206	0,0660	0,766	276	0,172	0,914	267	0,859	251	0,1859	231,87	317,64	0,65	
362		S1-4	40	310	310	450		450	0,0147		202,87	0,1627	0,0925	0,766	386	0,242	0,879	257	0,859	251	0,1859	231,87	317,64	0,87	
363		S1-5		365	365			0	0,0000		184,91	0,1483	0,0832	0,766	347	0,217	0,891	260	0,859	251	0,1859	231,87	317,64	0,80	
364		S1-6	49	420	420			0	0,0000		163,59	0,1312	0,0724	0,766	302	0,189	0,905	264	0,859	251	0,1859	231,87	317,64	0,71	
365		S2-1		464	464			0	0,0000		190,02	0,1336	0,0744	0,733	354	0,203	0,899	262	0,870	254	0,1653	235,06	322,00	0,81	
366		S2-2	45	344	344			0	0,0000		169,73	0,1194	0,0655	0,733	312	0,179	0,911	266	0,870	254	0,1653	235,06	322,00	0,72	
367		S2-3		354	354	266		266	0,0076		184,91	0,1301	0,0721	0,733	343	0,197	0,902	263	0,870	254	0,1653	235,06	322,00	0,79	
368		S2-4	55	408	408	279		279	0,0080		160,16	0,1126	0,0615	0,733	293	0,168	0,916	268	0,870	254	0,1653	235,06	322,00	0,68	
369		S2-5		310	310			0	0,0000		205,93	0,1448	0,0815	0,733	388	0,222	0,889	260	0,870	254	0,1653	235,06	322,00	0,88	
371		S3-1	44	359	359			0	0,0000		154,81	0,1132	0,0617	0,752	450	0,160	0,920	273	0,920	273	0,1108	151,47	204,69	1,02	BB
372		S3-2	50	381	381			0	0,0000		131,72	0,0963	0,0519	0,752	387	0,138	0,931	277	0,920	273	0,1108	151,47	204,69	0,87	
373		S3-3	47	420	420	300		300	0,0060		166,88	0,1254	0,0693	0,752	307	0,184	0,908	266	0,865	253	0,1760	234,28	320,93	0,71	
374		S3-4	58	398	398	350		350	0,0070		127,68	0,0959	0,0517	0,752	229	0,137	0,932	273	0,865	253	0,1760	234,28	320,93	0,54	
375		S3-5	45	360	360			0	0,0000		213,55	0,1672	0,0955	0,752	309	0,255	0,873	250	0,818	235	0,2242	286,28	397,62	0,75	
376		S3-6	46	360	360			0	0,0000		203,69	0,1595	0,0904	0,752	292	0,241	0,879	252	0,818	235	0,2242	286,28	397,62	0,71	
377		S4-1	53	585	585	183		183	0,0044		460,20	0,0835	0,0450	0,679	230	0,132	0,934	490	0,870	456	0,1530	843,67	648,98	0,55	
379		S4-3	45	415	415			0	0,0000		202,02	0,0913	0,0492	0,679	266	0,145	0,928	308	0,877	291	0,1460	323,02	389,18	0,63	
380		S4-4		381	381	317		317	0,0076		188,41	0,1101	0,0604	0,679	346	0,178	0,911	266	0,884	258	0,1394	238,66	326,93	0,79	
382		S4-6			250	300		300	0,0072		101,45	0,1289	0,0714	0,679	415	0,212	0,894	177	0,887	176	0,1361	107,12	214,24	0,95	
383		S5-1	47	476	476	411		411	0,0096		212,70	0,1213	0,0559	0,671	396	0,201	0,899	263	0,885	258	0,1363	239,05	271,64	0,89	
384		S5-2		400	400			0	0,0000		207,92	0,1186	0,0600	0,671	386	0,196	0,902	263	0,885	258	0,1363	239,05	298,81	0,87	
385		S5-3	45	341	341	300		300	0,0070		177,97	0,1015	0,0553	0,671	325	0,165	0,918	268	0,885	258	0,1363	239,05	327,46	0,74	
389		S7-1			473			0	0,0000		205,25	0,1544	0,0646	0,725	256	0,242	0,879	244	0,795	221	0,2363	314,22	332,51	0,65	
390		S7-2			473			0	0,0000		194,10	0,1460	0,0606	0,725	240	0,227	0,886	246	0,795	221	0,2363	314,22	332,51	0,62	
391		S7-3			473	372		372	0,0104		232,94	0,1752	0,0750	0,725	297	0,281	0,859	239	0,795	221	0,2363	314,22	332,51	0,74	
392		S7-4			473	375		375	0,0131		258,55	0,1945	0,0851	0,725	337	0,319	0,840	234	0,795	221	0,2363	314,22	332,51	0,82	
393		S7-5			473			0	0,0000		287,66	0,2164	0,0973	0,725	385	0,365	0,817	227	0,795	221	0,2363	314,22	332,51	0,92	
394		S7-6		388	388			0	0,0000		293,52	0,2208	0,0999	0,725	396	0,375	0,813	226	0,795	221	0,2363	314,22	332,51	0,93	
395		S8-1	58	376	376			0	0,0000		198,63	0,1358	0,0758	0,725	371	0,209	0,896	261	0,873	255	0,1611	235,69	322,86	0,84	
396		S8-2	50	354	354			0	0,0000		183,23	0,1252	0,0692	0,725	339	0,191	0,905	264	0,873	255	0,1611	235,69	322,86	0,78	
397		S8-3	45	398	398			0	0,0000		226,01	0,1545	0,0879	0,725	430	0,242	0,879	257	0,873	255	0,1611	235,69	322,86	0,96	
398		S8-4	45		365			0	0,0000		194,03	0,1326	0,0738	0,725	362	0,204	0,898	262	0,873	255	0,1611	235,69	322,86	0,82	
399		S8-5			365			0	0,0000		211,12	0,1443	0,0813	0,725	398	0,224	0,888	259	0,873	255	0,1611	235,69	322,86	0,90	
400		S8-6		442	442			0	0,0000		207,25	0,1417	0,0796	0,725	390	0,219	0,890	260	0,873	255	0,1611	235,69	322,86	0,88	
420	Krefeld; Thurston (1966)	26-1			883		341	341	0,0183		365,12	0,1816	0,0667	0,847	352	0,243	0,879	400	0,879	400	0,1809	363,82	206,10	1,00	BB
422		29b-1			883		341	341	0,0130		282,67	0,1498	0,0535	0,857	266	0,194	0,903	412	0,877	400	0,1850	348,96	197,68	0,81	
425		29a-2			883		372	372	0,0144		382,40	0,2053	0,0769	0,859	378	0,278	0,861	392	0,859	391	0,2079	387,13	219,30	0,99	
426		29b-2			883		372	372	0,0129		357,27	0,1723	0,0629	0,843	344	0,231	0,884	403	0,878	400	0,1811	375,52	212,72	0,95	
427		29c-2			883		372	372	0,0221		285,03	0,2357	0,0889	0,908	271	0,290	0,855	390	0,855	390	0,2252	272,33	154,27	1,05	BB
428		29d-2			883		372	372	0,0176		291,31	0,1912	0,0704	0,884	283	0,247	0,877	400	0,873	398	0,1956	298,01	168,81	0,98	
430		29f-2			883		372	372	0,0128		413,80	0,1976	0,0738	0,841	406	0,271	0,864	394	0,864	394	0,1971	412,67	233,77	1,00	BB
431		29g-2			883		372	372	0,0340		264,61	0,3359	0,1388	0,940	277	0,439	0,781	356	0,781	356	0,3220	253,70	143,71	1,04	BB
434		24.5-3			883			0	0,0000		410,66	0,2312	0,0883	0,865	406	0,311	0,845	385	0,845	385	0,2270	403,22	228,41	1,02	BB
444		39-3			883		276	276	0,0208		423,23	0,1979	0,0737	0,838	406	0,266	0,867	395	0,867	395	0,1934	413,77	234,39	1,02	BB

			sin2thp	thp	cotthp	vup	gamwp	cotthu	thu	nueu	betalb	VB	A2+A3	kon34	KONA4	KONA5	A4+A5	Differenz	AF	f1ck	fcm_cyl	fck	fc_prime	Vct_ACI	Vs_ACI	Vcal_sACI	Vmax_ACI	rat_smax	VACI_cal
-	-	-	-	[°]	-	-	-	-	[°]	-	-	-	-	-	-	-	-	-		MPa	MPa	MPa	MPa	kN	kN	kN	kN		kN
359	Kong;	S1-1	0,02	7,85	7,26	0,14	0,55	0,00	0,00	0,00	0,29		0	1	0	0	0	0		54,7	61,6	57,6	59,2	93,28	63,69	156,97	466,41	0,337	156,97
360	Rangan (1997)	S1-2	0,02	7,85	7,26	0,14	0,49	0,00	0,00	0,00	0,27		0	1	0	0	0	0		54,7	61,6	57,6	59,2	93,28	63,69	156,97	466,41	0,337	156,97
361		S1-3	0,02	7,85	7,26	0,14	0,49	6,42	8,85	0,35	0,27		1	1	0	1	1	0		54,7	61,6	57,6	59,2	93,28	63,69	156,97	466,41	0,337	156,97
362		S1-4	0,02	7,85	7,26	0,14	0,68	6,27	9,05	0,48	0,36		1	1	0	1	1	0		54,7	61,6	57,6	59,2	93,28	63,69	156,97	466,41	0,337	156,97
363		S1-5	0,02	7,85	7,26	0,14	0,61	0,00	0,00	0,00	0,33		0	1	0	0	0	0		54,7	61,6	57,6	59,2	93,28	63,69	156,97	466,41	0,337	156,97
364		S1-6	0,02	7,85	7,26	0,14	0,54	0,00	0,00	0,00	0,29		0	1	0	0	0	0		54,7	61,6	57,6	59,2	93,28	63,69	156,97	466,41	0,337	156,97
365		S2-1	0,01	5,99	9,53	0,10	0,72	0,00	0,00	0,00	0,32		0	1	0	0	0	0		62,9	70,2	66,2	67,8	99,84	42,46	142,30	499,21	0,285	142,30
366		S2-2	0,01	6,57	8,69	0,11	0,58	0,00	0,00	0,00	0,28		0	1	0	0	0	0		62,9	70,2	66,2	67,8	99,84	50,95	150,79	499,21	0,302	150,79
367		S2-3	0,02	7,35	7,76	0,13	0,57	9,44	6,05	0,55	0,31		1	1	0	1	1	0		62,9	70,2	66,2	67,8	99,84	63,69	163,53	499,21	0,328	163,53
368		S2-4	0,02	7,35	7,76	0,13	0,48	7,68	7,42	0,38	0,27		1	1	0	1	1	0		62,9	70,2	66,2	67,8	99,84	63,69	163,53	499,21	0,328	163,53
369		S2-5	0,02	8,49	6,70	0,15	0,56	0,00	0,00	0,00	0,34		0	1	0	0	0	0		62,9	70,2	66,2	67,8	99,84	84,92	184,76	499,21	0,370	184,76
371		S3-1	0,01	6,43	8,87	0,11	0,55	0,00	0,00	0,00	0,33		0	1	0	0	0	0		58,2	65,3	61,3	62,9	97,78	46,24	144,02	488,92	0,295	144,02
372		S3-2	0,01	6,43	8,87	0,11	0,47	0,00	0,00	0,00	0,28		0	1	0	0	0	0		58,2	65,3	61,3	62,9	97,78	46,24	144,02	488,92	0,295	144,02
373		S3-3	0,01	6,43	8,87	0,11	0,62	11,63	4,92	0,65	0,28		1	1	0	1	1	0		58,2	65,3	61,3	62,9	96,47	45,61	142,08	482,34	0,295	142,08
374		S3-4	0,01	6,43	8,87	0,11	0,46	7,43	7,66	0,31	0,22		1	1	0	1	1	0		58,2	65,3	61,3	62,9	96,47	45,61	142,08	482,34	0,295	142,08
375		S3-5	0,01	6,43	8,87	0,11	0,86	0,00	0,00	0,00	0,19		0	1	0	0	0	0		58,2	65,3	61,3	62,9	94,49	44,68	139,17	472,46	0,295	139,17
376		S3-6	0,01	6,43	8,87	0,11	0,81	0,00	0,00	0,00	0,18		0	1	0	0	0	0		58,2	65,3	61,3	62,9	94,49	44,68	139,17	472,46	0,295	139,17
377		S4-1	0,01	6,69	8,52	0,12	0,39	10,29	5,55	0,37	0,03		1	1	0	1	1	0		76,5	84,5	80,5	82,1	197,20	114,29	311,49	985,99	0,316	311,49
379		S4-3	0,01	6,69	8,52	0,12	0,43	0,00	0,00	0,00	0,13		0	1	0	0	0	0		76,5	84,5	80,5	82,1	124,94	72,41	197,36	624,71	0,316	197,36
380		S4-4	0,01	6,69	8,52	0,12	0,52	7,99	7,13	0,39	0,23		1	1	0	1	1	0		76,5	84,5	80,5	82,1	109,89	63,69	173,58	549,45	0,316	173,58
382		S4-6	0,01	6,69	8,52	0,12	0,62	9,97	5,73	0,58	0,43		1	1	0	1	1	0		76,5	84,5	80,5	82,1	74,51	43,19	117,70	372,57	0,316	117,70
383		S5-1	0,01	6,61	8,63	0,11	0,49	5,85	9,70	0,27	0,32		1	1	1	0	1	0		78,4	86,6	82,6	84,2	111,24	63,69	174,93	556,21	0,315	174,93
384		S5-2	0,01	6,61	8,63	0,11	0,52	0,00	0,00	0,00	0,27		0	1	0	0	0	0		78,4	86,6	82,6	84,2	111,24	63,69	174,93	556,21	0,315	174,93
385		S5-3	0,01	6,61	8,63	0,11	0,48	7,91	7,20	0,36	0,27		1	1	0	1	1	0		78,4	86,6	82,6	84,2	111,24	63,69	174,93	556,21	0,315	174,93
389		S7-1	0,01	5,90	9,68	0,10	0,63	0,00	0,00	0,00	0,17		0	1	0	0	0	0		65,0	72,4	68,4	70,0	96,60	40,42	137,03	483,02	0,284	137,03
390		S7-2	0,01	6,47	8,82	0,11	0,54	0,00	0,00	0,00	0,16		0	1	0	0	0	0		65,0	72,4	68,4	70,0	96,60	48,51	145,11	483,02	0,300	145,11
391		S7-3	0,02	7,23	7,88	0,12	0,60	7,23	7,87	0,44	0,19		1	1	1	0	1	0		65,0	72,4	68,4	70,0	96,60	60,64	157,24	483,02	0,326	157,24
392		S7-4	0,02	8,09	7,03	0,14	0,61	6,52	8,72	0,45	0,21		1	1	1	0	1	0		65,0	72,4	68,4	70,0	96,60	75,80	172,40	483,02	0,357	172,40
393		S7-5	0,02	8,65	6,57	0,15	0,65	0,00	0,00	0,00	0,23		0	1	0	0	0	0		65,0	72,4	68,4	70,0	96,60	86,62	183,23	483,02	0,379	183,23
394		S7-6	0,03	9,35	6,07	0,16	0,62	0,00	0,00	0,00	0,24		0	1	0	0	0	0		65,0	72,4	68,4	70,0	96,60	101,06	197,66	483,02	0,409	197,66
395		S8-1	0,01	5,91	9,66	0,10	0,74	0,00	0,00	0,00	0,33		0	1	0	0	0	0		64,8	72,2	68,2	69,8	101,33	42,46	143,79	506,65	0,284	143,79
396		S8-2	0,01	6,47	8,81	0,11	0,62	0,00	0,00	0,00	0,30		0	1	0	0	0	0		64,8	72,2	68,2	69,8	101,33	50,95	152,28	506,65	0,301	152,28
397		S8-3	0,02	7,24	7,87	0,13	0,70	0,00	0,00	0,00	0,37		0	1	0	0	0	0		64,8	72,2	68,2	69,8	101,33	63,69	165,02	506,65	0,326	165,02
398		S8-4	0,02	7,24	7,87	0,13	0,59	0,00	0,00	0,00	0,32		0	1	0	0	0	0		64,8	72,2	68,2	69,8	101,33	63,69	165,02	506,65	0,326	165,02
399		S8-5	0,02	8,10	7,02	0,14	0,58	0,00	0,00	0,00	0,35		0	1	0	0	0	0		64,8	72,2	68,2	69,8	101,33	79,61	180,94	506,65	0,357	180,94
400		S8-6	0,02	8,67	6,56	0,15	0,53	0,00	0,00	0,00	0,34		0	1	0	0	0	0		64,8	72,2	68,2	69,8	101,33	90,99	192,31	506,65	0,380	192,31
420	Krefeld; Thurston (1966)	26-1	0,02	7,78	7,32	0,13	0,50	3,64	15,35	0,21	0,30		1	1	1	0	1	0		34,3	40,1	36,1	37,7	118,09	64,65	182,74	590,47	0,309	182,74
422		29b-1	0,01	6,55	8,71	0,11	0,47	4,11	13,66	0,19	0,24		1	1	1	0	1	0		32,0	37,7	33,7	35,3	114,14	43,10	157,24	570,71	0,276	157,24
425		29a-2	0,01	6,89	8,28	0,12	0,65	5,35	10,59	0,34	0,33		1	1	1	0	1	0		31,5	37,2	33,2	34,8	113,36	47,02	160,38	566,79	0,283	160,38
426		29b-2	0,01	6,53	8,74	0,11	0,56	4,87	11,61	0,26	0,28		1	1	1	0	1	0		35,5	41,4	37,4	39,0	120,02	47,02	167,04	600,10	0,278	167,04
427		29c-2	0,02	8,56	6,65	0,15	0,60	4,02	13,98	0,30	0,34		1	1	1	0	1	0		19,1	24,1	20,1	21,7	89,63	47,02	136,65	448,14	0,305	136,65
428		29d-2	0,02	7,62	7,48	0,13	0,54	4,00	14,03	0,24	0,29		1	1	1	0	1	0		25,1	30,4	26,4	28,0	101,75	47,02	148,77	508,74	0,292	148,77
430		29f-2	0,01	6,49	8,79	0,11	0,66	5,77	9,84	0,35	0,33		1	1	1	0	1	0		35,9	41,8	37,8	39,4	120,66	47,02	167,67	603,28	0,278	167,67
431		29g-2	0,03	10,62	5,33	0,18	0,77	4,08	13,76	0,48	0,47		1	1	1	0	1	0		11,1	15,7	11,7	13,3	70,17	47,02	117,19	350,85	0,334	117,19
434		24.5-3	0,02	7,97	7,15	0,14	0,64	0,00	0,00	0,00	0,37		0	1	0	0	0	0		29,9	35,4	31,4	33,0	110,51	59,91	170,42	552,56	0,308	170,42
444		39-3	0,02	8,29	6,86	0,14	0,52	3,54	15,76	0,23	0,33		1	1	1	0	1	0		36,8	42,7	38,7	40,3	122,02	78,05	200,07	610,10	0,328	200,07

			gmod_ACI
-	-	-	
359	Kong;	S1-1	1,454
360	Rangan (1997)	S1-2	1,327
361		S1-3	1,313
362		S1-4	1,770
363		S1-5	1,614
364		S1-6	1,428
365		S2-1	1,829
366		S2-2	1,542
367		S2-3	1,549
368		S2-4	1,342
369		S2-5	1,527
371		S3-1	1,453
372		S3-2	1,236
373		S3-3	1,609
374		S3-4	1,231
375		S3-5	2,131
376		S3-6	2,033
377		S4-1	1,136
379		S4-3	1,233
380		S4-4	1,487
382		S4-6	1,724
383		S5-1	1,382
384		S5-2	1,486
385		S5-3	1,394
389		S7-1	1,585
390		S7-2	1,415
391		S7-3	1,568
392		S7-4	1,587
393		S7-5	1,661
394		S7-6	1,571
395		S8-1	1,892
396		S8-2	1,648
397		S8-3	1,876
398		S8-4	1,611
399		S8-5	1,598
400		S8-6	1,476
420	Krefeld; Thurston (1966)	26-1	1,132
422		29b-1	1,018
425		29a-2	1,351
426		29b-2	1,212
427		29c-2	1,182
428		29d-2	1,109
430		29f-2	1,398
431		29g-2	1,279
434		24.5-3	1,365
444		39-3	1,198

				E	b	bw	h	hf	hh,top	hw	hft	hh,bot	bft	Ac	zc2	aa	af	ba	L	c	a	kap	cc
	-	-	-	SI / A	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm²	mm	mm	mm	mm	mm	mm	mm	-	mm
Leonhardt_1962_002_ET2	483	Leonhardt; Walther;	ET2	SI	300,0	150,0	350,0	75,0		275,0				63750	150,7	100,0	100,0	200,0	3000,0	900,0	1050,0	3,50	
Leonhardt_1962_003_ET3	484	DAfStb 151 (1962)	ET3	SI	300,0	100,0	350,0	75,0		275,0				50000	133,8	100,0	100,0	200,0	3000,0	900,0	1050,0	3,50	
Leonhardt_1962_004_ET4	485		ET4	SI	300,0	50,0	350,0	75,0		175,0	100,0		100,0	41250	127,7	100,0	100,0	200,0	3000,0	900,0	1050,0	3,50	
Leonhardt_1962_002_T1 (r)	490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	SI	1500,0	100,0	900,0	150,0		600,0	150,0		500,0	362500	291,7	200,0	200,0	125,0	6000,0	1000,0	2500,0	3,03	
Leonhardt_1963_003_TA1	491	Leonhardt; Walther	TA1	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	150,0	3000,0	750,0	1125,0	3,00	
Leonhardt_1963_004_TA2	492	DAfStb 156 (1963)	TA2	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	150,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_005_TA3	493		TA3	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	150,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_006_TA4	494		TA4	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	150,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_009_TA11	497		TA11	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_010_TA12	498		TA12	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_011_TA13	499		TA13	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_012_TA14	500		TA14	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_013_TA15	501		TA15	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_014_TA6	502		TA6	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	150,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_015_TA16	503		TA16	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Levi_1989/1993_001_RC 30 A1	504	Levi;	RC 30 A1	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_002_RC 30 A2	505	Marro	RC 30 A2	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_003_RC 60 A1	506	(1989/1993)	RC 60 A1	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_004_RC 60 A2	507		RC 60 A2	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_005_RC 60 B1	508		RC 60 B1	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_006_RC 60 B2	509		RC 60 B2	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_007_RC 70 B1	510		RC 70 B1	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Lynberg_1974_001_5A-0	511	Lyngberg	5A-0	SI	700,0	120,0	600,0	90,0	10,0	360,0	110,0	30,0	380,0	159600	261,1			425,0	5000,0	2000,0	1500,0	2,78	20,0
Lynberg_1974_002_5B-0	512	(1974)	5B-0	SI	700,0	120,0	600,0	90,0	10,0	360,0	110,0	30,0	380,0	159600	261,1			425,0	5000,0	2000,0	1500,0	2,78	20,0
Maruyama_1988_001_RS2-WD	523	Maruyama K., Rizkalla S. H. (1988)	RS2-WD	SI	650,0	86,3	490,0	60,0	50,0	380,0				90550	168,0			250,0	3000,0	700,0	1150,0	2,89	
Moayer_1974_002_P20	525	Moayer; Regan (1974)	P20	A	599,4	149,9	320,0	81,3		238,8				84503	108,4	76,2	26,7	304,8	1981,2	26,7	977,3	3,50	
Özden_1967_002_T6	534	Özden (1967)	T6	SI	400,0	110,0	355,0	80,0		275,0				62250	126,3	120,0	130,0	250,0	3000,0	900,0	1050,0	3,52	19,0
Özden_1967_003_T7	535		T7	SI	400,0	110,0	355,0	80,0		275,0				62250	126,3	120,0	130,0	250,0	3000,0	900,0	1050,0	3,52	19,0
Özden_1967_004_T9	536		T9	SI	400,0	110,0	355,0	80,0		275,0				62250	126,3	120,0	130,0	250,0	3000,0	900,0	1050,0	3,52	19,0
Ozcebe_1999_003_TS56	553	Ozcebe; Ersoy;	TS56	SI	150,0	150,0	360,0							54000	180,0			100,0	3500,0	400,0	1550,0	5,00	
Ozcebe_1999_006_TS59	556	Tankut (1999)	TS59	SI	150,0	150,0	360,0							54000	180,0			100,0	3500,0	400,0	1550,0	5,00	
Ozcebe_1999_009_TS36	559		TS36	SI	150,0	150,0	360,0							54000	180,0			100,0	2260,0	400,0	930,0	3,00	
Ozcebe_1999_011_TH39	561		TH39	SI	150,0	150,0	360,0							54000	180,0			100,0	2260,0	400,0	930,0	3,00	
Ozcebe_1999_012_TS39	562		TS39	SI	150,0	150,0	360,0							54000	180,0			100,0	2260,0	400,0	930,0	3,00	
Petersson_1972_001_V1	564	Petersson (1972)	V1	SI	175,0	175,0	353,0							61775	176,5	175,0	100,0	300,0	2600,0	100,0	1150,0	3,57	
Petersson_1972_002_VL1	565		VL1	SI	301,0	120,0	307,0	63,0		175,0	63,0		301,0	60732	151,9	300,0	100,0	300,0	2200,0	100,0	900,0	3,24	

			ds	d	ns	dst	fR	As	alphaa	rhos	rhosw	fsy	esy	ftk	ftk/fsy	euk	ds2	ns2	dst2	As2	fsy2	dw	nsw	Asw	frw	sw	sw/h		zw1	zw2	rhow	fyw
-	-	-	mm	mm	-	mm	-	mm²	-	%	%	MPa	‰	MPa	-	‰	mm	-	mm	mm²	MPa	mm	-	mm²	-	mm	-	-	mm	mm	%	MPa
483	Leonhardt; Walther;	ET2	300,0	300,0	4	20,0	r	1256,6	1,00	1,3963	2,7925	419,9	2,10	524,84	1,25			2	8,0	100,5	456	6,0	2	56,5	0	110,0	0,31	0,37	270	0	0,3427	313,9
484	DAfStb 151 (1962)	ET3	300,0	300,0	4	20,0	r	1256,6	1,00	1,3963	4,1888	419,9	2,10	524,84	1,25			2	8,0	100,5	456	6,0	2	56,5	0	110,0	0,31	0,37	270	0	0,5141	313,9
485		ET4	300,0	300,0	4	20,0	r	1256,6	1,00	1,3963	8,3776	419,9	2,10	524,84	1,25			2	8,0	100,5	456	6,0	2	56,5	0	110,0	0,31	0,37	270	0	1,0282	313,9
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	825,0	825,0	16	26,0	r	8464,0	0,01	0,6840	10,2594	464,8	2,32	549,17	1,18			11	10,0	863,9		12,0	2	232,0	r	80,0	0,09	0,10	765	0	2,9000	426,6
491	Leonhardt; Walther	TA1	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426	12,0	2	226,2	r	113,0	0,26	0,30	315	0	1,2511	426,7
492	DAfStb 156 (1963)	TA2	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426	10,0	2	157,1	r	113,0	0,26	0,30	325	0	0,8688	425,8
493		TA3	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426	8,0	2	100,5	r	113,0	0,26	0,30	335	0	0,5560	417,9
494		TA4	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426	6,0	2	56,5	r	113,0	0,26	0,30	345	0	0,3128	440,5
497		TA11	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426	8,0	2	100,5	r	113,0	0,26	0,30	335	0	0,5560	417,9
498		TA12	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426	6,0	2	56,5	r	113,0	0,26	0,30	345	0	0,3128	440,5
499		TA13	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426	12,0	2	226,2	r	113,0	0,26	0,30	315	0	1,2511	426,7
500		TA14	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426	10,0	2	157,1	r	113,0	0,26	0,30	325	0	0,8688	425,8
501		TA15	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	390,4	1,95	518,95	1,33			4	10,0	314,2	426	8,0	2	100,5	r	113,0	0,26	0,30	335	0	0,5560	417,9
502		TA6	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426	8,0	2	100,5	r	113,0	0,26	0,30	335	0	0,5560	417,9
503		TA16	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	390,4	1,95	518,95	1,33			4	10,0	314,2	426	8,0	2	100,5	r	113,0	0,26	0,30	335	0	0,5560	417,9
504	Levi;	RC 30 A1	940,0	940,0	20	24,0	r	9047,8	1,00	1,0695	8,0211	500,0	2,50		0,00							12,0	2	226,2	r.ass	225,0	0,21	0,24	880	0	0,8378	480,0
505	Marro	RC 30 A2	940,0	940,0	20	24,0	r	9047,8	1,00	1,0695	8,0211	500,0	2,50		0,00							12,0	2	226,2	r.ass	225,0	0,21	0,24	880	0	0,8378	480,0
506	(1989/1993)	RC 60 A1	940,0	940,0	20	26,0	r	10618,6	1,00	1,2552	9,4136	450,0	2,25		0,00							12,0	2	226,2	r.ass	225,0	0,21	0,24	880	0	0,8378	480,0
507		RC 60 A2	940,0	940,0	20	26,0	r	10618,6	1,00	1,2552	9,4136	450,0	2,25		0,00							12,0	2	226,2	r.ass	225,0	0,21	0,24	880	0	0,8378	480,0
508		RC 60 B1	940,0	940,0	20	30,0	r	14137,2	1,00	1,6711	12,5329	470,0	2,35		0,00							12,0	2	226,2	r.ass	150,0	0,14	0,16	880	0	1,2566	480,0
509		RC 60 B2	940,0	940,0	20	30,0	r	14137,2	1,00	1,6711	12,5329	470,0	2,35		0,00							12,0	2	226,2	r.ass	150,0	0,14	0,16	880	0	1,2566	480,0
510		RC 70 B1	940,0	940,0	20	30,0	r	14137,2	1,00	1,6711	12,5329	470,0	2,35		0,00							12,0	2	226,2	r.ass	150,0	0,14	0,16	880	0	1,2566	480,0
511	Lyngberg	5A-0	540,0	540,0	8	20,0	r	2513,3	1,00	0,6649	3,8785	616,3	3,08	725,00	1,18			6	10,0	471,2		8,0	2	100,5	r	160,0	0,27	0,30	500	500	0,5236	674,0
512	(1974)	5B-0	540,0	540,0	8	20,0	r	2513,3	1,00	0,6649	3,8785	623,1	3,12	733,00	1,18			6	10,0	471,2		8,0	2	100,5	r	160,0	0,27	0,30	500	500	0,5236	647,0
523	Maruyama K., Rizkalla S. H. (1988)	RS2-WD	450,0	397,5	2	16,0	r	402,1	0,70	0,1375	1,0355	400,0	2,00		0,00			2	9,5	142,5		6,2	1	30,2	r	152,0	0,31	0,38	419	0	0,2302	645,0
525	Moayer; Regan (1974)	P20	279,4	279,4	4	15,9	r	791,7	1,00	0,4727	1,8909	641,4	3,21	937,93	1,46							5,6	2	48,5	0	152,4	0,48	0,55	252	0	0,2123	310,3
534	Özden (1967)	T6	298,0	298,0	4	20,0	r	1256,6	1,00	1,0542	3,8335	441,5	2,21	448,32	1,02							8,0	2	100,5	r	150,0	0,42	0,50	258	260	0,6093	272,7
535		T7	298,0	298,0	4	20,0	r	1256,6	1,00	1,0542	3,8335	447,3	2,24	448,32	1,00							8,0	2	100,5	r	150,0	0,42	0,50	258	260	0,6093	272,7
536		T9	298,0	298,0	4	20,0	r	1256,6	1,00	1,0542	3,8335	444,4	2,22	448,32	1,01							8,0	2	100,5	r	150,0	0,42	0,50	258	260	0,6093	272,7
553	Özcebe; Ersoy;	TS56	310,0	310,0	8	16,0	r	1608,5	0,01	3,4591	3,4591	450,0	2,25	720,00	1,60			2	10,0	157,1	410	4,0	2	25,1	r	70,0	0,19	0,23	290	0	0,2394	255,0
556	Tankut (1999)	TS59	310,0	310,0	4	20,0	r	2060,9	0,01	4,4320	4,4320	434,8	2,17	695,61	1,60			2	10,0	157,1	410	4,0	2	25,1	r	60,0	0,17	0,19	290	0	0,2793	255,0
559		TS36	310,0	310,0	6	16,0	r	1206,4	0,01	2,5943	2,5943	450,0	2,25	720,00	1,60			2	10,0	157,1	410	4,0	2	25,1	r	70,0	0,19	0,23	290	0	0,2394	255,0
561		TH39	310,0	310,0	2	20,0	r	1432,6	0,01	3,0808	3,0808	439,0	2,20	702,46	1,60			2	10,0	157,1	410	4,0	2	25,1	r	80,0	0,22	0,26	290	0	0,2094	255,0
562		TS39	310,0	310,0	2	20,0	r	1432,6	0,01	3,0808	3,0808	439,0	2,20	702,46	1,60			2	10,0	157,1	410	4,0	2	25,1	r	60,0	0,17	0,19	290	0	0,2793	255,0
564	Petersson (1972)	V1	322,0	322,0	3	16,0	r	603,2	1,00	1,0704	1,0704	977,0	4,89		0,00			2	10,0	157,1	463	6,0	2	56,5	0	150,0	0,42	0,47	292	0	0,2154	323,0
565		VL1	278,0	278,0	5	16,0	r	1005,3	1,00	1,2014	3,0135	990,0	4,95		0,00			2	16,0	402,1	635	8,0	2	100,5	0	100,0	0,33	0,36	238	0	0,8378	366,0

			fyw_rhow_100	fwtk	fwtk/fyw	ewuk	diaa	f1c	Method	fcwu	f1cttest	Method	betacttest	f1ctmcal	betactcal	oms	oml	omwy	g	Vg	F	Vu_F+g,rep	Vu_Rep	Vu_gF
-	-	-	MPa	MPa	-	‰	mm	MPa	-	MPa	MPa	-	-	MPa	-	-	-	-	kN/m	kN	kN	kN	kN	kN
483	Leonhardt; Walther;	ET2	1,08	376,70	1,20		30,0	22,56	pr	18,05	2,47	fl	0,1093	2,21	0,0978	0,2598	0,2598	0,0596	1,53	1,94	129,00	131,94	130,94	130,94
484	DAfStb 151 (1962)	ET3	1,61	376,70	1,20		30,0	22,56	pr	18,05	2,47	fl	0,1093	2,21	0,0978	0,2598	0,2598	0,0894	1,20	1,17	125,08	127,53	126,25	126,25
485		ET4	3,23	376,70	1,20		30,0	22,56	pr	18,05	2,47	fl	0,1093	2,21	0,0978	0,2598	0,2598	0,1788	0,99	0,97	97,12	99,08	98,08	98,08
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	12,37	536,42	1,26		30,0	23,01	cu	18,41	2,38	fl	0,1036	2,24	0,0974	0,1381	0,1381	0,6719	8,70	15,23	784,80		800,03	800,03
491	Leonhardt; Walther	TA1	5,34	536,61	1,26		30,0	16,68	pr	13,34	2,06	fl	0,1235	1,72	0,1029	0,1849	0,1849	0,4002	3,23	3,02	335,50	340,90	338,53	338,53
492	DAfStb 156 (1963)	TA2	3,70	537,59	1,26		30,0	16,68	pr	13,34	2,06	fl	0,1235	1,72	0,1029	0,1849	0,1849	0,2773	3,23	2,82	319,32	324,71	322,14	322,14
493		TA3	2,32	525,82	1,26		30,0	16,19	pr	12,95	1,82	fl	0,1127	1,67	0,1034	0,1906	0,1906	0,1794	3,23	2,82	272,23	277,62	275,05	275,05
494		TA4	1,38	569,96	1,29		30,0	16,19	pr	12,95	1,82	fl	0,1127	1,67	0,1034	0,1906	0,1906	0,1064	3,23	2,82	229,55	234,95	232,38	232,38
497		TA11	2,32	525,82	1,26		30,0	24,92	pr	19,93	2,41	fl	0,0969	2,39	0,0958	0,1238	0,1238	0,1166	3,23	2,82	335,50	340,90	338,32	338,32
498		TA12	1,38	569,96	1,29		30,0	24,92	pr	19,93	2,41	fl	0,0969	2,39	0,0958	0,1238	0,1238	0,0691	3,23	2,82	264,87	270,27	267,69	267,69
499		TA13	5,34	536,61	1,26		30,0	21,29	pr	17,03	2,18	fl	0,1023	2,11	0,0989	0,1449	0,1449	0,3135	3,23	2,82	343,35	348,75	346,17	346,17
500		TA14	3,70	537,59	1,26		30,0	21,29	pr	17,03	2,18	fl	0,1023	2,11	0,0989	0,1449	0,1449	0,2172	3,23	2,82	333,54	338,94	336,36	336,36
501		TA15	2,32	525,82	1,26		30,0	20,31	pr	16,25	2,35	fl	0,1159	2,03	0,0998	0,1450	0,1450	0,1430	3,23	2,82	292,34	297,73	295,16	295,16
502		TA6	2,32	525,82	1,26		30,0	15,70	pr	12,56	1,77	fl	0,1125	1,63	0,1038	0,1965	0,1965	0,1851	3,23	2,82	232,50	237,89	235,32	235,32
503		TA16	2,32	525,82	1,26		30,0	20,31	pr	16,25	2,35	fl	0,1159	2,03	0,0998	0,1450	0,1450	0,1430	3,23	2,82	293,32	298,71	296,14	296,14
504	Levi;	RC 30 A1	4,02		0,00			23,75	cyl	19,00	0,00		0,0000	2,30	0,0968	0,2252	0,2252	0,2116	6,67	19,35			676,00	695,35
505	Marro	RC 30 A2	4,02		0,00			23,75	cyl	19,00	0,00		0,0000	2,30	0,0968	0,2252	0,2252	0,2116	6,67	19,35			688,00	707,35
506	(1989/1993)	RC 60 A1	4,02		0,00			44,65	cyl	35,72	0,00		0,0000	3,71	0,0830	0,1265	0,1265	0,1126	6,67	19,35			990,00	1009,35
507		RC 60 A2	4,02		0,00			44,65	cyl	35,72	0,00		0,0000	3,71	0,0830	0,1265	0,1265	0,1126	6,67	19,35			938,00	957,35
508		RC 60 B1	6,03		0,00			47,50	cyl	38,00	0,00		0,0000	3,88	0,0816	0,1653	0,1653	0,1587	6,67	19,35			1181,00	1200,35
509		RC 60 B2	6,03		0,00			47,50	cyl	38,00	0,00		0,0000	3,88	0,0816	0,1653	0,1653	0,1587	6,67	19,35			1239,00	1258,35
510		RC 70 B1	6,03		0,00			57,00	cyl	45,60	0,00		0,0000	4,27	0,0748	0,1378	0,1378	0,1323	6,67	19,35			1330,00	1349,35
511	Lyngberg	5A-0	3,53	760,00	1,13			24,42	cyl	19,53	2,03	sp	0,0829	2,35	0,0962	0,1678	0,1678	0,1807	3,83	6,70			435,00	441,70
512	(1974)	5B-0	3,39	734,00	1,13			25,27	cyl	20,22	2,68	sp	0,1061	2,41	0,0955	0,1639	0,1639	0,1676	3,83	6,70			435,00	441,70
523	Maruyama K., Rizkalla S. H. (1988)	RS2-WD	1,48	645,00	1,00			38,10	cyl	30,48	0,00		0,0000	3,30	0,0866	0,0144	0,0646	0,0487	2,17	2,01			126,00	128,01
525	Moayer; Regan (1974)	P20	0,66	427,59	1,38		19,0	38,66	cyl	30,92	0,00		0,0000	3,33	0,0863	0,0784	0,0784	0,0213	2,03	1,02			120,10	121,11
534	Özden (1967)	T6	1,66		0,00		16,0	31,69	cyl	25,35	0,00		0,0000	2,87	0,0907	0,1469	0,1469	0,0655	1,49	1,46			137,34	138,80
535		T7	1,66		0,00		16,0	28,98	cyl	23,19	0,00		0,0000	2,68	0,0926	0,1627	0,1627	0,0717	1,49	1,46			141,26	142,72
536		T9	1,66		0,00		16,0	28,98	cyl	23,19	0,00		0,0000	2,68	0,0926	0,1616	0,1616	0,0717	1,49	1,46			154,02	155,47
553	Özcebe; Ersoy;	TS56	0,61	360,00	1,41		15,0	57,95	cyl	46,36	0,00		0,0000	4,29	0,0740	0,2686	0,2686	0,0132	1,30	1,26			129,20	130,46
556	Tankut (1999)	TS59	0,71	360,00	1,41		15,0	77,90	cyl	62,32	0,00		0,0000	4,76	0,0612	0,2473	0,2473	0,0114	1,30	1,26			125,40	126,66
559		TS36	0,61	360,00	1,41		15,0	71,25	cyl	57,00	0,00		0,0000	4,62	0,0648	0,1639	0,1639	0,0107	1,30	0,86			155,90	156,76
561		TH39	0,53	360,00	1,41		15,0	69,35	cyl	55,48	0,00		0,0000	4,57	0,0659	0,1950	0,1950	0,0096	1,30	0,86			142,90	143,76
562		TS39	0,71	360,00	1,41		15,0	69,35	cyl	55,48	0,00		0,0000	4,57	0,0659	0,1950	0,1950	0,0128	1,30	0,86			179,20	180,06
564	Petersson (1972)	V1	0,70		0,00		16,0	36,10	cyl	28,88	0,00		0,0000	3,17	0,0878	0,2897	0,2897	0,0241	1,48	0,93	100,00		100,93	100,93
565		VL1	3,07		0,00		16,0	40,85	cyl	32,68	0,00		0,0000	3,47	0,0850	0,2912	0,2912	0,0938	1,46	0,73	190,00		190,73	190,73

			betar_meas	xr_meas	xr	sigswmeas	sigswass	sigsw	omwu	andbr	Mu	muu	vutest	kapc	sigsl	xsitest	zetatest	z_test	zeta	z_	muflex	Mu_flex	Vu_flex	betaflex	BB
-	-	-	[°]	mm	mm	MPa	MPa	MPa	-	-	kNm	-	-	-	MPa	-	-	mm	-	mm	-	-	kN	-	-
483	Leonhardt; Walther;	ET2	41	233	233	314		314	0,0596		137,49	0,2257	0,1881	0,910	420	0,286	0,857	257	0,857	257	0,2227	135,68	129,22	1,01	BB
484	DAfStb 151 (1962)	ET3			525	314		314	0,0894		132,56	0,2176	0,2707	0,910	408	0,278	0,861	258	0,857	257	0,2227	135,68	129,22	0,98	
485		ET4			525	314		314	0,1788		102,99	0,1691	0,4042	0,910	305	0,207	0,896	269	0,857	257	0,2227	135,68	129,22	0,76	
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)			1250	315		315	0,4964		2000,06	0,0851	0,5540	0,908	301	0,099	0,951	784	0,924	762	0,1276	2998,92	1199,57	0,67	
491	Leonhardt; Walther	TA1			563	349		349	0,3275		380,84	0,1692	0,4694	0,933	409	0,198	0,901	338	0,901	338	0,1666	375,13	333,45	1,02	BB
492	DAfStb 156 (1963)	TA2			625	417		417	0,2715		402,67	0,1789	0,4467	0,933	409	0,198	0,901	338	0,901	338	0,1666	375,13	300,11	1,07	BB
493		TA3			625	418		418	0,1794		343,81	0,1573	0,3902	0,935	372	0,185	0,907	340	0,898	337	0,1711	373,97	299,18	0,92	
494		TA4			625	440		440	0,1064		290,47	0,1329	0,3240	0,935	309	0,154	0,923	346	0,898	337	0,1711	373,97	299,18	0,78	
497		TA11			625	418		418	0,1166		422,91	0,1257	0,3038	0,900	409	0,137	0,931	349	0,931	349	0,1153	387,77	310,21	1,09	BB
498		TA12			625	440		440	0,0691		334,62	0,0995	0,2378	0,900	349	0,117	0,941	353	0,931	349	0,1153	387,77	310,21	0,86	
499		TA13			625	348		348	0,2556		432,72	0,1506	0,3679	0,915	409	0,158	0,921	345	0,921	345	0,1334	383,42	306,73	1,13	BB
500		TA14			625	425		425	0,2167		420,45	0,1463	0,3575	0,915	409	0,158	0,921	345	0,921	345	0,1334	383,42	306,73	1,097	BB
501		TA15			625	418		418	0,1430		368,95	0,1346	0,3288	0,919	390	0,158	0,921	345	0,921	345	0,1335	366,06	292,85	1,01	BB
502		TA6			625	393		393	0,1742		294,15	0,1388	0,3397	0,937	314	0,161	0,919	345	0,895	336	0,1759	372,74	298,19	0,79	
503		TA16			625	418		418	0,1430		370,18	0,1350	0,3298	0,919	390	0,158	0,921	345	0,921	345	0,1335	366,06	292,85	1,01	BB
504	Levi;	RC 30 A1			1900	480		480	0,2116		2568,80	0,1360	0,3435	0,905	329	0,164	0,918	863	0,876	823	0,1971	3723,48	979,86	0,69	
505	Marro	RC 30 A2			1900	480		480	0,2116		2614,40	0,1384	0,3502	0,905	335	0,167	0,917	862	0,876	823	0,1971	3723,48	979,86	0,70	
506	(1989/1993)	RC 60 A1			1900	480		480	0,1126		3762,00	0,1059	0,2640	0,821	405	0,139	0,931	875	0,923	868	0,1168	4145,79	1091,00	0,91	
507		RC 60 A2			1900	480		480	0,1126		3564,40	0,1004	0,2491	0,821	382	0,131	0,935	879	0,923	868	0,1168	4145,79	1091,00	0,86	
508		RC 60 B1			1900	480		480	0,1587		4487,80	0,1188	0,2994	0,810	367	0,159	0,920	865	0,898	844	0,1485	5608,32	1475,87	0,80	
509		RC 60 B2			1900	480		480	0,1587		4708,20	0,1246	0,3156	0,810	387	0,168	0,916	861	0,898	844	0,1485	5608,32	1475,87	0,84	
510		RC 70 B1			1900	480		480	0,1323		5054,00	0,1115	0,2806	0,772	413	0,157	0,922	866	0,911	856	0,1255	5688,41	1496,95	0,89	
511	Lyngberg	5A-0			750	674		674	0,1807		652,50	0,1309	0,3731	0,902	522	0,158	0,921	497	0,907	490	0,1522	758,58	505,72	0,86	
512	(1974)	5B-0			750		647	647	0,1676		652,50	0,1265	0,3594	0,899	520	0,152	0,924	499	0,909	491	0,1490	768,48	512,32	0,85	
523	Maruyama K., Rizkalla S. H. (1988)	RS2-WD			575			0	0,0000		144,90	0,0370	0,1233	0,848	235	0,045	0,978	389	0,962	382	0,0621	243,05	211,34	0,60	
525	Moayer; Regan (1974)	P20			489		310	310	0,0213		117,37	0,0649	0,0966	0,845	553	0,080	0,960	268	0,954	266	0,0748	135,30	138,44	0,87	
534	Özden (1967)	T6	39		525	273	255	255	0,0325		144,21	0,1281	0,1796	0,873	418	0,159	0,920	274	0,916	273	0,1345	151,41	144,20	0,95	
535		T7	42		525	273		273	0,0717		148,33	0,1441	0,2041	0,884	435	0,179	0,911	271	0,908	271	0,1477	152,10	144,86	0,98	
536		T9			525	273		273	0,0717		161,72	0,1571	0,2230	0,884	444	0,183	0,909	271	0,909	271	0,1469	151,20	144,00	1,07	BB
553	Özcebe; Ersoy;	TS56			775			0	0,0000		200,26	0,2397	0,0726	0,768	450	0,350	0,825	256	0,825	256	0,2217	185,16	119,46	1,08	BB
556	Tankut (1999)	TS59			775			0	0,0000		194,37	0,1731	0,0508	0,688	357	0,295	0,853	264	0,820	254	0,2029	227,85	147,00	0,85	
559		TS36			465			0	0,0000		144,99	0,1412	0,0662	0,715	436	0,222	0,889	276	0,885	274	0,1451	149,01	160,22	0,97	
561		TH39			465			0	0,0000		132,90	0,1329	0,0617	0,723	333	0,205	0,898	278	0,865	268	0,1687	168,66	181,36	0,79	
562		TS39			465			0	0,0000		166,66	0,1667	0,0801	0,723	433	0,266	0,867	269	0,865	268	0,1687	168,66	181,36	0,99	
564	Petersson (1972)	V1			575		323	323	0,0241		116,07	0,1772	0,0703	0,856	677	0,235	0,883	284	0,831	267	0,2407	157,63	137,07	0,74	
565		VL1			450		366	366	0,0938		171,66	0,1806	0,1995	0,837	700	0,246	0,877	244	0,826	230	0,2405	228,53	253,93	0,75	

			sin2thp	thp	cotthp	vup	gamwp	cotthu	thu	nueu	betalb	VB	A2+A3	kon34	KONA4	KONA5	A4+A5	Differenz	AF	f1ck	fcm_cyl	fck	fc_prime	Vct_ACI	Vs_ACI	Vcal_sACI	Vmax_ACI	rat_smax	VACI_cal
-	-	-	-	[°]	-	-	-	-	[°]	-	-	-	-	-	-	-	-	-		MPa	MPa	MPa	MPa	kN	kN	kN	kN		kN
483	Leonhardt; Walther;	ET2	0,06	14,13	3,97	0,24	0,79	3,16	17,59	0,52	0,73		1	1	1	0	1	0		18,8	23,8	19,8	21,4	34,54	48,41	82,95	172,68	0,480	82,95
484	DAfStb 151 (1962)	ET3	0,09	17,40	3,19	0,29	0,95	3,03	18,27	0,73	0,71		1	1	1	0	1	0		18,8	23,8	19,8	21,4	23,02	48,41	71,44	115,12	0,621	71,44
485		ET4	0,18	25,02	2,14	0,38	1,05	2,26	23,87	0,87	0,55		1	0	0	0	0	1		18,8	23,8	19,8	21,4	11,51	48,41	59,93	57,56	1,041	57,56
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	0,67	55,06	0,70	0,47	1,18	1,12	41,86	0,89	0,01		1	1	1	0	1	0		19,2	24,2	20,2	21,8	64,01	1020,61	1084,63	320,07	3,389	320,07
491	Leonhardt; Walther	TA1	0,40	39,24	1,22	0,49	0,96	1,43	34,90	0,80	1,84	VB	0	1	0	0	0	0	AF	12,9	17,6	13,6	15,2	38,79	320,33	359,12	193,97	1,851	193,97
492	DAfStb 156 (1963)	TA2	0,28	31,77	1,61	0,45	1,00	1,65	31,29	0,81	1,75	VB	0	1	0	0	0	0	AF	12,9	17,6	13,6	15,2	38,79	221,94	260,73	193,97	1,344	193,97
493		TA3	0,18	25,06	2,14	0,38	1,02	2,17	24,70	0,82	1,53	VB	0	1	0	0	0	0	AF	12,4	17,0	13,0	14,6	38,13	139,42	177,55	190,64	0,931	177,55
494		TA4	0,11	19,04	2,90	0,31	1,05	3,05	18,18	0,87	1,30	VB	0	1	0	0	0	0	AF	12,4	17,0	13,0	14,6	38,13	82,66	120,79	190,64	0,634	120,79
497		TA11	0,12	19,96	2,75	0,32	0,95	2,61	21,00	0,73	0,66		1	1	1	0	1	0		21,1	26,2	22,2	23,8	48,65	139,42	188,07	243,23	0,773	188,07
498		TA12	0,07	15,24	3,67	0,25	0,94	3,44	16,21	0,71	0,53		1	1	1	0	1	0		21,1	26,2	22,2	23,8	48,65	82,66	131,31	243,23	0,540	131,31
499		TA13	0,31	34,05	1,48	0,46	0,79	1,44	34,79	0,63	0,77		1	1	1	0	1	0		17,5	22,4	18,4	20,0	44,58	320,33	364,90	222,88	1,637	222,88
500		TA14	0,22	27,78	1,90	0,41	0,87	1,65	31,22	0,65	0,75		1	1	1	0	1	0		17,5	22,4	18,4	20,0	44,58	221,94	266,51	222,88	1,196	222,88
501		TA15	0,14	22,22	2,45	0,35	0,94	2,30	23,51	0,72	0,69		1	1	1	0	1	0		16,5	21,4	17,4	19,0	43,41	139,42	182,83	217,05	0,842	182,83
502		TA6	0,19	25,48	2,10	0,39	0,87	1,95	27,15	0,67	1,35	VB	0	1	0	0	0	0	AF	11,9	16,5	12,5	14,1	37,45	139,42	176,87	187,25	0,945	176,87
503		TA16	0,14	22,22	2,45	0,35	0,94	2,31	23,44	0,72	0,69		1	1	1	0	1	0		16,5	21,4	17,4	19,0	43,41	139,42	182,83	217,05	0,842	182,83
504	Levi;	RC 30 A1	0,21	27,39	1,93	0,41	0,84	1,62	31,64	0,62	0,42		1	1	1	0	1	0		20,0	25,0	21,0	22,6	89,07	453,60	542,66	445,33	1,219	445,33
505	Marro	RC 30 A2	0,21	27,39	1,93	0,41	0,86	1,65	31,14	0,63	0,42		1	1	1	0	1	0		20,0	25,0	21,0	22,6	89,07	453,60	542,66	445,33	1,219	445,33
506	(1989/1993)	RC 60 A1	0,11	19,60	2,81	0,32	0,84	2,35	23,09	0,59	0,34		1	1	1	0	1	0		40,9	47,0	43,0	44,6	125,12	453,60	578,71	625,59	0,925	578,71
507		RC 60 A2	0,11	19,60	2,81	0,32	0,79	2,21	24,32	0,53	0,33		1	1	1	0	1	0		40,9	47,0	43,0	44,6	125,12	453,60	578,71	625,59	0,925	578,71
508		RC 60 B1	0,16	23,48	2,30	0,37	0,82	1,89	27,93	0,58	0,34		1	1	1	0	1	0		43,7	50,0	46,0	47,6	129,26	680,39	809,65	646,29	1,253	646,29
509		RC 60 B2	0,16	23,48	2,30	0,37	0,86	1,99	26,70	0,63	0,36		1	1	1	0	1	0		43,7	50,0	46,0	47,6	129,26	680,39	809,65	646,29	1,253	646,29
510		RC 70 B1	0,13	21,33	2,56	0,34	0,83	2,12	25,24	0,58	0,35		1	1	1	0	1	0		53,2	60,0	56,0	57,6	142,19	680,39	822,58	710,95	1,157	710,95
511	Lyngberg	5A-0	0,18	25,15	2,13	0,38	0,97	2,06	25,84	0,76	0,46		1	1	0	1	1	0		20,6	25,7	21,7	23,3	51,95	228,68	280,63	259,76	1,080	259,76
512	(1974)	5B-0	0,17	24,16	2,23	0,37	0,96	2,14	25,00	0,75	0,44		1	1	0	1	1	0		21,5	26,6	22,6	24,2	52,95	219,52	272,47	264,73	1,029	264,73
523	Maruyama K., Rizkalla S. H. (1988)	RS2-WD	0,05	12,75	4,42	0,22	0,57	0,00	0,00	0,00	0,55		0	0	0	0	0	0		34,3	40,1	36,1	37,7	39,60	57,65	97,25	198,02	0,491	97,25
525	Moayer; Regan (1974)	P20	0,02	8,39	6,78	0,14	0,67	4,53	12,44	0,37	0,34		1	1	1	0	1	0		34,9	40,7	36,7	38,3	43,03	27,59	70,62	215,16	0,328	70,62
534	Özden (1967)	T6	0,07	14,83	3,78	0,25	0,73	5,53	10,24	0,82	0,48		1	1	1	0	1	0		27,9	33,4	29,4	31,0	30,29	54,47	84,76	151,45	0,560	84,76
535		T7	0,07	15,53	3,60	0,26	0,79	2,85	19,34	0,52	0,53		1	1	1	0	1	0		25,2	30,5	26,5	28,1	28,87	54,47	83,33	144,33	0,577	83,33
536		T9	0,07	15,53	3,60	0,26	0,86	3,11	17,81	0,61	0,58		1	1	1	0	1	0		25,2	30,5	26,5	28,1	28,87	54,47	83,33	144,33	0,577	83,33
553	Özcebe; Ersoy;	TS56	0,01	6,59	8,66	0,11	0,64	0,00	0,00	0,00	0,00		0	1	0	0	0	0		54,2	61,0	57,0	58,6	59,12	28,38	87,50	295,61	0,296	87,50
556	Tankut (1999)	TS59	0,01	6,14	9,30	0,11	0,48	0,00	0,00	0,00	0,00		0	1	0	0	0	0		74,1	82,0	78,0	79,6	68,91	33,11	102,02	344,53	0,296	102,02
559		TS36	0,01	5,94	9,61	0,10	0,64	0,00	0,00	0,00	0,01		0	1	0	0	0	0		67,5	75,0	71,0	72,6	65,81	28,38	94,19	329,03	0,286	94,19
561		TH39	0,01	5,63	10,14	0,10	0,63	0,00	0,00	0,00	0,01		0	1	0	0	0	0		65,6	73,0	69,0	70,6	64,89	24,83	89,73	324,47	0,277	89,73
562		TS39	0,01	6,51	8,77	0,11	0,71	0,00	0,00	0,00	0,01		0	1	0	0	0	0		65,6	73,0	69,0	70,6	64,89	33,11	98,01	324,47	0,302	98,01
564	Petersson (1972)	V1	0,02	8,93	6,36	0,15	0,46	2,92	18,93	0,18	0,37		1	1	1	0	1	0		32,3	38,0	34,0	35,6	55,84	39,21	95,05	279,21	0,340	95,05
565		VL1	0,09	17,84	3,11	0,29	0,68	2,13	25,19	0,41	0,40		1	1	1	0	1	0		37,1	43,0	39,0	40,6	35,30	102,29	137,59	176,52	0,779	137,59

			gmod_ACI
-	-	-	
483	Leonhardt; Walther;	ET2	1,579
484	DAfStb 151 (1962)	ET3	1,767
485		ET4	1,704
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	2,500
491	Leonhardt; Walther	TA1	1,745
492	DAfStb 156 (1963)	TA2	1,661
493		TA3	1,549
494		TA4	1,924
497		TA11	1,799
498		TA12	2,039
499		TA13	1,553
500		TA14	1,509
501		TA15	1,614
502		TA6	1,330
503		TA16	1,620
504	Levi;	RC 30 A1	1,518
505	Marro	RC 30 A2	1,545
506	(1989/1993)	RC 60 A1	1,711
507		RC 60 A2	1,621
508		RC 60 B1	1,827
509		RC 60 B2	1,917
510		RC 70 B1	1,871
511	Lyngberg	5A-0	1,675
512	(1974)	5B-0	1,643
523	Maruyama K., Rizkalla S. H. (1988)	RS2-WD	1,296
525	Moayer; Regan (1974)	P20	1,701
534	Özden (1967)	T6	1,620
535		T7	1,695
536		T9	1,848
553	Ozcebe; Ersoy;	TS56	1,477
556	Tankut (1999)	TS59	1,229
559		TS36	1,655
561		TH39	1,593
562		TS39	1,828
564	Petersson (1972)	V1	1,062
565		VL1	1,386

				E	b	bw	h	hf	hh,top	hw	hft	hh,bot	bft	Ac	zc2	aa	af	ba	L	c	a	kap	cc
	-	-	-	SI / A	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm²	mm	mm	mm	mm	mm	mm	mm	-	mm
Regan_1971_012_R21	593	Regan (1971)	R21	SI	152,0	152,0	305,0							46360	152,5	60,0	75,0	300,0	1828,8	75,0	876,9	3,45	25,4
Regan_1971_024_T6	605		T6	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_027_T9	608		T9	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_032_T17	613		T17	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	3657,6	50,0	1803,8	7,10	25,4
Regan_1971_034_T20	615		T20	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	2743,2	50,0	1346,6	5,30	25,4
Regan_1971_039_T26	620		T26	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_040_T27	621		T27	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_044_T32	625		T32	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_048_T36	629		T36	SI	610,0	152,0	305,0	142,0		163,0				111396	104,9	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_049_T37	630		T37	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_050_T38	631		T38	SI	610,0	152,0	305,0	142,0		163,0				111396	104,9	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_056_W1	637		W1	SI	610,0	64,0	305,0	76,0		127,0	64,0	38,0	254,0	76786	113,3	75,0	50,0	305,0	1830,0	50,0	890,0	3,33	25,4
Regan_1971_058_W3	639		W3	SI	610,0	64,0	305,0	76,0		127,0	64,0	38,0	254,0	76786	113,3	75,0	50,0	305,0	1830,0	50,0	890,0	3,33	25,4
Regan_1971_059_W5	640		W5	SI	610,0	64,0	305,0	76,0		127,0	64,0	38,0	254,0	76786	113,3	75,0	50,0	305,0	1830,0	50,0	890,0	3,33	25,4
Regan_1971_061_W7	642		W7	SI	610,0	64,0	305,0	76,0		127,0	64,0	38,0	254,0	76786	113,3	75,0	50,0	305,0	1830,0	50,0	890,0	3,33	25,4
Rehm_1978_001_RsIS / BQ II 0	643	Rehm; Eligehausen;	RsIS / BQ II 0	SI	900,0	250,0	510,0	125,0		385,0				208750	180,1				3870,0	1290,0	1290,0	3,00	
Rehm_1978_004_RnIIS	646	Neubert (1978)	RnIIS	SI	950,0	450,0	600,0	150,0		450,0				345000	251,1				4932,0	1644,0	1644,0	3,00	
Reineck_1991_001_Stb III	647	Reineck (1991)	Stb III	SI	838,0	77,0	648,0	107,0		441,0	100,0		837,0	207323	318,2	120,0	160,0	700,0	5200,0	160,0	2520,0	4,27	
Reineck_1991_002_Stb I	648		Stb I	SI	837,0	77,0	650,0	100,0		440,0	110,0		838,0	209760	333,1	120,0	160,0	700,0	5190,0	160,0	2515,0	4,27	
Roller_1990_007_7	658	Roller; Russel (1990)	7	A	457,2	457,2	870,0							397741	435,0	266,7	133,4	152,4	4572,0	133,4	2219,3	2,91	
Roller_1990_009_9	660		9	A	457,2	457,2	870,0							397741	435,0	266,7	133,4	152,4	4572,0	133,4	2219,3	2,91	
Roller_1990_010_10	661		10	A	457,2	457,2	870,0							397741	435,0	266,7	133,4	152,4	4572,0	133,4	2219,3	2,91	
Sarsam_1992_010_BS4-H	671	Sarsam;	BS4-H	SI	180,0	180,0	270,0							48600	135,0	100,0	100,0	150,0	1565,0	400,0	582,5	2,50	25,0
Sarsam_1992_013_CS3-H	674	Al-Musawi (1992)	CS3-H	SI	180,0	180,0	270,0							48600	135,0	100,0	100,0	150,0	1565,0	400,0	582,5	2,50	25,0
Sarsam_1992_014_CS4-H	675		CS4-H	SI	180,0	180,0	270,0							48600	135,0	100,0	100,0	150,0	1565,0	400,0	582,5	2,50	25,0
Shin_1999_010_MHB 2.5-50	685	Shin; Lee;	MHB 2.5-50	SI	125,0	125,0	250,0							31250	125,0		22,5	100,0	1075,0	22,5	526,3	2,45	
Shin_1999_022_HB 2.5-50	697	Moon; Gosh (1999)	HB 2.5-50	SI	125,0	125,0	250,0							31250	125,0		22,5	100,0	1075,0	22,5	526,3	2,45	
Soerensen_1974_003_T-23	702	Soerensen (1974)	T-23	SI	400,0	110,0	355,0	80,0		275,0				62250	126,3	100,0	100,0	250,0	3000,0	900,0	1050,0	3,52	26,0
Soerensen_1974_007_T-2-B	706		T-2-B	SI	400,0	110,0	355,0	80,0		275,0				62250	126,3	100,0	100,0	250,0	3000,0	900,0	1050,0	3,52	26,0
Stroband_1997_002_2	711	Stroband (1997)	2	SI	500,0	90,0	700,0	125,0	25,0	400,0	75,0	75,0	500,0	165500	335,5	242,3	91,4	346,2	5000,0	69,2	1654,3	2,56	
Stroband_1997_003_3	712		3	SI	500,0	90,0	700,0	125,0	25,0	400,0	75,0	75,0	500,0	165500	335,5	242,3	91,4	346,2	5000,0	69,2	1654,3	2,56	
Stroband_1997_004_4	713		4	SI	500,0	90,0	700,0	125,0	25,0	400,0	75,0	75,0	500,0	165500	335,5	242,3	91,4	346,2	5000,0	69,2	1654,3	2,50	
Stroband_1997_005_5	714		5	SI	500,0	90,0	700,0	125,0	25,0	400,0	75,0	75,0	500,0	165500	335,5	242,3	91,4	346,2	5000,0	69,2	1654,3	2,55	
Taylor_1966_001_ST-2-C	783	Taylor, R. (1966)	ST-2-C	A	304,8	114,3	304,8	76,2		228,6				49355	118,8			762,0	2133,6	304,8	914,4	3,56	25,4
Taylor_1966_004_HSS-1-B	786		HSS-1-B	A	304,8	114,3	304,8	76,2		228,6				49355	118,8			533,4	2590,8	304,8	1143,0	4,44	25,4
Yoon_1996_003_N2-N	803	Yoon; Cook;	N2-N	SI	375,0	375,0	750,0							281250	375,0	150,0	75,0	350,0	4300,0	75,0	2112,5	3,23	40,0
Yoon_1996_006_M2-N	806	Mitchell (1996)	M2-N	SI	375,0	375,0	750,0							281250	375,0	150,0	75,0	350,0	4300,0	75,0	2112,5	3,23	40,0
Yoon_1996_008_H2-S	808		H2-S	SI	375,0	375,0	750,0							281250	375,0	150,0	75,0	350,0	4300,0	75,0	2112,5	3,23	40,0
Yoon_1996_009_H2-N	809		H2-N	SI	375,0	200,0	750,0							281250	375,0	150,0	75,0	350,0	4300,0	75,0	2112,5	3,23	40,0
Rosenbusch_2003_002_1.4-1	817	Rosenbusch (2003)	1.4/1	SI	200,0	200,0	300,0							60000	150,0	100,0	50,0	100,0	1800,0		875,0	3,37	36,0
Rosenbusch_2003_004_1.7-1	819		1.7/1	SI	200,0	200,0	300,0							60000	150,0	100,0	50,0	100,0	1800,0		875,0	3,37	32,0
Kautsch_2010_001_A2	820	Kautsch (2010)	A2	SI	655,0	155,0	820,0	150,0	20,0	650,0			155,0	207100	282,6	100,0	150,0	200,0	4000,0		1925,0	2,68	20,0
Kautsch_2010_002_B2	821		B2	SI	655,0	155,0	820,0	150,0	20,0	650,0			155,0	207100	282,6	100,0	150,0	200,0	4000,0		1925,0	2,68	20,0
Tanimura_2005_031_41	874	Tanimura; Sato (2005)	41	SI	300,0	300,0	450,0							135000	225,0	100,0	100,0	300,0	2400,0	400,0	1000,0	2,50	
	157																						
Plain	68																						
Ripp	89																						

			ds	d	ns	dst	fR	As	alphaa	rhos	rhosw	fsy	esy	ftk	ftk/fsy	euk	ds2	ns2	dst2	As2	fsy2	dw	nsw	Asw	frw	sw	sw/h		zw1	zw2	rhow	fyw
-	-	-	mm	mm	-	mm	-	mm²	-	%	%	MPa	‰	MPa	-	‰	mm	-	mm	mm²	MPa	mm	-	mm²	-	mm	-	-	mm	mm	%	MPa
593	Regan (1971)	R21	254,0	254,0	8	16,0	r	1608,5	1,00	4,1662	4,1662	630,0	3,15	950,00	1,51			2	9,5	141,8	270	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0
605		T6	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	9,5	2	142,5	0	114,0	0,37	0,45	206	203	0,8224	270,0
608		T9	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0
613		T17	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0
615		T20	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0
620		T26	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0
621		T27	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0
625		T32	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	9,5	2	142,5	0	114,0	0,37	0,45	206	203	0,8224	270,0
629		T36	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0
630		T37	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	9,5	2	142,5	0	114,0	0,37	0,45	206	203	0,8224	270,0
631		T38	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270	9,5	2	142,5	0	114,0	0,37	0,45	206	203	0,8224	270,0
637		W1	267,0	267,0	8	16,0	r	1608,5	1,00	0,9876	9,4130	630,0	3,15	950,00	1,51			4	10,0	314,2	275	10,0	2	157,1	r	114,0	0,37	0,43	217	216	2,1530	704,0
639		W3	267,0	267,0	8	16,0	r	1608,5	1,00	0,9876	9,4130	630,0	3,15	950,00	1,51			4	10,0	314,2	275	10,0	2	157,1	r	114,0	0,37	0,43	217	216	2,1530	704,0
640		W5	267,0	267,0	8	16,0	r	1608,5	1,00	0,9876	9,4130	630,0	3,15	950,00	1,51			4	10,0	314,2	275	10,0	2	157,1	r	76,0	0,25	0,28	217	216	3,2294	704,0
642		W7	267,0	267,0	8	16,0	r	1608,5	1,00	0,9876	9,4130	630,0	3,15	950,00	1,51			4	10,0	314,2	275	8,0	2	100,5	r	76,0	0,25	0,28	227	216	2,0668	691,0
643	Rehm; Eligehausen;	RsIS / BQ II 0	430,0	430,0	8	26,0	r	4247,4	1,00	1,0975	3,9511	469,0	2,35	565,00	1,20	14		4	8,0	201,1		11,0	2	190,1	r	100,0	0,20	0,23	375	0	0,7603	560,0
646	Neubert (1978)	RnIIS	548,0	548,0	4	26,0	r	3355,2	1,00	0,6445	1,3606	746,0	3,73	1007,00	1,35							8,5	2	113,5	r	75,0	0,13	0,14	506	0	0,3363	557,0
647	Reineck (1991)	Stb III	590,0	590,0	4	28,0	r	6157,5	1,00	1,2454	13,5539	514,6	2,57	605,00	1,18			5	28,0	7081,1	514	8,0	2	99,4	r	108,0	0,17	0,18	550	0	1,1953	515,0
648		Stb I	589,0	589,0	5	28,0	r	7081,1	1,00	1,4364	15,6134	514,0	2,57	603,78	1,17			4	28,0	6157,5	515	10,0	2	162,8	r	126,0	0,19	0,21	539	0	1,6780	549,0
658	Roller; Russel (1990)	7	762,0	762,0	8	32,3	r	6538,1	1,00	1,8767	1,8767	472,4	2,36	741,24	1,57							9,5	2	142,5	r	196,9	0,23	0,26	714	0	0,1583	445,2
660		9	762,0	762,0	10	32,3	r	8172,7	1,00	2,3459	2,3459	472,4	2,36	741,24	1,57							9,5	2	142,5	r	196,9	0,23	0,26	714	0	0,1583	445,2
661		10	762,0	762,0	10	35,8	r	10073,9	1,00	2,8916	2,8916	431,0	2,16	690,55	1,60							9,5	2	142,5	r	133,4	0,15	0,18	714	0	0,2337	445,2
671	Sarsam;	BS4-H	233,0	233,0	2	25,0	r	1182,8	0,70	2,8202	2,8202	502,3	2,51		0,00		34,0	2	10,0	157,1	450	4,0	2	25,1	0	75,0	0,28	0,32	213	183	0,1862	820,0
674	Al-Musawi (1992)	CS3-H	233,0	233,0	3	25,0	r	1470,0	0,70	3,5051	3,5051	543,0	2,72		0,00		34,0	2	10,0	157,1	450	4,0	2	25,1	0	100,0	0,37	0,43	213	183	0,1396	820,0
675		CS4-H	233,0	233,0	3	25,0	r	1470,0	0,70	3,5051	3,5051	543,0	2,72		0,00		34,0	2	10,0	157,1	450	4,0	2	25,1	0	75,0	0,28	0,32	213	183	0,1862	820,0
685	Shin; Lee;	MHB 2.5-50	215,0	215,0	2	25,4	r	1013,4	0,70	3,7708	3,7708	410,1	2,05		0,00			2	9,5	142,5	392	6,4	2	63,3	r	86,0	0,34	0,40	183	0	0,5892	365,9
697	Moon; Gosh (1999)	HB 2.5-50	215,0	215,0	2	25,4	r	1013,4	0,70	3,7708	3,7708	410,1	2,05		0,00			2	9,5	142,5	392	6,4	2	63,3	r	86,0	0,34	0,40	183	0	0,5892	365,9
702	Soerensen (1974)	T-23	298,0	298,0	4	20,0	r	1256,6	1,00	1,0542	3,8335	419,9	2,10	716,13	1,71	15						6,0	2	56,5	0	150,0	0,42	0,50	268	246	0,3427	349,2
706		T-2-B	298,0	298,0	4	20,0	r	1256,6	1,00	1,0542	3,8335	457,1	2,29	765,18	1,67	15	36,0	4	8,0	201,1	262	5,0	2	39,3	0	117,0	0,33	0,39	273	246	0,3051	397,3
711	Stroband (1997)	2	647,1	647,1	6	25,0	r	3573,6	1,00	1,1045	6,1359	600,0	3,00		0,00			4	10,0	314,2		8,0	2	100,5	r	165,0	0,24	0,25	607	0	0,6770	600,0
712		3	647,1	647,1	6	32,0	r	5227,6	1,00	1,6157	8,9759	560,0	2,80		0,00			4	10,0	314,2		12,0	2	226,2	r	165,0	0,24	0,25	587	0	1,5232	630,0
713		4	661,5	661,5	8	32,0	r	6434,0	1,00	1,9452	10,8064	560,0	2,80		0,00			4	10,0	314,2		12,0	2	226,2	r	95,0	0,14	0,14	602	0	2,6456	630,0
714		5	650,0	650,0	8	32,0	r	6836,1	1,00	2,1034	11,6856	560,0	2,80		0,00			4	10,0	314,2		12,0	2	226,2	r	65,0	0,09	0,10	590	0	3,8666	630,0
783	Taylor, R. (1966)	ST-2-C	257,2	257,2	5	15,9	r	989,7	1,00	1,2625	3,3668	441,4	1,75		0,00			2	22,2	775,9	278	6,4	2	63,3	0	90,0	0,30	0,35	0	210	0,6157	289,7
786		HSS-1-B	257,2	257,2	5	15,9	r	989,7	1,00	1,2625	3,3668	441,4	1,75		0,00			2	22,2	775,9	278	5,2	2	41,8	0	112,0	0,37	0,44	0	210	0,3262	437,9
803	Yoon; Cook;	N2-N	655,0	655,0	10	29,9	r	7021,5	1,00	2,8586	2,8586	400,0	2,00		0,00			2	10,0	157,1		9,5	2	141,8	r	325,0	0,43	0,50	608	575	0,1163	430,0
806	Mitchell (1996)	M2-N	655,0	655,0	10	29,9	r	7021,5	1,00	2,8586	2,8586	400,0	2,00		0,00			2	10,0	157,1		9,5	2	141,8	r	230,0	0,31	0,35	608	575	0,1644	430,0
808		H2-S	655,0	655,0	10	29,9	r	7021,5	1,00	2,8586	2,8586	400,0	2,00		0,00			2	10,0	157,1		9,5	2	141,8	r	270,0	0,36	0,41	608	575	0,1400	430,0
809		H2-N	655,0	655,0	10	29,9	r	7021,5	1,00	2,8586	5,3600	400,0	2,00		0,00			2	10,0	157,1		9,5	2	141,8	r	160,0	0,21	0,24	608	575	0,4430	430,0
817	Rosenbusch (2003)	1.4/1	260,0	260,0	3	28,0	r	1847,3	0,01	3,5524	3,5524	550,0	2,75		0,00		40,0	2	6,0	56,5	550	4,0	2	25,1	r	90,0	0,30	0,35	240	188	0,1396	680,0
819		1.7/1	260,0	260,0	3	28,0	r	1847,3	0,01	3,5524	3,5524	550,0	2,75		0,00		40,0	2	6,0	56,5	550	8,0	2	100,5	r	90,0	0,30	0,35	220	196	0,5585	550,0
820	Kautsch (2010)	A2	719,4	719,4		28,0	r	3195,0	0,01	0,6781	2,8655	468,1	2,34	648,18	1,38	18,0	75,0	12	12,0	1357,2	585	8,0	2	100,5	r	150,0	0,18	0,21	679	679	0,4324	605,0
821		B2	719,4	719,4		28,0	r	3195,0	0,01	0,6781	2,8655	468,1	2,34	648,18	1,38	11,0	75,0	12	12,0	1357,2	585	8,0	2	100,5	r	200,0	0,24	0,28	679	679	0,3243	605,0
874	Tanimura; Sato (2005)	41	400,0	400,0	4	29,0	r	2642,1	0,70	2,2017	2,2017	750,0	3,75					2	16,0	402,1		10,0	2	157,1		100,0	0,22	0,25	350	0	0,5236	388,0
157																																
68																																
89																																

			fyw_rhow_100	fwtk	fwtk/fyw	ewuk	diaa	f1c	Method	fcwu	f1cttest	Method	betacttest	f1ctmcal	betactcal	oms	oml	omwy	g	Vg	F	Vu_F+g,rep	Vu_Rep	Vu_gF
-	-	-	MPa	MPa	-	‰	mm	MPa	-	MPa	MPa	-	-	MPa	-	-	-	-	kN/m	kN	kN	kN	kN	kN
593	Regan (1971)	R21	1,13	460,00	1,64		19,0	45,08	cu	36,06	0,00		0,0000	3,73	0,0828	0,5823	0,5823	0,0312	1,11	0,53			153,00	153,53
605		T6	2,22	435,00	1,61		19,0	26,33	cu	21,06	0,00		0,0000	2,49	0,0947	0,2484	0,2484	0,1054	1,95	0,91			205,00	205,91
608		T9	1,13	460,00	1,64		19,0	20,25	cu	16,20	0,00		0,0000	2,02	0,0998	0,3230	0,3230	0,0695	1,95	0,91			154,00	154,91
613		T17	1,13	460,00	1,64		19,0	33,53	cu	26,82	0,00		0,0000	3,00	0,0894	0,1951	0,1951	0,0420	1,95	1,81			134,00	135,81
615		T20	1,13	460,00	1,64		19,0	30,08	cu	24,06	0,00		0,0000	2,76	0,0918	0,2175	0,2175	0,0468	1,95	1,36			138,00	139,36
620		T26	1,13	460,00	1,64		19,0	49,28	cu	39,42	0,00		0,0000	3,98	0,0808	0,1327	0,1327	0,0286	1,95	0,91			179,00	179,91
621		T27	1,13	460,00	1,64		19,0	12,90	cu	10,32	0,00		0,0000	1,36	0,1056	0,5070	0,5070	0,1091	1,95	0,91			132,00	132,91
625		T32	2,22	435,00	1,61		19,0	26,70	cu	21,36	0,00		0,0000	2,52	0,0944	0,2450	0,2450	0,1040	1,95	0,91			216,00	216,91
629		T36	1,13	460,00	1,64		19,0	21,30	cu	17,04	0,00		0,0000	2,11	0,0989	0,3071	0,3071	0,0661	2,67	1,26			179,00	180,26
630		T37	2,22	435,00	1,61		19,0	28,50	cu	22,80	0,00		0,0000	2,65	0,0930	0,2295	0,2295	0,0974	1,95	0,91			209,00	209,91
631		T38	2,22	435,00	1,61		19,0	29,18	cu	23,34	0,00		0,0000	2,70	0,0925	0,2242	0,2242	0,0951	2,67	1,26			238,00	239,26
637		W1	15,16	1010,00	1,43		19,0	28,41	cyl	22,72	2,70	sp	0,0951	2,64	0,0931	0,2190	0,2190	0,6670	1,84	0,87			160,00	160,87
639		W3	15,16	1010,00	1,43		19,0	43,23	cyl	34,58	3,31	sp	0,0766	3,62	0,0837	0,1439	0,1439	0,4383	1,84	0,87			199,00	199,87
640		W5	22,74	1010,00	1,43		19,0	32,59	cyl	26,07	2,92	sp	0,0895	2,94	0,0901	0,1909	0,1909	0,8722	1,84	0,87			190,00	190,87
642		W7	14,28	827,00	1,20		19,0	32,59	cyl	26,07	3,02	sp	0,0925	2,94	0,0901	0,1909	0,1909	0,5479	1,84	0,87			162,00	162,87
643	Rehm; Eligehausen;	RsIS / BQ II 0	4,26	644,00	1,15	11	32,0	23,23	cu	18,59	0,00		0,0000	2,26	0,0972	0,2216	0,2216	0,2291	5,01	6,46			594,78	601,24
646	Neubert (1978)	RnIIS	1,87	628,00	1,13	13	32,0	21,03	cu	16,82	0,00		0,0000	2,08	0,0991	0,2287	0,2287	0,1113	8,28	13,61			744,73	758,34
647	Reineck (1991)	Stb III	6,16	539,00	1,05	20	8,0	57,57	cu	46,05	4,32	sp	0,0750	4,28	0,0744	0,1113	0,1113	0,1337	4,98	6,67	530,00		536,67	536,67
648		Stb I	9,21	630,00	1,15	19	8,0	57,57	cu	46,05	4,32	sp	0,0750	4,28	0,0744	0,1283	0,1283	0,2000	5,03	6,73	675,00		681,73	681,73
658	Roller; Russel (1990)	7	0,70	576,83	1,30		12,7	68,79	cyl	55,03	3,79	sp	0,0551	4,56	0,0663	0,1289	0,1289	0,0128	9,55	11,23			787,83	799,06
660		9	0,70	576,83	1,30		12,7	119,04	cyl	95,24	5,07	sp	0,0426	5,52	0,0464	0,0931	0,0931	0,0074	9,55	11,23			749,40	760,63
661		10	1,04	576,83	1,30		12,7	119,04	cyl	95,24	5,07	sp	0,0426	5,52	0,0464	0,1047	0,1047	0,0109	9,55	11,23			1172,14	1183,37
671	Sarsam;	BS4-H	1,53		0,00		12,0	76,10	cyl	60,88	0,00		0,0000	4,73	0,0621	0,1862	0,1862	0,0251	1,17	0,57			206,90	207,47
674	Al-Musawi (1992)	CS3-H	1,14		0,00		12,0	70,49	cyl	56,39	0,00		0,0000	4,60	0,0653	0,2700	0,2700	0,0203	1,17	0,57			247,20	247,77
675		CS4-H	1,53		0,00		12,0	71,92	cyl	57,53	0,00		0,0000	4,63	0,0644	0,2647	0,2647	0,0265	1,17	0,57			220,70	221,27
685	Shin; Lee;	MHB 2.5-50	2,16		0,00		19,0	49,40	cyl	39,52	0,00		0,0000	3,99	0,0807	0,3130	0,3130	0,0546	0,75	0,21	149,11		149,32	149,32
697	Moon; Gosh (1999)	HB 2.5-50	2,16		0,00		13,0	69,35	cyl	55,48	0,00		0,0000	4,57	0,0659	0,2230	0,2230	0,0389	0,75	0,21	149,11		149,32	149,32
702	Soerensen (1974)	T-23	1,20	470,88	1,35	20	32,0	32,53	cyl	26,02	0,00		0,0000	2,93	0,0901	0,1361	0,1361	0,0460	1,49	1,46	139,30		140,76	140,76
706		T-2-B	1,21	519,93	1,31	14	32,0	23,67	cyl	18,94	0,00		0,0000	2,29	0,0969	0,2036	0,2036	0,0640	1,49	1,46	129,49		130,95	130,95
711	Stroband (1997)	2	4,06		0,00		16,0	83,00	pr	66,40	6,12	sp	0,0737	4,87	0,0587	0,0798	0,0798	0,0612	3,97	3,42			548,00	551,42
712		3	9,60		0,00		16,0	83,90	pr	67,12	6,30	sp	0,0751	4,89	0,0583	0,1078	0,1078	0,1430	3,97	3,42			891,00	894,42
713		4	16,67		0,00		16,0	84,80	pr	67,84	6,75	sp	0,0796	4,91	0,0579	0,1285	0,1285	0,2457	3,97	3,42			1221,00	1224,42
714		5	24,36		0,00		16,0	84,70	pr	67,76	6,21	sp	0,0733	4,91	0,0579	0,1391	0,1391	0,3595	3,97	3,42			1330,00	1333,42
783	Taylor, R. (1966)	ST-2-C	1,78		0,00		19,1	18,29	cu	17,86	0,00		0,0000	1,86	0,0980	0,3046	0,3046	0,0999	1,18	0,56	104,97		105,53	105,53
786		HSS-1-B	1,43		0,00		19,1	18,53	cu	18,09	0,00		0,0000	1,88	0,0978	0,3008	0,3008	0,0790	1,18	0,65	86,33		86,98	86,98
803	Yoon; Cook;	N2-N	0,50		0,00		20,0	34,20	cyl	27,36	2,79	sp	0,0816	3,04	0,0890	0,3343	0,3343	0,0183	6,75	7,38			483,00	490,38
806	Mitchell (1996)	M2-N	0,71		0,00		10,0	63,65	cyl	50,92	4,14	sp	0,0650	4,44	0,0697	0,1796	0,1796	0,0139	6,75	7,38			689,00	696,38
808		H2-S	0,60		0,00		10,0	82,65	cyl	66,12	5,49	sp	0,0664	4,86	0,0588	0,1383	0,1383	0,0091	6,75	7,38			598,00	605,38
809		H2-N	1,90		0,00		10,0	82,65	cyl	66,12	5,49	sp	0,0664	4,86	0,0588	0,1383	0,1383	0,0288	6,75	7,38			721,00	728,38
817	Rosenbusch (2003)	1.4/1	0,95		0,00		16,0	45,50	cu	36,40			0,0000	3,76	0,0826	0,4294	0,4294	0,0261	1,44	0,61	170,00		170,61	170,61
819		1.7/1	3,07		0,00		16,0	47,85	cu	38,28			0,0000	3,90	0,0815	0,4083	0,4083	0,0802	1,44	0,46	190,00		190,46	190,46
820	Kautsch (2010)	A2	2,62	631,00	1,04		16,0	29,30	cyl	23,44	2,51	sp	0,0857	2,71	0,0924	0,1083	0,1083	0,1116	4,97	4,60	652,50		657,10	657,10
821		B2	1,96	631,00	1,04		16,0	29,83	cyl	23,86	2,48	sp	0,0833	2,74	0,0920	0,1064	0,1064	0,0822	4,97	5,22	566,00		571,22	571,22
874	Tanimura; Sato (2005)	41	2,03					19,57	cyl	15,66			0,0000	1,97	0,1004	0,8438	0,8438	0,1298	3,24	2,27			324,00	326,27
157																								
68																								
89																								

			betar_meas	xr_meas	xr	sigswmeas	sigswass	sigsw	omwu	andbr	Mu	muu	vutest	kapc	sigsl	xsitest	zetatest	z_test	zeta	z_	muflex	Mu_flex	Vu_flex	betaflex	BB
-	-	-	[°]	mm	mm	MPa	MPa	MPa	-	-	kNm	-	-	-	MPa	-	-	mm	-	mm	-	-	kN	-	-
593	Regan (1971)	R21			438			0	0,0000		134,17	0,3035	0,1456	0,820	435	0,491	0,755	192	0,711	181	0,3367	148,82	169,72	0,90	
605		T6			445	270		270	0,1054		182,33	0,1760	0,2835	0,895	502	0,221	0,889	226	0,861	219	0,2139	221,65	249,22	0,82	
608		T9			445	280		280	0,0695		136,97	0,1719	0,2749	0,919	374	0,209	0,896	227	0,824	209	0,2662	212,16	238,55	0,65	
613		T17			902	280		280	0,0420		241,71	0,1832	0,1458	0,866	630	0,225	0,887	225	0,887	225	0,1731	228,40	126,62	1,06	BB
615		T20			673	280		280	0,0468		185,83	0,1570	0,1649	0,880	505	0,198	0,901	229	0,876	223	0,1906	225,58	167,52	0,82	
620		T26			445	280		280	0,0286		159,20	0,0821	0,1243	0,803	412	0,108	0,946	240	0,917	233	0,1218	236,12	265,48	0,67	
621		T27			445	280		280	0,1091		117,40	0,2313	0,3862	0,948	335	0,284	0,858	218	0,733	186	0,3715	188,59	212,05	0,62	
625		T32			445	270		270	0,1040		192,11	0,1828	0,2962	0,893	532	0,231	0,884	225	0,863	219	0,2114	222,10	249,72	0,86	
629		T36			445	280		280	0,0661		159,20	0,1899	0,3084	0,915	442	0,235	0,882	224	0,832	211	0,2555	214,19	240,83	0,74	
630		T37			445	270		270	0,0974		185,88	0,1657	0,2651	0,886	508	0,209	0,896	227	0,870	221	0,1998	224,06	251,92	0,83	
631		T38			445	270		270	0,0951		211,68	0,1844	0,2996	0,883	588	0,237	0,882	224	0,873	222	0,1957	224,73	252,68	0,94	
637		W1			445	320		320	0,3032		142,40	0,1153	0,4430	0,886	356	0,140	0,930	248	0,876	234	0,1920	237,13	266,44	0,60	
639		W3			445			0	0,0000		177,11	0,0942	0,3585	0,827	439	0,121	0,939	251	0,913	244	0,1314	247,02	277,55	0,72	
640		W5			445	368		368	0,4559		169,10	0,1193	0,4607	0,870	425	0,148	0,926	247	0,890	238	0,1700	240,86	270,63	0,70	
642		W7			445			0	0,0000		144,18	0,1018	0,3879	0,870	358	0,125	0,938	250	0,890	238	0,1700	240,86	270,63	0,60	
643	Rehm; Eligehausen;	RsIS / BQ II 0			645	560		560	0,2291		767,27	0,1985	0,3391	0,907	469	0,244	0,878	377	0,878	377	0,1945	751,96	582,92	1,02	BB
646	Neubert (1978)	RnIIS			822	557		557	0,1113		1224,34	0,2041	0,2051	0,916	746	0,250	0,875	480	0,875	480	0,2001	1200,41	730,18	1,02	BB
647	Reineck (1991)	Stb III	24		1260	515		515	0,1337		1352,40	0,0805	0,2715	0,770	394	0,111	0,945	557	0,928	547	0,1033	1734,31	688,22	0,78	
648		Stb I	29		1258	549		549	0,2000		1714,56	0,1026	0,3516	0,770	443	0,144	0,928	547	0,917	540	0,1176	1965,35	781,45	0,87	
658	Roller; Russel (1990)	7			1110		445	445	0,0128		1748,45	0,0957	0,0442	0,725	378	0,142	0,929	708	0,911	694	0,1174	2144,36	966,22	0,82	
660		9			1110		445	445	0,0074		1663,16	0,0526	0,0239	0,524	282	0,106	0,947	722	0,911	694	0,0848	2680,57	1207,83	0,62	
661		10			1110		445	445	0,0109		2601,35	0,0823	0,0387	0,524	371	0,172	0,914	696	0,900	686	0,0942	2978,08	1341,88	0,87	
671	Sarsam;	BS4-H			291		820	820	0,0251		120,52	0,1621	0,0936	0,696	502	0,268	0,866	202	0,866	202	0,1612	119,90	205,84	1,01	BB
674	Al-Musawi (1992)	CS3-H			291		820	820	0,0203		143,99	0,2090	0,1270	0,718	511	0,354	0,823	192	0,812	189	0,2192	151,02	259,26	0,95	
675		CS4-H			291		820	820	0,0265		128,56	0,1829	0,1078	0,712	442	0,303	0,849	198	0,814	190	0,2155	151,44	259,98	0,85	
685	Shin; Lee;	MHB 2.5-50			263			0	0,0000		78,58	0,2753	0,1747	0,802	410	0,390	0,805	173	0,805	173	0,2520	71,92	136,66	1,09	BB
697	Moon; Gosh (1999)	HB 2.5-50			263	366		366	0,0389		78,58	0,1961	0,1184	0,723	410	0,309	0,846	182	0,846	182	0,1886	75,56	143,58	1,04	BB
702	Soerensen (1974)	T-23			525	349		349	0,0460		147,80	0,1279	0,1790	0,870	420	0,156	0,922	275	0,922	275	0,1254	144,93	138,03	1,02	BB
706		T-2-B			525			0	0,0000		137,50	0,1635	0,2345	0,905	408	0,201	0,900	268	0,888	264	0,1807	151,94	144,71	0,90	
711	Stroband (1997)	2			827		600	600	0,0612		906,56	0,0522	0,1477	0,668	409	0,081	0,959	621	0,940	608	0,0751	1304,59	788,60	0,69	
712		3			827		630	630	0,1430		1473,99	0,0839	0,2445	0,664	467	0,135	0,932	603	0,919	595	0,0991	1740,67	1052,20	0,85	
713		4			827			0	0,0000		2019,91	0,1089	0,3324	0,661	522	0,181	0,909	602	0,903	597	0,1160	2151,88	1300,77	0,94	
714		5			827			0	0,0000		2200,23	0,1230	0,3744	0,661	552	0,208	0,896	583	0,895	582	0,1244	2226,67	1345,98	0,99	
783	Taylor, R. (1966)	ST-2-C			457			0	0,0000		96,49	0,2616	0,2935	0,927	441	0,329	0,836	215	0,836	215	0,2545	93,88	102,67	1,03	BB
786		HSS-1-B			572			0	0,0000		99,42	0,2662	0,2384	0,926	441	0,325	0,838	215	0,838	215	0,2519	94,09	82,32	1,06	BB
803	Yoon; Cook;	N2-N			1056	430		430	0,0183		1020,34	0,1854	0,0819	0,863	253	0,245	0,878	575	0,806	528	0,2696	1483,37	702,19	0,69	
806	Mitchell (1996)	M2-N			1056	430		430	0,0139		1455,51	0,1421	0,0617	0,745	354	0,213	0,893	585	0,879	576	0,1580	1617,96	765,90	0,90	
808		H2-S			1056	430		430	0,0091		1263,28	0,0950	0,0399	0,669	298	0,154	0,923	605	0,897	587	0,1241	1649,54	780,85	0,77	
809		H2-N			1056	430		430	0,0288		1523,11	0,1145	0,0919	0,669	366	0,189	0,906	593	0,897	587	0,1241	1649,54	780,85	0,92	
817	Rosenbusch (2003)	1.4/1	33	450	450	0		0	0		149,29	0,2427	0,1101	0,818	380	0,362	0,819	213	0,738	192	0,3167	194,83	222,66	0,77	
819		1.7/1	31	556	556	0		0	0		166,65	0,2576	0,1194	0,809	433	0,398	0,801	208	0,748	194	0,3052	197,46	225,67	0,84	
820	Kautsch (2010)	A2	45	1000	1000	605		605	0,1116		1264,91	0,1274	0,2679	0,883	468	0,123	0,939	675	0,939	675	0,1017	1009,81	524,58	1,253	BB
821		B2	38	875	875	605		605	0,0822		1099,60	0,1088	0,2285	0,881	468	0,121	0,940	676	0,940	676	0,1000	1010,83	525,11	1,09	BB
874	Tanimura; Sato (2005)	41			500			0	0,0000		324,00	0,3449	0,1731	0,922	6	0,007	0,996	399	0,704	282	0,3838	360,53	360,53	0,90	
157																									
68																									
89																									

			sin2thp	thp	cotthp	vup	gamwp	cotthu	thu	nueu	betalb	VB	A2+A3	kon34	KONA4	KONA5	A4+A5	Differenz	AF	f1ck	fcm_cyl	fck	fc_prime	Vct_ACI	Vs_ACI	Vcal_sACI	Vmax_ACI	rat_smax	VACI_cal
-	-	-	-	[°]	-	-	-	-	[°]	-	-	-	-	-	-	-	-	-		MPa	MPa	MPa	MPa	kN	kN	kN	kN		kN
593	Regan (1971)	R21	0,03	10,18	5,57	0,17	0,84	0,00	0,00	0,00	0,25		0	1	0	0	0	0		41,3	47,4	43,4	45,0	43,04	43,47	86,51	215,19	0,402	86,51
605		T6	0,11	18,95	2,91	0,31	0,92	2,69	20,40	0,69	0,46		1	1	1	0	1	0		22,5	27,7	23,7	25,3	32,26	85,73	117,99	161,30	0,731	117,99
608		T9	0,07	15,29	3,66	0,25	1,08	3,96	14,19	0,93	0,43		1	1	1	0	1	0		16,5	21,3	17,3	18,9	27,89	43,47	71,36	139,45	0,512	71,36
613		T17	0,04	11,82	4,78	0,20	0,73	3,47	16,06	0,44	0,25		1	1	1	0	1	0		29,7	35,3	31,3	32,9	36,77	43,47	80,25	183,87	0,436	80,25
615		T20	0,05	12,49	4,51	0,21	0,78	3,52	15,84	0,50	0,28		1	1	1	0	1	0		26,3	31,7	27,7	29,3	34,69	43,47	78,16	173,43	0,451	78,16
620		T26	0,03	9,73	5,83	0,17	0,75	4,35	12,94	0,46	0,24		1	1	1	0	1	0		45,5	51,9	47,9	49,5	45,10	43,47	88,57	225,51	0,393	88,57
621		T27	0,11	19,29	2,86	0,31	1,24	3,54	15,78	1,18	0,58		0	1	0	0	0	0		9,1	13,6	9,6	11,2	21,44	43,47	64,91	107,20	0,606	64,91
625		T32	0,10	18,81	2,94	0,31	0,97	2,85	19,34	0,76	0,48		1	1	1	0	1	0		22,9	28,1	24,1	25,7	32,51	85,73	118,24	162,56	0,727	118,24
629		T36	0,07	14,90	3,76	0,25	1,24	4,67	12,09	1,20	0,48		0	1	0	0	0	0		17,5	22,4	18,4	20,0	28,69	43,47	72,16	143,46	0,503	72,16
630		T37	0,10	18,18	3,04	0,30	0,89	2,72	20,17	0,66	0,44		1	1	1	0	1	0		24,7	30,0	26,0	27,6	33,69	85,73	119,42	168,44	0,709	119,42
631		T38	0,10	17,97	3,08	0,29	1,02	3,15	17,62	0,83	0,49		1	1	1	0	1	0		25,4	30,7	26,7	28,3	34,12	85,73	119,85	170,60	0,703	119,85
637		W1	0,67	54,76	0,71	0,47	0,94	1,46	34,39	0,76	0,29		1	0	0	0	0	1		24,6	29,9	25,9	27,5	14,88	259,00	273,88	74,42	3,680	74,42
639		W3	0,44	41,46	1,13	0,50	0,72	0,00	0,00	0,00	0,26		0	0	0	0	0	0		39,4	45,5	41,5	43,1	18,63	259,00	277,63	93,16	2,980	93,16
640		W5	0,87	69,05	0,38	0,33	1,38	1,01	44,70	0,74	0,31		1	0	0	0	0	1		28,8	34,3	30,3	31,9	16,03	388,50	404,53	80,15	5,047	80,15
642		W7	0,55	47,75	0,91	0,50	0,78	0,00	0,00	0,00	0,26		0	0	0	0	0	0		28,8	34,3	30,3	31,9	16,03	244,05	260,08	80,15	3,245	80,15
643	Rehm; Eligehausen;	RsIS / BQ II 0	0,23	28,60	1,83	0,42	0,81	1,48	34,04	0,58	2,00	VB	0	1	0	0	0	0	AF	19,4	24,5	20,5	22,1	83,85	457,68	541,53	419,24	1,292	419,24
646	Neubert (1978)	RnIIS	0,11	19,49	2,83	0,31	0,65	1,84	28,49	0,39	2,69	VB	0	1	0	0	0	0	AF	17,2	22,1	18,1	19,7	181,94	461,88	643,83	909,71	0,708	643,83
647	Reineck (1991)	Stb III	0,13	21,44	2,55	0,34	0,80	2,03	26,21	0,55	0,10		1	0	0	0	0	1		53,8	60,6	56,6	58,2	57,56	279,65	337,22	287,81	1,172	287,81
648		Stb I	0,20	26,57	2,00	0,40	0,88	1,76	29,63	0,65	0,12		1	0	0	0	0	1		53,8	60,6	56,6	58,2	57,46	417,80	475,27	287,32	1,654	287,32
658	Roller; Russel (1990)	7	0,01	6,50	8,78	0,11	0,39	3,45	16,15	0,13	0,71		1	1	1	0	1	0		65,0	72,4	68,4	70,0	484,17	245,58	729,75	2420,86	0,301	729,75
660		9	0,01	4,94	11,58	0,09	0,28	3,22	17,24	0,07	0,44		1	1	1	0	1	0		115,2	125,3	121,3	122,9	641,51	245,58	887,09	3207,54	0,277	887,09
661		10	0,01	6,00	9,51	0,10	0,37	3,54	15,79	0,12	0,63		1	1	1	0	1	0		115,2	125,3	121,3	122,9	641,51	362,53	1004,03	3207,54	0,313	1004,03
671	Sarsam;	BS4-H	0,03	9,11	6,24	0,16	0,60	3,73	15,01	0,30	0,63		1	1	0	1	1	0		72,3	80,1	76,1	77,7	61,40	64,02	125,43	307,01	0,409	125,43
674	Al-Musawi (1992)	CS3-H	0,02	8,19	6,95	0,14	0,90	6,25	9,08	0,65	0,64		1	1	0	1	1	0		66,7	74,2	70,2	71,8	59,03	48,02	107,04	295,13	0,363	107,04
675		CS4-H	0,03	9,37	6,06	0,16	0,67	4,06	13,83	0,37	0,57		1	1	0	1	1	0		68,1	75,7	71,7	73,3	59,64	64,02	123,66	298,19	0,415	123,66
685	Shin; Lee;	MHB 2.5-50	0,05	13,51	4,16	0,23	0,77	0,00	0,00	0,00	0,90		0	1	0	0	0	0		45,6	52,0	48,0	49,6	31,44	57,94	89,38	157,18	0,569	89,38
697	Moon; Gosh (1999)	HB 2.5-50	0,04	11,37	4,97	0,19	0,61	3,05	18,17	0,32	0,77		1	1	0	1	1	0		65,6	73,0	69,0	70,6	37,51	57,94	95,45	187,53	0,509	95,45
702	Soerensen (1974)	T-23	0,05	12,38	4,55	0,21	0,85	3,89	14,41	0,59	0,49		1	1	1	0	1	0		28,7	34,2	30,2	31,8	30,72	39,23	69,95	153,60	0,455	69,95
706		T-2-B	0,06	14,66	3,82	0,24	0,96	0,00	0,00	0,00	0,59		0	1	0	0	0	0		19,9	24,9	20,9	22,5	25,84	39,74	65,57	129,18	0,508	65,57
711	Stroband (1997)	2	0,06	14,32	3,92	0,24	0,62	2,41	22,50	0,33	0,25		1	0	0	0	0	1		79,2	87,4	83,4	85,0	89,17	236,56	325,73	445,83	0,731	325,73
712		3	0,14	22,22	2,45	0,35	0,70	1,71	30,32	0,45	0,36		1	0	0	0	0	1		80,1	88,3	84,3	85,9	89,66	558,88	648,55	448,31	1,447	448,31
713		4	0,25	29,71	1,75	0,43	0,77	0,00	0,00	0,00	0,35		0	0	0	0	0	0		81,0	89,3	85,3	86,9	92,16	992,33	1084,49	460,82	2,353	460,82
714		5	0,36	36,84	1,33	0,48	0,78	0,00	0,00	0,00	0,38		0	0	0	0	0	0		80,9	89,2	85,2	86,8	90,50	1425,03	1515,53	452,51	3,349	452,51
783	Taylor, R. (1966)	ST-2-C	0,10	18,42	3,00	0,30	0,98	0,00	0,00	0,00	0,16		0	1	0	0	0	0		14,5	19,3	15,3	16,9	20,05	52,42	72,47	100,23	0,723	72,47
786		HSS-1-B	0,08	16,32	3,41	0,27	0,88	0,00	0,00	0,00	0,19		0	1	0	0	0	0		14,7	19,5	15,5	17,1	20,19	41,99	62,19	100,95	0,616	62,19
803	Yoon; Cook;	N2-N	0,02	7,77	7,33	0,13	0,61	4,48	12,58	0,31	0,30		1	1	1	0	1	0		30,4	36,0	32,0	33,6	236,48	122,86	359,33	1182,38	0,304	359,33
806	Mitchell (1996)	M2-N	0,01	6,77	8,43	0,12	0,53	4,44	12,68	0,23	0,29		1	1	1	0	1	0		59,9	67,0	63,0	64,6	327,90	173,60	501,49	1639,48	0,306	501,49
808		H2-S	0,01	5,48	10,43	0,09	0,42	4,38	12,86	0,15	0,22		1	1	1	0	1	0		78,9	87,0	83,0	84,6	375,24	147,88	523,12	1876,18	0,279	523,12
809		H2-N	0,03	9,77	5,81	0,17	0,55	3,19	17,40	0,26	0,27		1	1	1	0	1	0		78,9	87,0	83,0	84,6	200,13	249,55	449,68	1000,63	0,449	449,68
817	Rosenbusch (2003)	1.4/1	0,03	9,29	6,11	0,16	0,69	0,00	0,00	0,00	0,01		0	1	0	0	0	0		41,7	47,9	43,9	45,5	58,26	49,37	107,63	291,28	0,369	107,63
819		1.7/1	0,08	16,46	3,39	0,27	0,44	0,00	0,00	0,00	0,01		0	1	0	0	0	0		44,1	50,4	46,4	48,0	59,82	159,73	219,55	299,09	0,734	219,55
820	Kautsch (2010)	A2	0,11	19,52	2,82	0,31	0,85	2,40	22,62	0,60	0,02		1	1	0	1	1	0		25,5	30,8	26,8	28,4	98,76	291,68	390,44	493,81	0,791	390,44
821		B2	0,08	16,66	3,34	0,27	0,83	2,78	19,79	0,57	0,02		1	1	0	1	1	0		26,0	31,4	27,4	29,0	99,73	218,76	318,49	498,65	0,639	318,49
874	Tanimura; Sato (2005)	41	0,13	21,11	2,59	0,34	0,52	0,00	0,00	0,00	0,63		0	1	0	0	0	0		15,8	20,6	16,6	18,2	85,03	243,79	328,82	425,14	0,773	328,82
157													87	141	56	24	80	7											m
68													55,4	89,8	35,7	15,3	51,0	4,5											s
89																													s²
																													v=s/m
																											gmod-ACI <	6	0,05
																											%	3,8	0,95

			gmod_ACI
-	-	-	
593	Regan (1971)	R21	1,769
605		T6	1,737
608		T9	2,158
613		T17	1,670
615		T20	1,766
620		T26	2,021
621		T27	2,034
625		T32	1,827
629		T36	2,481
630		T37	1,750
631		T38	1,986
637		W1	2,150
639		W3	2,136
640		W5	2,371
642		W7	2,021
643	Rehm; Eligehausen;	RsIS / BQ II 0	1,419
646	Neubert (1978)	RnIIS	1,157
647	Reineck (1991)	Stb III	1,865
648		Stb I	2,373
658	Roller; Russel (1990)	7	1,080
660		9	0,845
661		10	1,167
671	Sarsam;	BS4-H	1,650
674	Al-Musawi (1992)	CS3-H	2,309
675		CS4-H	1,785
685	Shin; Lee;	MHB 2.5-50	1,671
697	Moon; Gosh (1999)	HB 2.5-50	1,564
702	Soerensen (1974)	T-23	2,012
706		T-2-B	1,997
711	Stroband (1997)	2	1,682
712		3	1,987
713		4	2,650
714		5	2,939
783	Taylor, R. (1966)	ST-2-C	1,456
786		HSS-1-B	1,399
803	Yoon; Cook;	N2-N	1,344
806	Mitchell (1996)	M2-N	1,374
808		H2-S	1,143
809		H2-N	1,603
817	Rosenbusch (2003)	1.4/1	1,585
819		1.7/1	0,867
820	Kautsch (2010)	A2	1,683
821		B2	1,794
874	Tanimura; Sato (2005)	41	0,985
157			1,592
68			0,407
89			0,166
			0,256
			0,922
			2,263

ACI-DAfStb Evalution Database vsw-RC_A2b-A3b_sl - K. - H. Reineck, ILEK, University of Stuttgart

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
					Querschnittswerte - Section properties											Laststellung und Geometrie - loading and geometry							
	Nr.	Lit.	Bez.	Einheiten	b	b_w	h	h_f	$h_{h,top}$	h_w	h_{ft}	$h_{h,bot}$	x	A_c	z_{c2}	a_A	a_F	b_A	L	c	a	a/d	c_c
				E	b	bw	h	hf	hh,top	hw	hft	hh,bot	bft	Ac	zc2	aa	af	ba	L	c	a	kap	cc
	-	-	-	SI / A	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm²	mm	mm	mm	mm	mm	mm	mm	-	mm
Ahmad_1996_008_NHW-3b	24	Ahmad; Xie; Yu (1996)	NHW-3b	A	127,0	127,0	254,0							32258	127,0	50,8	31,8	127,0	1188,7	31,8	578,5	2,92	28,6
Ahmad_1996_009_NHW-4	25		NHW-4	A	127,0	127,0	254,0							32258	127,0	50,8	31,8	127,0	1585,0	31,8	776,6	3,92	28,6
Angelakos_1999_001_DB0.530M	40	Angelakos; Bentz;	DB0.530M	SI	300,0	300,0	1000,0							300000	500,0	152,0	76,0	308,8	5400,0	76,0	2662,0	2,88	
Angelakos_1999_004_DB140M	42	Collins P. (1999)	DB140M	SI	300,0	300,0	1000,0							300000	500,0	152,0	76,0	308,8	5400,0	76,0	2662,0	2,88	
Bhal_1968_001_B1S	62	Bhal (1968)	B1S	SI	240,0	240,0	350,0							84000	175,0	180,0	180,0	300,0	1800,0	180,0	810,0	2,70	
Bhal_1968_002_B2S	63		B2S	SI	240,0	240,0	650,0							156000	325,0	180,0	240,0	300,0	3600,0	240,0	1680,0	2,80	
Bhal_1968_003_B3S	64		B3S	SI	240,0	240,0	950,0							228000	475,0	210,0	270,0	300,0	5400,0	270,0	2565,0	2,85	
Bhal_1968_004_B4S	65		B4S	SI	240,0	240,0	1250,0							300000	625,0	210,0	300,0	300,0	7200,0	300,0	3450,0	2,88	
Cladera_2002_010_H 100/2	132	Cladera; Mari (2002)	H 100/2	SI	200,0	200,0	400,0							80000	200,0	150,0	75,0	220,0	2160,0	75,0	1042,5	2,95	25,0
Hamadi_1976_002_GT-2	243	Hamadi (1976)	GT-2	SI	500,0	120,0	400,0	100,0		300,0				86000	133,7	150,0	75,0	250,0	2500,0	75,0	1212,5	3,46	22,0
Hamadi_1976_004_GT-4	245	Hamadi, Regan (1980)	GT-4	SI	500,0	120,0	400,0	100,0		300,0				86000	133,7	150,0	75,0	250,0	2500,0	75,0	1212,5	3,46	22,0
Hegger_2001_001_SVB 3b	247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	SI	220,0	180,0	500,0							100000	241,7			200,0	3000,0		1500,0	3,33	
Kong_1997_003_S1-3	361	Kong; Rangan (1997)	S1-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_004_S1-4	362		S1-4	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_009_S2-3	367		S2-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_010_S2-4	368		S2-4	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_015_S3-3	373		S3-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,49	35,0
Kong_1997_016_S3-4	374		S3-4	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,49	35,0
Kong_1997_019_S4-1	377		S4-1	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	350,0	3100,0	500,0	1300,0	2,48	35,0
Kong_1997_022_S4-4	380		S4-4	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	470,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_024_S4-6	382		S4-6	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	450,0	1500,0	500,0	500,0	2,53	35,0
Kong_1997_025_S5-1	383		S5-1	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	320,0	2260,0	500,0	880,0	3,01	35,0
Kong_1997_027_S5-3	385		S5-3	SI	250,0	250,0	350,0							87500	175,0	100,0	100,0	370,0	1960,0	500,0	730,0	2,50	35,0
Kong_1997_033_S7-3	391		S7-3	SI	250,0	250,0	350,0							87500	175,0	100,0	50,0	380,0	1940,0	50,0	945,0	3,40	35,0
Kong_1997_034_S7-4	392		S7-4	SI	250,0	250,0	350,0							87500	175,0	100,0	50,0	380,0	1940,0	50,0	945,0	3,40	35,0
Krefeld_1966_010_26-1	420	Krefeld; Thurston (1966)	26-1	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_012_29b-1	422		29b-1	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_015_29a-2	425		29a-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_016_29b-2	426		29b-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_017_29c-2	427		29c-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_018_29d-2	428		29d-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_020_29f-2	430		29f-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_021_29g-2	431		29g-2	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Krefeld_1966_034_39-3	444		39-3	A	254,0	254,0	508,0							129032	254,0	190,5	127,0	304,8	3657,6	127,0	1765,3	3,87	
Leonhardt_1962_002_ET2	483	Leonhardt; Walther;	ET2	SI	300,0	150,0	350,0	75,0		275,0				63750	150,7	100,0	100,0	200,0	3000,0	900,0	1050,0	3,50	
Leonhardt_1962_003_ET3	484	DAfStb 151 (1962)	ET3	SI	300,0	100,0	350,0	75,0		275,0				50000	133,8	100,0	100,0	200,0	3000,0	900,0	1050,0	3,50	
Leonhardt_1962_004_ET4	485		ET4	SI	300,0	50,0	350,0	75,0		175,0	100,0		100,0	41250	127,7	100,0	100,0	200,0	3000,0	900,0	1050,0	3,50	
Leonhardt_1962_002_T1 (r)	490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	SI	1500,0	100,0	900,0	150,0		600,0	150,0		500,0	362500	291,7	200,0	200,0	125,0	6000,0	1000,0	2500,0	3,03	
Leonhardt_1963_009_TA11	497	Leonhardt; Walther	TA11	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_010_TA12	498	DAfStb 156 (1963)	TA12	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_011_TA13	499		TA13	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_012_TA14	500		TA14	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_013_TA15	501		TA15	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	
Leonhardt_1963_015_TA16	503		TA16	SI	960,0	160,0	440,0	80,0		360,0				134400	134,3	120,0	120,0	290,0	3000,0	500,0	1250,0	3,33	

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2	3	4	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43
			Längsbewehrung (Zugbewehrung) - longitudinal tensile reinforcement														compr. reinf.				
Nr.	Lit.	Bez.	d_s	d	ns	Φ_{st}	f_{rs}	A_s	α_a	ρ_s	ρ_{sv}	f_{sy}	ε_{sy}	f_{tk}	f_{tk}/f_{sy}	ε_{uk}	d_{s2}	n_{s2}	Φ_{st2}	A_{s2}	f_{sy2}
			ds	d	ns	dst	fR	As	alphaa	rhos	rhosw	fsy	esy	ftk	ftk/fsy	euk	ds2	ns2	dst2	As2	fsy2
-	-	-	mm	mm	-	mm	-	mm²	-	%	%	MPa	‰	MPa	-	‰	mm	-	mm	mm²	MPa
24	Ahmad; Xie; Yu (1996)	NHW-3b	198,1	198,1	4	19,1	r	1140,1	1,00	4,5311	4,5311	420,7	2,10		0,00		38,1	2	12,7	253,4	421
25		NHW-4	198,1	198,1	4	19,1	r	1140,1	1,00	4,5311	4,5311	420,7	2,10		0,00		38,1	2	12,7	253,4	421
40	Angelakos; Bentz;	DB0.530M	925,0	925,0	2	29,9	r	1400,0	1,00	0,5045	0,5045	550,0	2,75	710,00	1,29		75,0	2		400,0	483
42	Collins P. (1999)	DB140M	925,0	925,0	4	29,9	r	2800,0	1,00	1,0090	1,0090	550,0	2,75	710,00	1,29		75,0	2		400,0	483
62	Bhal (1968)	B1S	300,0	300,0	2	24,0	r	904,8	1,00	1,2566	1,2566	425,8	2,13	513,06	1,21	15	25,0	2	10,0	157,1	443
63		B2S	600,0	600,0	4	24,0	r	1809,6	1,00	1,2566	1,2566	425,8	2,13	513,06	1,21	15	25,0	2	10,0	157,1	443
64		B3S	900,0	900,0	6	24,0	r	2714,3	1,00	1,2566	1,2566	425,8	2,13	513,06	1,21	15	25,0	2	10,0	157,1	443
65		B4S	1200,0	1200,0	8	24,0	r	3619,1	1,00	1,2566	1,2566	425,8	2,13	513,06	1,21	15	25,0	2	10,0	157,1	443
132	Cladera; Mari (2002)	H 100/2	353,0	353,0	2	32,0	r	1608,5	1,00	2,2783	2,2783	500,0	2,50		0,00		35,0	2	8,0	100,5	540
243	Hamadi (1976)	GT-2	350,0	350,0	4	20,0	r	1256,6	1,00	0,7181	2,9920	400,0	2,00	565,00	1,41			4	12,0	452,4	350
245	Hamadi, Regan (1980)	GT-4	350,0	350,0	4	20,0	r	1256,6	1,00	0,7181	2,9920	655,0	3,28	820,00	1,25			4	12,0	452,4	350
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	450,0	450,0	6	25,0	r	2945,2	1,00	2,9750	3,6361	532,0	2,66	673,00	1,27						
361	Kong; Rangan (1997)	S1-3	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462
362		S1-4	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462
367		S2-3	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462
368		S2-4	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462
373		S3-3	293,0	293,0	2	36,0	r	2046,0	0,70	2,7932	2,7932	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462
374		S3-4	293,0	293,0	2	36,0	r	2046,0	0,70	2,7932	2,7932	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462
377		S4-1	524,0	524,0	4	36,0	r	4092,0	0,70	3,1237	3,1237	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462
380		S4-4	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462
382		S4-6	198,0	198,0	3	24,0	r	1380,0	0,70	2,7879	2,7879	442,0	2,21	544,00	1,23		41,0	2	12,0	224,0	462
383		S5-1	292,0	292,0	2	36,0	r.	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	452
385		S5-3	292,0	292,0	2	36,0	r	2046,0	0,70	2,8027	2,8027	452,0	2,26	559,00	1,24		41,0	2	12,0	224,0	462
391		S7-3	278,0	278,0	4	32,0	r	3284,0	0,70	4,7252	4,7252	433,0	2,17	541,00	1,25		47,0	2	24,0	920,0	442
392		S7-4	278,0	278,0	4	32,0	r	3284,0	0,70	4,7252	4,7252	433,0	2,17	541,00	1,25		47,0	2	24,0	920,0	442
420	Krefeld; Thurston (1966)	26-1	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	352,4	1,76		0,00						
422		29b-1	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	338,6	1,69		0,00						
425		29a-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	383,4	1,92		0,00						
426		29b-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	364,1	1,82		0,00						
427		29c-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	271,0	1,36		0,00						
428		29d-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	290,3	1,45		0,00						
430		29f-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	406,2	2,03		0,00						
431		29g-2	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	276,6	1,38		0,00						
444		39-3	455,7	455,7	4	28,7	r	2578,9	1,00	2,2282	2,2282	406,2	2,03		0,00						
483	Leonhardt; Walther;	ET2	300,0	300,0	4	20,0	r	1256,6	1,00	1,3963	2,7925	419,9	2,10	524,84	1,25			2	8,0	100,5	456
484	DAfStb 151 (1962)	ET3	300,0	300,0	4	20,0	r	1256,6	1,00	1,3963	4,1888	419,9	2,10	524,84	1,25			2	8,0	100,5	456
485		ET4	300,0	300,0	4	20,0	r	1256,6	1,00	1,3963	8,3776	419,9	2,10	524,84	1,25			2	8,0	100,5	456
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	825,0	825,0	16	26,0	r	8464,0	0,01	0,6840	10,2594	464,8	2,32	549,17	1,18			11	10,0	863,9	
497	Leonhardt; Walther	TA11	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426
498	DAfStb 156 (1963)	TA12	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426
499		TA13	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426
500		TA14	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	409,1	2,05	535,63	1,31			4	10,0	314,2	426
501		TA15	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	390,4	1,95	518,95	1,33			4	10,0	314,2	426
503		TA16	375,0	375,0	6	24,0	r	2714,3	1,00	0,7540	4,5239	390,4	1,95	518,95	1,33			4	10,0	314,2	426

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2	3	4	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67
			Bügelbewehrung - Stirrups															ckfestigkeit - compressive compressiv								
Nr.	Lit.	Bez.	Φw	n_{sw}	A_{sw}	f_{rw}	s_w	s_w / h	sw/d	z_{w1}	z_{w2}	ρ_w	f_{yw}	$f_{yw} \cdot \rho_w/100$	f_{wtk}	f_{wk} / f_{yw}	ε_{wuk}	Φa	f_{1c}		f_{cwu}	$f_{1ct,test}$		$\beta_{ct,test}$	$f_{1ct,m,cal}$	$\beta_{ct,cal}$
			dw	nsw	Asw	frw	sw	sw/h		zw1	zw2	rhow	fyw	fyw_rhow_100	fwtk	fwtk/fyw	ewuk	diaa	f1c	Method	fcwu	f1cttest	Method	betacttest	f1ctmcal	betactcal
-	-	-	mm	-	mm²	-	mm	-	-	mm	mm	%	MPa	MPa	MPa	-	‰	mm	MPa	-	MPa	MPa	-	-	MPa	-
24	Ahmad; Xie; Yu (1996)	NHW-3b	6,4	2	63,3	0	63,5	0,25	0,32	166	141	0,7854	324,1	2,55		0,00		19,1	99,99	cyl	80,00	0,00		0,0000	5,20	0,0520
25		NHW-4	6,4	2	63,3	0	99,1	0,39	0,50	166	141	0,5035	324,1	1,63		0,00		19,1	95,74	cyl	76,59	0,00		0,0000	5,12	0,0535
40	Angelakos; Bentz;	DB0.530M	9,5	1	71,3	r	300,0	0,30	0,32	877	0	0,0792	508,0	0,40	778,00	1,53		10,0	30,40	cyl	24,32	0,00		0,0000	2,78	0,0916
42	Collins P. (1999)	DB140M	9,5	1	71,3	r	300,0	0,30	0,32	877	0	0,0792	508,0	0,40	778,00	1,53		10,0	36,10	cyl	28,88	0,00		0,0000	3,17	0,0878
62	Bhal (1968)	B1S	6,0	2	56,5	r	160,0	0,46	0,53	270	0	0,1473	441,5	0,65	567,51	1,29		30,0	25,88	cu	20,70	2,26	fl	0,0873	2,46	0,0950
63		B2S	6,0	2	56,5	r	160,0	0,25	0,27	570	0	0,1473	441,5	0,65	567,51	1,29		30,0	24,18	cu	19,34	2,39	fl	0,0988	2,33	0,0964
64		B3S	6,0	2	56,5	r	160,0	0,17	0,18	870	0	0,1473	441,5	0,65	567,51	1,29		30,0	25,34	cu	20,27	2,67	fl	0,1055	2,42	0,0955
65		B4S	6,0	2	56,5	r	160,0	0,13	0,13	1170	0	0,1473	441,5	0,65	567,51	1,29		30,0	24,49	cu	19,59	2,14	fl	0,0875	2,36	0,0962
132	Cladera; Mari (2002)	H 100/2	6,0	2	56,5	r	165,0	0,41	0,47	323	303	0,1714	530,0	0,91	680,00	1,28		12,0	82,65	cyl	66,12	0,00		0,0000	4,86	0,0588
243	Hamadi (1976)	GT-2	6,0	2	56,5	r	125,0	0,31	0,36	320	306	0,3770	320,0	1,21	430,00	1,34		20,0	23,63	cu	18,90	0,00		0,0000	2,29	0,0969
245	Hamadi, Regan (1980)	GT-4	8,0	2	100,5	r	160,0	0,40	0,46	310	306	0,5236	480,0	2,51	720,00	1,50		20,0	29,43	cu	23,54	0,00		0,0000	2,72	0,0923
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	8,0	2	100,5	r	100,0	0,20	0,22	410	0	0,5585	514,0	2,87		0,00			66,50	cyl	53,20	0,00		0,0000	4,51	0,0678
361	Kong; Rangan (1997)	S1-3	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0	0,87	614,00	1,08		7,0	58,51	cyl	46,81	0,00		0,0000	4,31	0,0736
362		S1-4	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0	0,87	614,00	1,08		7,0	58,51	cyl	46,81	0,00		0,0000	4,31	0,0736
367		S2-3	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0	0,87	614,00	1,08		7,0	66,70	cyl	53,36	3,25	sp	0,0487	4,51	0,0676
368		S2-4	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0	0,87	614,00	1,08		7,0	66,70	cyl	53,36	3,25	sp	0,0487	4,51	0,0676
373		S3-3	4,0	2	24,6	0	100,0	0,29	0,34	273	223	0,0985	632,0	0,62	687,00	1,09		7,0	62,01	cyl	49,61	3,28	sp	0,0528	4,40	0,0709
374		S3-4	4,0	2	24,6	0	100,0	0,29	0,34	273	223	0,0985	632,0	0,62	687,00	1,09		7,0	62,01	cyl	49,61	3,28	sp	0,0528	4,40	0,0709
377		S4-1	4,9	2	38,3	0	100,0	0,29	0,19	499	454	0,1533	569,0	0,87	614,00	1,08		7,0	80,32	cyl	64,25	3,72	sp	0,0463	4,82	0,0600
380		S4-4	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0	0,87	614,00	1,08		7,0	80,32	cyl	64,25	3,72	sp	0,0463	4,82	0,0600
382		S4-6	4,9	2	38,3	0	100,0	0,29	0,51	173	128	0,1533	569,0	0,87	614,00	1,08		7,0	80,32	cyl	64,25	3,72	sp	0,0463	4,82	0,0600
383		S5-1	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0	0,87	614,00	1,08		7,0	82,25	cyl	65,80	4,80	sp	0,0583	4,86	0,0590
385		S5-3	4,9	2	38,3	0	100,0	0,29	0,34	267	222	0,1533	569,0	0,87	614,00	1,08		7,0	82,25	cyl	65,80	4,80	sp	0,0583	4,86	0,0590
391		S7-3	4,9	2	38,3	0	100,0	0,29	0,36	253	208	0,1533	569,0	0,87	614,00	1,08		7,0	68,82	cyl	55,05	3,68	sp	0,0535	4,56	0,0663
392		S7-4	4,9	2	38,3	0	80,0	0,23	0,29	253	208	0,1917	569,0	1,09	614,00	1,08		7,0	68,82	cyl	55,05	3,68	sp	0,0535	4,56	0,0663
420	Krefeld; Thurston (1966)	26-1	6,4	2	63,3	0	152,4	0,30	0,33	424	0	0,1636	341,4	0,56		0,00			38,13	cyl	30,50	0,00		0,0000	3,30	0,0866
422		29b-1	6,4	2	63,3	0	228,6	0,45	0,50	424	0	0,1091	341,4	0,37		0,00			35,77	cyl	28,62	0,00		0,0000	3,15	0,0880
425		29a-2	6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4	0,41		0,00			35,31	cyl	28,25	0,00		0,0000	3,12	0,0883
426		29b-2	6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4	0,41		0,00			39,31	cyl	31,45	0,00		0,0000	3,38	0,0859
427		29c-2	6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4	0,41		0,00			22,93	cyl	18,34	0,00		0,0000	2,24	0,0975
428		29d-2	6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4	0,41		0,00			28,89	cyl	23,11	0,00		0,0000	2,68	0,0927
430		29f-2	6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4	0,41		0,00			39,70	cyl	31,76	0,00		0,0000	3,40	0,0857
431		29g-2	6,4	2	63,3	r	228,6	0,45	0,50	424	0	0,1091	372,4	0,41		0,00			14,94	cyl	11,95	0,00		0,0000	1,56	0,1043
444		39-3	9,5	2	141,9	r	228,6	0,45	0,50	408	0	0,2444	275,9	0,67		0,00			40,56	cyl	32,44	0,00		0,0000	3,45	0,0852
483	Leonhardt; Walther;	ET2	6,0	2	56,5	0	110,0	0,31	0,37	270	0	0,3427	313,9	1,08	376,70	1,20		30,0	22,56	pr	18,05	2,47	fl	0,1093	2,21	0,0978
484	DAfStb 151 (1962)	ET3	6,0	2	56,5	0	110,0	0,31	0,37	270	0	0,5141	313,9	1,61	376,70	1,20		30,0	22,56	pr	18,05	2,47	fl	0,1093	2,21	0,0978
485		ET4	6,0	2	56,5	0	110,0	0,31	0,37	270	0	1,0282	313,9	3,23	376,70	1,20		30,0	22,56	pr	18,05	2,47	fl	0,1093	2,21	0,0978
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	12,0	2	232,0	r	80,0	0,09	0,10	765	0	2,9000	426,6	12,37	536,42	1,26		30,0	23,01	cu	18,41	2,38	fl	0,1036	2,24	0,0974
497	Leonhardt; Walther	TA11	8,0	2	100,5	r	113,0	0,26	0,30	335	0	0,5560	417,9	2,32	525,82	1,26		30,0	24,92	pr	19,93	2,41	fl	0,0969	2,39	0,0958
498	DAfStb 156 (1963)	TA12	6,0	2	56,5	r	113,0	0,26	0,30	345	0	0,3128	440,5	1,38	569,96	1,29		30,0	24,92	pr	19,93	2,41	fl	0,0969	2,39	0,0958
499		TA13	12,0	2	226,2	r	113,0	0,26	0,30	315	0	1,2511	426,7	5,34	536,61	1,26		30,0	21,29	pr	17,03	2,18	fl	0,1023	2,11	0,0989
500		TA14	10,0	2	157,1	r	113,0	0,26	0,30	325	0	0,8688	425,8	3,70	537,59	1,26		30,0	21,29	pr	17,03	2,18	fl	0,1023	2,11	0,0989
501		TA15	8,0	2	100,5	r	113,0	0,26	0,30	335	0	0,5560	417,9	2,32	525,82	1,26		30,0	20,31	pr	16,25	2,35	fl	0,1159	2,03	0,0998
503		TA16	8,0	2	100,5	r	113,0	0,26	0,30	335	0	0,5560	417,9	2,32	525,82	1,26		30,0	20,31	pr	16,25	2,35	fl	0,1159	2,03	0,0998

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2	3	4	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87
			mech. reinforcement ratios			Versuch - Test results														Bruchschnittgrößen		
Nr.	Lit.	Bez.	ω_x	ω_l	ω_{wy}	g	V_g	F	$V_{u,g+F,Rep}$	$V_{u,Rep}$	V_{u_g+F}	β_r	x_r	$\sigma_{sl,a}$	$\sigma_{sw,meas}$	$\sigma_{sw,ass}$	σ_{sw}	ω_{wu}	andere Bruchart	M_u	μ_u	$\upsilon_{u,test}$
			oms	oml	omwy	g	Vg	F	Vu_F+g,rep	Vu_Rep	Vu_gF	betar_meas	xr_meas	xr	sigswmeas	sigswass	sigsw	omwu	andbr	Mu	muu	vutest
-	-	-	-	-	-	kN/m	kN	kN	kN	kN	kN	[°]	mm	mm	MPa	MPa	MPa	-	-	kNm	-	-
24	Ahmad; Xie; Yu (1996)	NHW-3b	0,1906	0,1906	0,0318	0,77	0,24	122,54		122,78	122,78			289	324		324	0,0318		71,03	0,1425	0,0707
25		NHW-4	0,1991	0,1991	0,0213	0,77	0,31	93,72		94,03	94,03			388	324		324	0,0213		73,03	0,1530	0,0571
40	Angelakos; Bentz;	DB0.530M	0,0913	0,0913	0,0165	7,20	9,86	256,00		265,86	265,86			1331	508		508	0,0165		707,71	0,0907	0,0416
42	Collins P. (1999)	DB140M	0,1537	0,1537	0,0139	7,20	9,50	270,00		279,50	279,50		1380	1380	508		508	0,0139		744,04	0,0803	0,0367
62	Bhal (1968)	B1S	0,2067	0,2067	0,0314	2,02	1,17	127,53		128,70	128,70		321	321	438		438	0,0312		104,25	0,1865	0,0976
63		B2S	0,2213	0,2213	0,0336	3,74	1,87	247,70		249,57	249,57		1300	1300	441		441	0,0336		419,29	0,2007	0,1021
64		B3S	0,2111	0,2111	0,0321	5,47	3,97	365,42		369,39	369,39		1975	1975	441		441	0,0321		947,48	0,1923	0,0956
65		B4S	0,2185	0,2185	0,0332	7,20	13,50	459,11		472,61	472,61			1725	441		441	0,0332		1630,50	0,1926	0,0953
132	Cladera; Mari (2002)	H 100/2	0,1378	0,1378	0,0137	1,92	1,07			225,55	226,62			521	530		530	0,0137		235,14	0,1142	0,0533
243	Hamadi (1976)	GT-2	0,1216	0,1216	0,0638	2,06	1,33			148,50	149,83			606		320	320	0,0638		180,06	0,1244	0,2005
245	Hamadi, Regan (1980)	GT-4	0,1598	0,1598	0,1067	2,06	1,33			227,50	228,83			606		480	480	0,1067		275,84	0,1530	0,2530
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	0,2380	0,2380	0,0540	2,40	1,80			427,00	428,80			750		514	514	0,0540		640,50	0,2162	0,1183
361	Kong; Rangan (1997)	S1-3	0,2165	0,2165	0,0186	2,10	1,31			206,10	207,41	41	354	354	314		314	0,0103		150,45	0,1206	0,0660
362		S1-4	0,2165	0,2165	0,0186	2,10	1,41			277,90	279,31	40	310	310	450		450	0,0147		202,87	0,1627	0,0925
367		S2-3	0,1899	0,1899	0,0164	2,10	1,31			253,30	254,61		354	354	266		266	0,0076		184,91	0,1301	0,0721
368		S2-4	0,1899	0,1899	0,0164	2,10	1,20			219,40	220,60	55	408	408	279		279	0,0080		160,16	0,1126	0,0615
373		S3-3	0,2036	0,2036	0,0126	2,10	1,18			228,60	229,78	47	420	420	300		300	0,0060		166,88	0,1254	0,0693
374		S3-4	0,2036	0,2036	0,0126	2,10	1,22			174,90	176,12	58	398	398	350		350	0,0070		127,68	0,0959	0,0517
377		S4-1	0,1758	0,1758	0,0136	2,10	2,03			354,00	356,03	53	585	585	183		183	0,0044		460,20	0,0835	0,0450
380		S4-4	0,1577	0,1577	0,0136	2,10	1,26			258,10	259,36		381	381	317		317	0,0076		188,41	0,1101	0,0604
382		S4-6	0,1534	0,1534	0,0136	2,10	1,05			202,90	203,95			250	300		300	0,0072		101,45	0,1289	0,0714
383		S5-1	0,1540	0,1540	0,0133	2,10	1,37			241,70	243,07	47	476	476	411		411	0,0096		212,70	0,1213	0,0559
385		S5-3	0,1540	0,1540	0,0133	2,10	1,34			243,80	245,14	45	341	341	300		300	0,0070		177,97	0,1015	0,0553
391		S7-3	0,2973	0,2973	0,0158	2,10	1,04			246,50	247,54			473	372		372	0,0104		232,94	0,1752	0,0750
392		S7-4	0,2973	0,2973	0,0198	2,10	1,04			273,60	274,64			473	375		375	0,0131		258,55	0,1945	0,0851
420	Krefeld; Thurston (1966)	26-1	0,2059	0,2059	0,0183	3,10	2,93			206,83	209,76			883		341	341	0,0183		365,12	0,1816	0,0667
422		29b-1	0,2109	0,2109	0,0130	3,10	2,93			160,13	163,06			883		341	341	0,0130		282,67	0,1498	0,0535
425		29a-2	0,2419	0,2419	0,0144	3,10	2,93			216,62	219,55			883		372	372	0,0144		382,40	0,2053	0,0769
426		29b-2	0,2064	0,2064	0,0129	3,10	2,93			202,38	205,31			883		372	372	0,0129		357,27	0,1723	0,0629
427		29c-2	0,2634	0,2634	0,0221	3,10	2,93			161,46	164,39			883		372	372	0,0221		285,03	0,2357	0,0889
428		29d-2	0,2239	0,2239	0,0176	3,10	2,93			165,02	167,95			883		372	372	0,0176		291,31	0,1912	0,0704
430		29f-2	0,2280	0,2280	0,0128	3,10	2,93			234,41	237,34			883		372	372	0,0128		413,80	0,1976	0,0738
431		29g-2	0,4125	0,4125	0,0340	3,10	2,93			149,90	152,83			883		372	372	0,0340		264,61	0,3359	0,1388
444		39-3	0,2232	0,2232	0,0208	3,10	2,93			239,75	242,68			883		276	276	0,0208		423,23	0,1979	0,0737
483	Leonhardt; Walther;	ET2	0,2598	0,2598	0,0596	1,53	1,94	129,00	131,94	130,94	130,94	41	233	233	314		314	0,0596		137,49	0,2257	0,1881
484	DAfStb 151 (1962)	ET3	0,2598	0,2598	0,0894	1,20	1,17	125,08	127,53	126,25	126,25			525	314		314	0,0894		132,56	0,2176	0,2707
485		ET4	0,2598	0,2598	0,1788	0,99	0,97	97,12	99,08	98,08	98,08			525	314		314	0,1788		102,99	0,1691	0,4042
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	0,1381	0,1381	0,6719	8,70	15,23	784,80		800,03	800,03			1250	315		315	0,4964		2000,06	0,0851	0,5540
497	Leonhardt; Walther	TA11	0,1238	0,1238	0,1166	3,23	2,82	335,50	340,90	338,32	338,32			625	418		418	0,1166		422,91	0,1257	0,3038
498	DAfStb 156 (1963)	TA12	0,1238	0,1238	0,0691	3,23	2,82	264,87	270,27	267,69	267,69			625	440		440	0,0691		334,62	0,0995	0,2378
499		TA13	0,1449	0,1449	0,3135	3,23	2,82	343,35	348,75	346,17	346,17			625	348		348	0,2556		432,72	0,1506	0,3679
500		TA14	0,1449	0,1449	0,2172	3,23	2,82	333,54	338,94	336,36	336,36			625	425		425	0,2167		420,45	0,1463	0,3575
501		TA15	0,1450	0,1450	0,1430	3,23	2,82	292,34	297,73	295,16	295,16			625	418		418	0,1430		368,95	0,1346	0,3288
503		TA16	0,1450	0,1450	0,1430	3,23	2,82	293,32	298,71	296,14	296,14			625	418		418	0,1430		370,18	0,1350	0,3298

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2	3	4	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
			Kontrolle Biegung												Querkraft - PT (mit ν=0,80)				Querkraft - PT im Versuch				anchorage		
Nr.	Lit.	Bez.	κ_c	σ_{sl}	ξ_{test}	ζ_{test}	z_{test}	ζ	z	μ_{flex}	μ_{u_flex}	$\nu_{u,flex}$	β_{flex}	Bbem	$\sin^2\theta_p$	θ_p	$\cot\theta_p$	υ_{uP}	γ_{wp}	$\cot\theta_u$	θ_u	ν_u	β_{lb}		
			kapc	sigsl	xsitest	zetatest	z_test	zeta	z_	muflex	Mu_flex	Vu_flex	betaflex	BB	sin2thp	thp	cotthp	vup	gamwp	cotthu	thu	nueu	betalb	VB	A2+A3
-	-	-	-	MPa	-	-	mm	-	mm	-	-	kN	-	-	-	[°]	-	-	-	-	[°]	-	-	-	-
24	Ahmad; Xie; Yu (1996)	NHW-3b	0,600	365	0,275	0,862	171	0,841	167	0,1603	79,93	138,17	0,89		0,03	10,28	5,52	0,18	0,40	2,22	24,22	0,15	0,73		1
25		NHW-4	0,617	378	0,290	0,855	169	0,839	166	0,1670	79,69	102,62	0,92		0,02	8,39	6,78	0,14	0,40	2,68	20,47	0,14	0,57		1
40	Angelakos; Bentz;	DB0.530M	0,878	550	0,104	0,948	877	0,948	877	0,0865	675,24	253,66	1,05	BB	0,02	7,39	7,71	0,13	0,33	2,51	21,70	0,10	0,80		1
42	Collins P. (1999)	DB140M	0,856	302	0,099	0,951	879	0,910	842	0,1399	1296,52	487,05	0,57		0,01	6,78	8,41	0,12	0,31	2,63	20,79	0,09	0,37		1
62	Bhal (1968)	B1S	0,896	426	0,231	0,885	265	0,885	265	0,1829	102,24	126,22	1,02	BB	0,03	10,21	5,55	0,17	0,56	3,13	17,71	0,27	0,69		1
63		B2S	0,903	426	0,245	0,878	527	0,878	527	0,1942	405,64	241,45	1,03	BB	0,03	10,56	5,36	0,18	0,57	3,04	18,22	0,28	0,56		1
64		B3S	0,899	426	0,235	0,883	794	0,883	794	0,1863	917,89	357,85	1,03	BB	0,03	10,32	5,49	0,18	0,54	2,98	18,55	0,25	0,47		1
65		B4S	0,902	426	0,242	0,879	1055	0,879	1055	0,1920	1625,11	471,05	1,00	BB	0,03	10,50	5,40	0,18	0,53	2,87	19,20	0,25	0,44		1
132	Cladera; Mari (2002)	H 100/2	0,669	457	0,188	0,906	320	0,897	317	0,1236	254,67	244,29	0,92		0,01	6,73	8,47	0,12	0,46	3,88	14,44	0,18	0,56		1
243	Hamadi (1976)	GT-2	0,906	400	0,134	0,933	327	0,933	327	0,1134	164,12	135,36	1,10	BB	0,06	14,63	3,83	0,24	0,82	3,14	17,66	0,56	0,57		1
245	Hamadi, Regan (1980)	GT-4	0,882	655	0,181	0,909	318	0,909	318	0,1453	261,99	216,08	1,05	BB	0,11	19,07	2,89	0,31	0,82	2,37	22,88	0,57	0,74		1
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	0,734	532	0,324	0,838	377	0,838	377	0,1994	590,78	393,85	1,08	BB	0,05	13,43	4,19	0,23	0,52	2,19	24,53	0,25	0,54		1
361	Kong; Rangan (1997)	S1-3	0,766	276	0,172	0,914	267	0,859	251	0,1859	231,87	317,64	0,65		0,02	7,85	7,26	0,14	0,49	6,42	8,85	0,35	0,27		1
362		S1-4	0,766	386	0,242	0,879	257	0,859	251	0,1859	231,87	317,64	0,87		0,02	7,85	7,26	0,14	0,68	6,27	9,05	0,48	0,36		1
367		S2-3	0,733	343	0,197	0,902	263	0,870	254	0,1653	235,06	322,00	0,79		0,02	7,35	7,76	0,13	0,57	9,44	6,05	0,55	0,31		1
368		S2-4	0,733	293	0,168	0,916	268	0,870	254	0,1653	235,06	322,00	0,68		0,02	7,35	7,76	0,13	0,48	7,68	7,42	0,38	0,27		1
373		S3-3	0,752	307	0,184	0,908	266	0,865	253	0,1760	234,28	320,93	0,71		0,01	6,43	8,87	0,11	0,62	11,63	4,92	0,65	0,28		1
374		S3-4	0,752	229	0,137	0,932	273	0,865	253	0,1760	234,28	320,93	0,54		0,01	6,43	8,87	0,11	0,46	7,43	7,66	0,31	0,22		1
377		S4-1	0,679	230	0,132	0,934	490	0,870	456	0,1530	843,67	648,98	0,55		0,01	6,69	8,52	0,12	0,39	10,29	5,55	0,37	0,03		1
380		S4-4	0,679	346	0,178	0,911	266	0,884	258	0,1394	238,66	326,93	0,79		0,01	6,69	8,52	0,12	0,52	7,99	7,13	0,39	0,23		1
382		S4-6	0,679	415	0,212	0,894	177	0,887	176	0,1361	107,12	214,24	0,95		0,01	6,69	8,52	0,12	0,62	9,97	5,73	0,58	0,43		1
383		S5-1	0,671	396	0,201	0,899	263	0,885	258	0,1363	239,05	271,64	0,89		0,01	6,61	8,63	0,11	0,49	5,85	9,70	0,27	0,32		1
385		S5-3	0,671	325	0,165	0,918	268	0,885	258	0,1363	239,05	327,46	0,74		0,01	6,61	8,63	0,11	0,48	7,91	7,20	0,36	0,27		1
391		S7-3	0,725	297	0,281	0,859	239	0,795	221	0,2363	314,22	332,51	0,74		0,02	7,23	7,88	0,12	0,60	7,23	7,87	0,44	0,19		1
392		S7-4	0,725	337	0,319	0,840	234	0,795	221	0,2363	314,22	332,51	0,82		0,02	8,09	7,03	0,14	0,61	6,52	8,72	0,45	0,21		1
420	Krefeld; Thurston (1966)	26-1	0,847	352	0,243	0,879	400	0,879	400	0,1809	363,82	206,10	1,00	BB	0,02	7,78	7,32	0,13	0,50	3,64	15,35	0,21	0,30		1
422		29b-1	0,857	266	0,194	0,903	412	0,877	400	0,1850	348,96	197,68	0,81		0,01	6,55	8,71	0,11	0,47	4,11	13,66	0,19	0,24		1
425		29a-2	0,859	378	0,278	0,861	392	0,859	391	0,2079	387,13	219,30	0,99		0,01	6,89	8,28	0,12	0,65	5,35	10,59	0,34	0,33		1
426		29b-2	0,843	344	0,231	0,884	403	0,878	400	0,1811	375,52	212,72	0,95		0,01	6,53	8,74	0,11	0,56	4,87	11,61	0,26	0,28		1
427		29c-2	0,908	271	0,290	0,855	390	0,855	390	0,2252	272,33	154,27	1,05	BB	0,02	8,56	6,65	0,15	0,60	4,02	13,98	0,30	0,34		1
428		29d-2	0,884	283	0,247	0,877	400	0,873	398	0,1956	298,01	168,81	0,98		0,02	7,62	7,48	0,13	0,54	4,00	14,03	0,24	0,29		1
430		29f-2	0,841	406	0,271	0,864	394	0,864	394	0,1971	412,67	233,77	1,00	BB	0,01	6,49	8,79	0,11	0,66	5,77	9,84	0,35	0,33		1
431		29g-2	0,940	277	0,439	0,781	356	0,781	356	0,3220	253,70	143,71	1,04	BB	0,03	10,62	5,33	0,18	0,77	4,08	13,76	0,48	0,47		1
444		39-3	0,838	406	0,266	0,867	395	0,867	395	0,1934	413,77	234,39	1,02	BB	0,02	8,29	6,86	0,14	0,52	3,54	15,76	0,23	0,33		1
483	Leonhardt; Walther;	ET2	0,910	420	0,286	0,857	257	0,857	257	0,2227	135,68	129,22	1,01	BB	0,06	14,13	3,97	0,24	0,79	3,16	17,59	0,52	0,73		1
484	DAfStb 151 (1962)	ET3	0,910	408	0,278	0,861	258	0,857	257	0,2227	135,68	129,22	0,98		0,09	17,40	3,19	0,29	0,95	3,03	18,27	0,73	0,71		1
485		ET4	0,910	305	0,207	0,896	269	0,857	257	0,2227	135,68	129,22	0,76		0,18	25,02	2,14	0,38	1,05	2,26	23,87	0,87	0,55		1
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	0,908	301	0,099	0,951	784	0,924	762	0,1276	2998,92	1199,57	0,67		0,67	55,06	0,70	0,47	1,18	1,12	41,86	0,89	0,01		1
497	Leonhardt; Walther	TA11	0,900	409	0,137	0,931	349	0,931	349	0,1153	387,77	310,21	1,09	BB	0,12	19,96	2,75	0,32	0,95	2,61	21,00	0,73	0,66		1
498	DAfStb 156 (1963)	TA12	0,900	349	0,117	0,941	353	0,931	349	0,1153	387,77	310,21	0,86		0,07	15,24	3,67	0,25	0,94	3,44	16,21	0,71	0,53		1
499		TA13	0,915	409	0,158	0,921	345	0,921	345	0,1334	383,42	306,73	1,13	BB	0,31	34,05	1,48	0,46	0,79	1,44	34,79	0,63	0,77		1
500		TA14	0,915	409	0,158	0,921	345	0,921	345	0,1334	383,42	306,73	1,097	BB	0,22	27,78	1,90	0,41	0,87	1,65	31,22	0,65	0,75		1
501		TA15	0,919	390	0,158	0,921	345	0,921	345	0,1335	366,06	292,85	1,01	BB	0,14	22,22	2,45	0,35	0,94	2,30	23,51	0,72	0,69		1
503		TA16	0,919	390	0,158	0,921	345	0,921	345	0,1335	366,06	292,85	1,01	BB	0,14	22,22	2,45	0,35	0,94	2,31	23,44	0,72	0,69		1

ACI-DAfStb Evalution Database vsw-RC_A2b-A																		
2	3	4	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126
Nr.	Lit.	Bez.	kon34	KONA4	KONA5	A4+A5	Differenz	f_{1ck} f1ck	$f_{cm,cyl}$ fcm_cyl	f_{ck} fck	f´c fc_prime	Vct_ACI	Vs_ACI	Vcal_sACI	Vmax_ACI	rat_smax	VACI_cal	gmod_ACI
-	-	-	-	-	-	-	-	MPa	MPa	MPa	MPa	kN	kN	kN	kN		kN	
24	Ahmad; Xie; Yu (1996)	NHW-3b	1	1	0	1	0	96,2	105,3	101,3	102,9	42,38	64,05	106,44	211,92	0,502	106,44	1,154
25		NHW-4	1	1	0	1	0	91,9	100,8	96,8	98,4	41,45	41,06	82,51	207,26	0,398	82,51	1,140
40	Angelakos; Bentz;	DB0.530M	1	0	1	1	0	26,6	32,0	28,0	29,6	250,76	111,61	362,37	1253,79	0,289	362,37	0,734
42	Collins P. (1999)	DB140M	1	0	1	1	0	32,3	38,0	34,0	35,6	275,00	111,61	386,61	1375,00	0,281	386,61	0,723
62	Bhal (1968)	B1S	1	0	1	1	0	22,1	27,2	23,2	24,8	59,60	46,81	106,41	298,02	0,357	106,41	1,209
63		B2S	1	0	1	1	0	20,4	25,5	21,5	23,1	114,83	93,61	208,45	574,17	0,363	208,45	1,197
64		B3S	1	0	1	1	0	21,5	26,7	22,7	24,3	176,75	140,42	317,17	883,75	0,359	317,17	1,165
65		B4S	1	0	1	1	0	20,7	25,8	21,8	23,4	231,28	187,23	418,51	1156,42	0,362	418,51	1,129
132	Cladera; Mari (2002)	H 100/2	1	1	0	1	0	78,9	87,0	83,0	84,6	107,85	64,12	171,97	539,27	0,319	171,97	1,312
243	Hamadi (1976)	GT-2	1	1	0	1	0	19,8	24,9	20,9	22,5	33,07	50,67	83,73	165,33	0,506	83,73	1,773
245	Hamadi, Regan (1980)	GT-4	1	1	0	1	0	25,6	31,0	27,0	28,6	37,29	105,56	142,85	186,46	0,766	142,85	1,593
247	Hegger, J.; Görtz, S.; Will, N.(2001)	SVB 3b	1	1	0	1	0	62,7	70,0	66,0	67,6	110,61	232,53	343,14	553,06	0,620	343,14	1,244
361	Kong; Rangan (1997)	S1-3	1	0	1	1	0	54,7	61,6	57,6	59,2	93,28	63,69	156,97	466,41	0,337	156,97	1,313
362		S1-4	1	0	1	1	0	54,7	61,6	57,6	59,2	93,28	63,69	156,97	466,41	0,337	156,97	1,770
367		S2-3	1	0	1	1	0	62,9	70,2	66,2	67,8	99,84	63,69	163,53	499,21	0,328	163,53	1,549
368		S2-4	1	0	1	1	0	62,9	70,2	66,2	67,8	99,84	63,69	163,53	499,21	0,328	163,53	1,342
373		S3-3	1	0	1	1	0	58,2	65,3	61,3	62,9	96,47	45,61	142,08	482,34	0,295	142,08	1,609
374		S3-4	1	0	1	1	0	58,2	65,3	61,3	62,9	96,47	45,61	142,08	482,34	0,295	142,08	1,231
377		S4-1	1	0	1	1	0	76,5	84,5	80,5	82,1	197,20	114,29	311,49	985,99	0,316	311,49	1,136
380		S4-4	1	0	1	1	0	76,5	84,5	80,5	82,1	109,89	63,69	173,58	549,45	0,316	173,58	1,487
382		S4-6	1	0	1	1	0	76,5	84,5	80,5	82,1	74,51	43,19	117,70	372,57	0,316	117,70	1,724
383		S5-1	1	1	0	1	0	78,4	86,6	82,6	84,2	111,24	63,69	174,93	556,21	0,315	174,93	1,382
385		S5-3	1	0	1	1	0	78,4	86,6	82,6	84,2	111,24	63,69	174,93	556,21	0,315	174,93	1,394
391		S7-3	1	1	0	1	0	65,0	72,4	68,4	70,0	96,60	60,64	157,24	483,02	0,326	157,24	1,568
392		S7-4	1	1	0	1	0	65,0	72,4	68,4	70,0	96,60	75,80	172,40	483,02	0,357	172,40	1,587
420	Krefeld; Thurston (1966)	26-1	1	1	0	1	0	34,3	40,1	36,1	37,7	118,09	64,65	182,74	590,47	0,309	182,74	1,132
422		29b-1	1	1	0	1	0	32,0	37,7	33,7	35,3	114,14	43,10	157,24	570,71	0,276	157,24	1,018
425		29a-2	1	1	0	1	0	31,5	37,2	33,2	34,8	113,36	47,02	160,38	566,79	0,283	160,38	1,351
426		29b-2	1	1	0	1	0	35,5	41,4	37,4	39,0	120,02	47,02	167,04	600,10	0,278	167,04	1,212
427		29c-2	1	1	0	1	0	19,1	24,1	20,1	21,7	89,63	47,02	136,65	448,14	0,305	136,65	1,182
428		29d-2	1	1	0	1	0	25,1	30,4	26,4	28,0	101,75	47,02	148,77	508,74	0,292	148,77	1,109
430		29f-2	1	1	0	1	0	35,9	41,8	37,8	39,4	120,66	47,02	167,67	603,28	0,278	167,67	1,398
431		29g-2	1	1	0	1	0	11,1	15,7	11,7	13,3	70,17	47,02	117,19	350,85	0,334	117,19	1,279
444		39-3	1	1	0	1	0	36,8	42,7	38,7	40,3	122,02	78,05	200,07	610,10	0,328	200,07	1,198
483	Leonhardt; Walther;	ET2	1	1	0	1	0	18,8	23,8	19,8	21,4	34,54	48,41	82,95	172,68	0,480	82,95	1,579
484	DAfStb 151 (1962)	ET3	1	1	0	1	0	18,8	23,8	19,8	21,4	23,02	48,41	71,44	115,12	0,621	71,44	1,767
485		ET4	0	0	0	0	1	18,8	23,8	19,8	21,4	11,51	48,41	59,93	57,56	1,041	57,56	1,704
490	Leonhardt; Walther; DAfStb 152 (1962)	T1 (r)	1	1	0	1	0	19,2	24,2	20,2	21,8	64,01	1020,61	1084,63	320,07	3,389	320,07	2,500
497	Leonhardt; Walther	TA11	1	1	0	1	0	21,1	26,2	22,2	23,8	48,65	139,42	188,07	243,23	0,773	188,07	1,799
498	DAfStb 156 (1963)	TA12	1	1	0	1	0	21,1	26,2	22,2	23,8	48,65	82,66	131,31	243,23	0,540	131,31	2,039
499		TA13	1	1	0	1	0	17,5	22,4	18,4	20,0	44,58	320,33	364,90	222,88	1,637	222,88	1,553
500		TA14	1	1	0	1	0	17,5	22,4	18,4	20,0	44,58	221,94	266,51	222,88	1,196	222,88	1,509
501		TA15	1	1	0	1	0	16,5	21,4	17,4	19,0	43,41	139,42	182,83	217,05	0,842	182,83	1,614
503		TA16	1	1	0	1	0	16,5	21,4	17,4	19,0	43,41	139,42	182,83	217,05	0,842	182,83	1,620

				E	b	bw	h	hf	hh,top	hw	hft	hh,bot	bft	Ac	zc2	aa	af	ba	L	c	a	kap	cc
	-	-	-	SI / A	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm²	mm	mm	mm	mm	mm	mm	mm	-	mm
Levi_1989/1993_001_RC 30 A1	504	Levi;	RC 30 A1	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_002_RC 30 A2	505	Marro	RC 30 A2	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_003_RC 60 A1	506	(1989/1993)	RC 60 A1	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_004_RC 60 A2	507		RC 60 A2	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_005_RC 60 B1	508		RC 60 B1	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_006_RC 60 B2	509		RC 60 B2	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Levi_1989/1993_007_RC 70 B1	510		RC 70 B1	SI	900,0	120,0	1050,0	130,0		700,0	220,0		350,0	278000	432,8			250,0	9600,0	2000,0	3800,0	4,04	
Lynberg_1974_001_5A-0	511	Lyngberg	5A-0	SI	700,0	120,0	600,0	90,0	10,0	360,0	110,0	30,0	380,0	159600	261,1			425,0	5000,0	2000,0	1500,0	2,78	20,0
Lynberg_1974_002_5B-0	512	(1974)	5B-0	SI	700,0	120,0	600,0	90,0	10,0	360,0	110,0	30,0	380,0	159600	261,1			425,0	5000,0	2000,0	1500,0	2,78	20,0
Moayer_1974_002_P20	525	Moayer; Regan (1974)	P20	A	599,4	149,9	320,0	81,3		238,8				84503	108,4	76,2	26,7	304,8	1981,2	26,7	977,3	3,50	
Özden_1967_002_T6	534	Özden (1967)	T6	SI	400,0	110,0	355,0	80,0		275,0				62250	126,3	120,0	130,0	250,0	3000,0	900,0	1050,0	3,52	19,0
Özden_1967_003_T7	535		T7	SI	400,0	110,0	355,0	80,0		275,0				62250	126,3	120,0	130,0	250,0	3000,0	900,0	1050,0	3,52	19,0
Özden_1967_004_T9	536		T9	SI	400,0	110,0	355,0	80,0		275,0				62250	126,3	120,0	130,0	250,0	3000,0	900,0	1050,0	3,52	19,0
Petersson_1972_001_V1	564	Petersson (1972)	V1	SI	175,0	175,0	353,0							61775	176,5	175,0	100,0	300,0	2600,0	100,0	1150,0	3,57	
Petersson_1972_002_VL1	565		VL1	SI	301,0	120,0	307,0	63,0		175,0	63,0		301,0	60732	151,9	300,0	100,0	300,0	2200,0	100,0	900,0	3,24	
Regan_1971_024_T6	605	Regan (1971)	T6	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_027_T9	608		T9	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_032_T17	613		T17	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	3657,6	50,0	1803,8	7,10	25,4
Regan_1971_034_T20	615		T20	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	2743,2	50,0	1346,6	5,30	25,4
Regan_1971_039_T26	620		T26	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_044_T32	625		T32	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_049_T37	630		T37	SI	610,0	152,0	305,0	76,0		229,0				81168	103,4	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_050_T38	631		T38	SI	610,0	152,0	305,0	142,0		163,0				111396	104,9	60,0	50,0	300,0	1828,8	50,0	889,4	3,50	25,4
Regan_1971_056_W1	637		W1	SI	610,0	64,0	305,0	76,0		127,0	64,0	38,0	254,0	76786	113,3	75,0	50,0	305,0	1830,0	50,0	890,0	3,33	25,4
Regan_1971_059_W5	640		W5	SI	610,0	64,0	305,0	76,0		127,0	64,0	38,0	254,0	76786	113,3	75,0	50,0	305,0	1830,0	50,0	890,0	3,33	25,4
Reineck_1991_001_Stb III	647	Reineck (1991)	Stb III	SI	838,0	77,0	648,0	107,0		441,0	100,0		837,0	207323	318,2	120,0	160,0	700,0	5200,0	160,0	2520,0	4,27	
Reineck_1991_002_Stb I	648		Stb I	SI	837,0	77,0	650,0	100,0		440,0	110,0		838,0	209760	333,1	120,0	160,0	700,0	5190,0	160,0	2515,0	4,27	
Roller_1990_007_7	658	Roller; Russel (1990)	7	A	457,2	457,2	870,0							397741	435,0	266,7	133,4	152,4	4572,0	133,4	2219,3	2,91	
Roller_1990_009_9	660		9	A	457,2	457,2	870,0							397741	435,0	266,7	133,4	152,4	4572,0	133,4	2219,3	2,91	
Roller_1990_010_10	661		10	A	457,2	457,2	870,0							397741	435,0	266,7	133,4	152,4	4572,0	133,4	2219,3	2,91	
Sarsam_1992_010_BS4-H	671	Sarsam;	BS4-H	SI	180,0	180,0	270,0							48600	135,0	100,0	100,0	150,0	1565,0	400,0	582,5	2,50	25,0
Sarsam_1992_013_CS3-H	674	Al-Musawi (1992)	CS3-H	SI	180,0	180,0	270,0							48600	135,0	100,0	100,0	150,0	1565,0	400,0	582,5	2,50	25,0
Sarsam_1992_014_CS4-H	675		CS4-H	SI	180,0	180,0	270,0							48600	135,0	100,0	100,0	150,0	1565,0	400,0	582,5	2,50	25,0
Shin_1999_022_HB 2.5-50	697	Shin; Lee; Moon; Gosh (1999)	HB 2.5-50	SI	125,0	125,0	250,0							31250	125,0		22,5	100,0	1075,0	22,5	526,3	2,45	
Soerensen_1974_003_T-23	702	Soerensen (1974)	T-23	SI	400,0	110,0	355,0	80,0		275,0				62250	126,3	100,0	100,0	250,0	3000,0	900,0	1050,0	3,52	26,0
Stroband_1997_002_2	711	Stroband (1997)	2	SI	500,0	90,0	700,0	125,0	25,0	400,0	75,0	75,0	500,0	165500	335,5	242,3	91,4	346,2	5000,0	69,2	1654,3	2,56	
Stroband_1997_003_3	712		3	SI	500,0	90,0	700,0	125,0	25,0	400,0	75,0	75,0	500,0	165500	335,5	242,3	91,4	346,2	5000,0	69,2	1654,3	2,56	
Yoon_1996_003_N2-N	803	Yoon; Cook;	N2-N	SI	375,0	375,0	750,0							281250	375,0	150,0	75,0	350,0	4300,0	75,0	2112,5	3,23	40,0
Yoon_1996_006_M2-N	806	Mitchell (1996)	M2-N	SI	375,0	375,0	750,0							281250	375,0	150,0	75,0	350,0	4300,0	75,0	2112,5	3,23	40,0
Yoon_1996_008_H2-S	808		H2-S	SI	375,0	375,0	750,0							281250	375,0	150,0	75,0	350,0	4300,0	75,0	2112,5	3,23	40,0
Yoon_1996_009_H2-N	809		H2-N	SI	375,0	200,0	750,0							281250	375,0	150,0	75,0	350,0	4300,0	75,0	2112,5	3,23	40,0
Kautsch_2010_001_A2	820	Kautsch (2010)	A2	SI	655,0	155,0	820,0	150,0	20,0	650,0			155,0	207100	282,6	100,0	150,0	200,0	4000,0		1925,0	2,68	20,0
Kautsch_2010_002_B2	821		B2	SI	655,0	155,0	820,0	150,0	20,0	650,0			155,0	207100	282,6	100,0	150,0	200,0	4000,0		1925,0	2,68	20,0
	87																						
Plain	35																						
Ripp	52																						

			ds	d	ns	dst	fR	As	alphaa	rhos	rhosw	fsy	esy	ftk	ftk/fsy	euk	ds2	ns2	dst2	As2	fsy2
-	-	-	mm	mm	-	mm	-	mm²	-	%	%	MPa	‰	MPa	-	‰	mm	-	mm	mm²	MPa
504	Levi;	RC 30 A1	940,0	940,0	20	24,0	r	9047,8	1,00	1,0695	8,0211	500,0	2,50		0,00						
505	Marro	RC 30 A2	940,0	940,0	20	24,0	r	9047,8	1,00	1,0695	8,0211	500,0	2,50		0,00						
506	(1989/1993)	RC 60 A1	940,0	940,0	20	26,0	r	10618,6	1,00	1,2552	9,4136	450,0	2,25		0,00						
507		RC 60 A2	940,0	940,0	20	26,0	r	10618,6	1,00	1,2552	9,4136	450,0	2,25		0,00						
508		RC 60 B1	940,0	940,0	20	30,0	r	14137,2	1,00	1,6711	12,5329	470,0	2,35		0,00						
509		RC 60 B2	940,0	940,0	20	30,0	r	14137,2	1,00	1,6711	12,5329	470,0	2,35		0,00						
510		RC 70 B1	940,0	940,0	20	30,0	r	14137,2	1,00	1,6711	12,5329	470,0	2,35		0,00						
511	Lyngberg	5A-0	540,0	540,0	8	20,0	r	2513,3	1,00	0,6649	3,8785	616,3	3,08	725,00	1,18			6	10,0	471,2	
512	(1974)	5B-0	540,0	540,0	8	20,0	r	2513,3	1,00	0,6649	3,8785	623,1	3,12	733,00	1,18			6	10,0	471,2	
525	Moayer; Regan (1974)	P20	279,4	279,4	4	15,9	r	791,7	1,00	0,4727	1,8909	641,4	3,21	937,93	1,46						
534	Özden (1967)	T6	298,0	298,0	4	20,0	r	1256,6	1,00	1,0542	3,8335	441,5	2,21	448,32	1,02						
535		T7	298,0	298,0	4	20,0	r	1256,6	1,00	1,0542	3,8335	447,3	2,24	448,32	1,00						
536		T9	298,0	298,0	4	20,0	r	1256,6	1,00	1,0542	3,8335	444,4	2,22	448,32	1,01						
564	Petersson (1972)	V1	322,0	322,0	3	16,0	r	603,2	1,00	1,0704	1,0704	977,0	4,89		0,00			2	10,0	157,1	463
565		VL1	278,0	278,0	5	16,0	r	1005,3	1,00	1,2014	3,0135	990,0	4,95		0,00			2	16,0	402,1	635
605	Regan (1971)	T6	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270
608		T9	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270
613		T17	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270
615		T20	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270
620		T26	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270
625		T32	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270
630		T37	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270
631		T38	254,0	254,0	8	16,0	r	1608,5	1,00	1,0381	4,1662	630,0	3,15	950,00	1,51			6	9,7	443,4	270
637		W1	267,0	267,0	8	16,0	r	1608,5	1,00	0,9876	9,4130	630,0	3,15	950,00	1,51			4	10,0	314,2	275
640		W5	267,0	267,0	8	16,0	r	1608,5	1,00	0,9876	9,4130	630,0	3,15	950,00	1,51			4	10,0	314,2	275
647	Reineck (1991)	Stb III	590,0	590,0	4	28,0	r	6157,5	1,00	1,2454	13,5539	514,6	2,57	605,00	1,18			5	28,0	7081,1	514
648		Stb I	589,0	589,0	5	28,0	r	7081,1	1,00	1,4364	15,6134	514,0	2,57	603,78	1,17			4	28,0	6157,5	515
658	Roller; Russel (1990)	7	762,0	762,0	8	32,3	r	6538,1	1,00	1,8767	1,8767	472,4	2,36	741,24	1,57						
660		9	762,0	762,0	10	32,3	r	8172,7	1,00	2,3459	2,3459	472,4	2,36	741,24	1,57						
661		10	762,0	762,0	10	35,8	r	10073,9	1,00	2,8916	2,8916	431,0	2,16	690,55	1,60						
671	Sarsam;	BS4-H	233,0	233,0	2	25,0	r	1182,8	0,70	2,8202	2,8202	502,3	2,51		0,00		34,0	2	10,0	157,1	450
674	Al-Musawi (1992)	CS3-H	233,0	233,0	3	25,0	r	1470,0	0,70	3,5051	3,5051	543,0	2,72		0,00		34,0	2	10,0	157,1	450
675		CS4-H	233,0	233,0	3	25,0	r	1470,0	0,70	3,5051	3,5051	543,0	2,72		0,00		34,0	2	10,0	157,1	450
697	Shin; Lee; Moon; Gosh (1999)	HB 2.5-50	215,0	215,0	2	25,4	r	1013,4	0,70	3,7708	3,7708	410,1	2,05		0,00			2	9,5	142,5	392
702	Soerensen (1974)	T-23	298,0	298,0	4	20,0	r	1256,6	1,00	1,0542	3,8335	419,9	2,10	716,13	1,71	15					
711	Stroband (1997)	2	647,1	647,1	6	25,0	r	3573,6	1,00	1,1045	6,1359	600,0	3,00		0,00			4	10,0	314,2	
712		3	647,1	647,1	6	32,0	r	5227,6	1,00	1,6157	8,9759	560,0	2,80		0,00			4	10,0	314,2	
803	Yoon; Cook;	N2-N	655,0	655,0	10	29,9	r	7021,5	1,00	2,8586	2,8586	400,0	2,00		0,00			2	10,0	157,1	
806	Mitchell (1996)	M2-N	655,0	655,0	10	29,9	r	7021,5	1,00	2,8586	2,8586	400,0	2,00		0,00			2	10,0	157,1	
808		H2-S	655,0	655,0	10	29,9	r	7021,5	1,00	2,8586	2,8586	400,0	2,00		0,00			2	10,0	157,1	
809		H2-N	655,0	655,0	10	29,9	r	7021,5	1,00	2,8586	5,3600	400,0	2,00		0,00			2	10,0	157,1	
820	Kautsch (2010)	A2	719,4	719,4		28,0	r	3195,0	0,01	0,6781	2,8655	468,1	2,34	648,18	1,38	18,0	75,0	12	12,0	1357,2	585
821		B2	719,4	719,4		28,0	r	3195,0	0,01	0,6781	2,8655	468,1	2,34	648,18	1,38	11,0	75,0	12	12,0	1357,2	585
87																					
35																					
52																					

			dw	nsw	Asw	frw	sw	sw/h		zw1	zw2	rhow	fyw	fyw_rhow_100	fwtk	fwtk/fyw	ewuk	diaa	f1c	Method	fcwu	f1cttest	Method	betacttest	f1ctmcal	betactcal
-	-	-	mm	-	mm²	-	mm	-	-	mm	mm	%	MPa	MPa	MPa	-	‰	mm	MPa	-	MPa	MPa	-	-	MPa	-
504	Levi;	RC 30 A1	12,0	2	226,2	r.ass	225,0	0,21	0,24	880	0	0,8378	480,0	4,02		0,00			23,75	cyl	19,00	0,00		0,0000	2,30	0,0968
505	Marro	RC 30 A2	12,0	2	226,2	r.ass	225,0	0,21	0,24	880	0	0,8378	480,0	4,02		0,00			23,75	cyl	19,00	0,00		0,0000	2,30	0,0968
506	(1989/1993)	RC 60 A1	12,0	2	226,2	r.ass	225,0	0,21	0,24	880	0	0,8378	480,0	4,02		0,00			44,65	cyl	35,72	0,00		0,0000	3,71	0,0830
507		RC 60 A2	12,0	2	226,2	r.ass	225,0	0,21	0,24	880	0	0,8378	480,0	4,02		0,00			44,65	cyl	35,72	0,00		0,0000	3,71	0,0830
508		RC 60 B1	12,0	2	226,2	r.ass	150,0	0,14	0,16	880	0	1,2566	480,0	6,03		0,00			47,50	cyl	38,00	0,00		0,0000	3,88	0,0816
509		RC 60 B2	12,0	2	226,2	r.ass	150,0	0,14	0,16	880	0	1,2566	480,0	6,03		0,00			47,50	cyl	38,00	0,00		0,0000	3,88	0,0816
510		RC 70 B1	12,0	2	226,2	r.ass	150,0	0,14	0,16	880	0	1,2566	480,0	6,03		0,00			57,00	cyl	45,60	0,00		0,0000	4,27	0,0748
511	Lyngberg	5A-0	8,0	2	100,5	r	160,0	0,27	0,30	500	500	0,5236	674,0	3,53	760,00	1,13			24,42	cyl	19,53	2,03	sp	0,0829	2,35	0,0962
512	(1974)	5B-0	8,0	2	100,5	r	160,0	0,27	0,30	500	500	0,5236	647,0	3,39	734,00	1,13			25,27	cyl	20,22	2,68	sp	0,1061	2,41	0,0955
525	Moayer; Regan (1974)	P20	5,6	2	48,5	0	152,4	0,48	0,55	252	0	0,2123	310,3	0,66	427,59	1,38		19,0	38,66	cyl	30,92	0,00		0,0000	3,33	0,0863
534	Özden (1967)	T6	8,0	2	100,5	r	150,0	0,42	0,50	258	260	0,6093	272,7	1,66		0,00		16,0	31,69	cyl	25,35	0,00		0,0000	2,87	0,0907
535		T7	8,0	2	100,5	r	150,0	0,42	0,50	258	260	0,6093	272,7	1,66		0,00		16,0	28,98	cyl	23,19	0,00		0,0000	2,68	0,0926
536		T9	8,0	2	100,5	r	150,0	0,42	0,50	258	260	0,6093	272,7	1,66		0,00		16,0	28,98	cyl	23,19	0,00		0,0000	2,68	0,0926
564	Petersson (1972)	V1	6,0	2	56,5	0	150,0	0,42	0,47	292	0	0,2154	323,0	0,70		0,00		16,0	36,10	cyl	28,88	0,00		0,0000	3,17	0,0878
565		VL1	8,0	2	100,5	0	100,0	0,33	0,36	238	0	0,8378	366,0	3,07		0,00		16,0	40,85	cyl	32,68	0,00		0,0000	3,47	0,0850
605	Regan (1971)	T6	9,5	2	142,5	0	114,0	0,37	0,45	206	203	0,8224	270,0	2,22	435,00	1,61		19,0	26,33	cu	21,06	0,00		0,0000	2,49	0,0947
608		T9	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0	1,13	460,00	1,64		19,0	20,25	cu	16,20	0,00		0,0000	2,02	0,0998
613		T17	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0	1,13	460,00	1,64		19,0	33,53	cu	26,82	0,00		0,0000	3,00	0,0894
615		T20	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0	1,13	460,00	1,64		19,0	30,08	cu	24,06	0,00		0,0000	2,76	0,0918
620		T26	6,3	2	62,3	0	102,0	0,33	0,40	223	203	0,4021	280,0	1,13	460,00	1,64		19,0	49,28	cu	39,42	0,00		0,0000	3,98	0,0808
625		T32	9,5	2	142,5	0	114,0	0,37	0,45	206	203	0,8224	270,0	2,22	435,00	1,61		19,0	26,70	cu	21,36	0,00		0,0000	2,52	0,0944
630		T37	9,5	2	142,5	0	114,0	0,37	0,45	206	203	0,8224	270,0	2,22	435,00	1,61		19,0	28,50	cu	22,80	0,00		0,0000	2,65	0,0930
631		T38	9,5	2	142,5	0	114,0	0,37	0,45	206	203	0,8224	270,0	2,22	435,00	1,61		19,0	29,18	cu	23,34	0,00		0,0000	2,70	0,0925
637		W1	10,0	2	157,1	r	114,0	0,37	0,43	217	216	2,1530	704,0	15,16	1010,00	1,43		19,0	28,41	cyl	22,72	2,70	sp	0,0951	2,64	0,0931
640		W5	10,0	2	157,1	r	76,0	0,25	0,28	217	216	3,2294	704,0	22,74	1010,00	1,43		19,0	32,59	cyl	26,07	2,92	sp	0,0895	2,94	0,0901
647	Reineck (1991)	Stb III	8,0	2	99,4	r	108,0	0,17	0,18	550	0	1,1953	515,0	6,16	539,00	1,05	20	8,0	57,57	cu	46,05	4,32	sp	0,0750	4,28	0,0744
648		Stb I	10,0	2	162,8	r	126,0	0,19	0,21	539	0	1,6780	549,0	9,21	630,00	1,15	19	8,0	57,57	cu	46,05	4,32	sp	0,0750	4,28	0,0744
658	Roller; Russel (1990)	7	9,5	2	142,5	r	196,9	0,23	0,26	714	0	0,1583	445,2	0,70	576,83	1,30		12,7	68,79	cyl	55,03	3,79	sp	0,0551	4,56	0,0663
660		9	9,5	2	142,5	r	196,9	0,23	0,26	714	0	0,1583	445,2	0,70	576,83	1,30		12,7	119,04	cyl	95,24	5,07	sp	0,0426	5,52	0,0464
661		10	9,5	2	142,5	r	133,4	0,15	0,18	714	0	0,2337	445,2	1,04	576,83	1,30		12,7	119,04	cyl	95,24	5,07	sp	0,0426	5,52	0,0464
671	Sarsam;	BS4-H	4,0	2	25,1	0	75,0	0,28	0,32	213	183	0,1862	820,0	1,53		0,00		12,0	76,10	cyl	60,88	0,00		0,0000	4,73	0,0621
674	Al-Musawi (1992)	CS3-H	4,0	2	25,1	0	100,0	0,37	0,43	213	183	0,1396	820,0	1,14		0,00		12,0	70,49	cyl	56,39	0,00		0,0000	4,60	0,0653
675		CS4-H	4,0	2	25,1	0	75,0	0,28	0,32	213	183	0,1862	820,0	1,53		0,00		12,0	71,92	cyl	57,53	0,00		0,0000	4,63	0,0644
697	Shin; Lee; Moon; Gosh (1999)	HB 2.5-50	6,4	2	63,3	r	86,0	0,34	0,40	183	0	0,5892	365,9	2,16		0,00		13,0	69,35	cyl	55,48	0,00		0,0000	4,57	0,0659
702	Soerensen (1974)	T-23	6,0	2	56,5	0	150,0	0,42	0,50	268	246	0,3427	349,2	1,20	470,88	1,35	20	32,0	32,53	cyl	26,02	0,00		0,0000	2,93	0,0901
711	Stroband (1997)	2	8,0	2	100,5	r	165,0	0,24	0,25	607	0	0,6770	600,0	4,06		0,00		16,0	83,00	pr	66,40	6,12	sp	0,0737	4,87	0,0587
712		3	12,0	2	226,2	r	165,0	0,24	0,25	587	0	1,5232	630,0	9,60		0,00		16,0	83,90	pr	67,12	6,30	sp	0,0751	4,89	0,0583
803	Yoon; Cook;	N2-N	9,5	2	141,8	r	325,0	0,43	0,50	608	575	0,1163	430,0	0,50		0,00		20,0	34,20	cyl	27,36	2,79	sp	0,0816	3,04	0,0890
806	Mitchell (1996)	M2-N	9,5	2	141,8	r	230,0	0,31	0,35	608	575	0,1644	430,0	0,71		0,00		10,0	63,65	cyl	50,92	4,14	sp	0,0650	4,44	0,0697
808		H2-S	9,5	2	141,8	r	270,0	0,36	0,41	608	575	0,1400	430,0	0,60		0,00		10,0	82,65	cyl	66,12	5,49	sp	0,0664	4,86	0,0588
809		H2-N	9,5	2	141,8	r	160,0	0,21	0,24	608	575	0,4430	430,0	1,90		0,00		10,0	82,65	cyl	66,12	5,49	sp	0,0664	4,86	0,0588
820	Kautsch (2010)	A2	8,0	2	100,5	r	150,0	0,18	0,21	679	679	0,4324	605,0	2,62	631,00	1,04		16,0	29,30	cyl	23,44	2,51	sp	0,0857	2,71	0,0924
821		B2	8,0	2	100,5	r	200,0	0,24	0,28	679	679	0,3243	605,0	1,96	631,00	1,04		16,0	29,83	cyl	23,86	2,48	sp	0,0833	2,74	0,0920
87																										
35																										
52																										

			oms	oml	omwy	g	Vg	F	Vu_F+g,rep	Vu_Rep	Vu_gF	betar_meas	xr_meas	xr	sigswmeas	sigswass	sigsw	omwu	andbr	Mu	muu	vutest
-	-	-	-	-	-	kN/m	kN	kN	kN	kN	kN	[°]	mm	mm	MPa	MPa	MPa	-	-	kNm	-	-
504	Levi;	RC 30 A1	0,2252	0,2252	0,2116	6,67	19,35			676,00	695,35			1900	480		480	0,2116		2568,80	0,1360	0,3435
505	Marro	RC 30 A2	0,2252	0,2252	0,2116	6,67	19,35			688,00	707,35			1900	480		480	0,2116		2614,40	0,1384	0,3502
506	(1989/1993)	RC 60 A1	0,1265	0,1265	0,1126	6,67	19,35			990,00	1009,35			1900	480		480	0,1126		3762,00	0,1059	0,2640
507		RC 60 A2	0,1265	0,1265	0,1126	6,67	19,35			938,00	957,35			1900	480		480	0,1126		3564,40	0,1004	0,2491
508		RC 60 B1	0,1653	0,1653	0,1587	6,67	19,35			1181,00	1200,35			1900	480		480	0,1587		4487,80	0,1188	0,2994
509		RC 60 B2	0,1653	0,1653	0,1587	6,67	19,35			1239,00	1258,35			1900	480		480	0,1587		4708,20	0,1246	0,3156
510		RC 70 B1	0,1378	0,1378	0,1323	6,67	19,35			1330,00	1349,35			1900	480		480	0,1323		5054,00	0,1115	0,2806
511	Lyngberg	5A-0	0,1678	0,1678	0,1807	3,83	6,70			435,00	441,70			750	674		674	0,1807		652,50	0,1309	0,3731
512	(1974)	5B-0	0,1639	0,1639	0,1676	3,83	6,70			435,00	441,70			750		647	647	0,1676		652,50	0,1265	0,3594
525	Moayer; Regan (1974)	P20	0,0784	0,0784	0,0213	2,03	1,02			120,10	121,11			489		310	310	0,0213		117,37	0,0649	0,0966
534	Özden (1967)	T6	0,1469	0,1469	0,0655	1,49	1,46			137,34	138,80	39		525	273	255	255	0,0325		144,21	0,1281	0,1796
535		T7	0,1627	0,1627	0,0717	1,49	1,46			141,26	142,72	42		525	273		273	0,0717		148,33	0,1441	0,2041
536		T9	0,1616	0,1616	0,0717	1,49	1,46			154,02	155,47			525	273		273	0,0717		161,72	0,1571	0,2230
564	Petersson (1972)	V1	0,2897	0,2897	0,0241	1,48	0,93	100,00		100,93	100,93			575		323	323	0,0241		116,07	0,1772	0,0703
565		VL1	0,2912	0,2912	0,0938	1,46	0,73	190,00		190,73	190,73			450		366	366	0,0938		171,66	0,1806	0,1995
605	Regan (1971)	T6	0,2484	0,2484	0,1054	1,95	0,91			205,00	205,91			445	270		270	0,1054		182,33	0,1760	0,2835
608		T9	0,3230	0,3230	0,0695	1,95	0,91			154,00	154,91			445	280		280	0,0695		136,97	0,1719	0,2749
613		T17	0,1951	0,1951	0,0420	1,95	1,81			134,00	135,81			902	280		280	0,0420		241,71	0,1832	0,1458
615		T20	0,2175	0,2175	0,0468	1,95	1,36			138,00	139,36			673	280		280	0,0468		185,83	0,1570	0,1649
620		T26	0,1327	0,1327	0,0286	1,95	0,91			179,00	179,91			445	280		280	0,0286		159,20	0,0821	0,1243
625		T32	0,2450	0,2450	0,1040	1,95	0,91			216,00	216,91			445	270		270	0,1040		192,11	0,1828	0,2962
630		T37	0,2295	0,2295	0,0974	1,95	0,91			209,00	209,91			445	270		270	0,0974		185,88	0,1657	0,2651
631		T38	0,2242	0,2242	0,0951	2,67	1,26			238,00	239,26			445	270		270	0,0951		211,68	0,1844	0,2996
637		W1	0,2190	0,2190	0,6670	1,84	0,87			160,00	160,87			445	320		320	0,3032		142,40	0,1153	0,4430
640		W5	0,1909	0,1909	0,8722	1,84	0,87			190,00	190,87			445	368		368	0,4559		169,10	0,1193	0,4607
647	Reineck (1991)	Stb III	0,1113	0,1113	0,1337	4,98	6,67	530,00		536,67	536,67	24		1260	515		515	0,1337		1352,40	0,0805	0,2715
648		Stb I	0,1283	0,1283	0,2000	5,03	6,73	675,00		681,73	681,73	29		1258	549		549	0,2000		1714,56	0,1026	0,3516
658	Roller; Russel (1990)	7	0,1289	0,1289	0,0128	9,55	11,23			787,83	799,06			1110		445	445	0,0128		1748,45	0,0957	0,0442
660		9	0,0931	0,0931	0,0074	9,55	11,23			749,40	760,63			1110		445	445	0,0074		1663,16	0,0526	0,0239
661		10	0,1047	0,1047	0,0109	9,55	11,23			1172,14	1183,37			1110		445	445	0,0109		2601,35	0,0823	0,0387
671	Sarsam;	BS4-H	0,1862	0,1862	0,0251	1,17	0,57			206,90	207,47			291		820	820	0,0251		120,52	0,1621	0,0936
674	Al-Musawi (1992)	CS3-H	0,2700	0,2700	0,0203	1,17	0,57			247,20	247,77			291		820	820	0,0203		143,99	0,2090	0,1270
675		CS4-H	0,2647	0,2647	0,0265	1,17	0,57			220,70	221,27			291		820	820	0,0265		128,56	0,1829	0,1078
697	Shin; Lee; Moon; Gosh (1999)	HB 2.5-50	0,2230	0,2230	0,0389	0,75	0,21	149,11		149,32	149,32			263	366		366	0,0389		78,58	0,1961	0,1184
702	Soerensen (1974)	T-23	0,1361	0,1361	0,0460	1,49	1,46	139,30		140,76	140,76			525	349		349	0,0460		147,80	0,1279	0,1790
711	Stroband (1997)	2	0,0798	0,0798	0,0612	3,97	3,42			548,00	551,42			827		600	600	0,0612		906,56	0,0522	0,1477
712		3	0,1078	0,1078	0,1430	3,97	3,42			891,00	894,42			827		630	630	0,1430		1473,99	0,0839	0,2445
803	Yoon; Cook;	N2-N	0,3343	0,3343	0,0183	6,75	7,38			483,00	490,38			1056	430		430	0,0183		1020,34	0,1854	0,0819
806	Mitchell (1996)	M2-N	0,1796	0,1796	0,0139	6,75	7,38			689,00	696,38			1056	430		430	0,0139		1455,51	0,1421	0,0617
808		H2-S	0,1383	0,1383	0,0091	6,75	7,38			598,00	605,38			1056	430		430	0,0091		1263,28	0,0950	0,0399
809		H2-N	0,1383	0,1383	0,0288	6,75	7,38			721,00	728,38			1056	430		430	0,0288		1523,11	0,1145	0,0919
820	Kautsch (2010)	A2	0,1083	0,1083	0,1116	4,97	4,60	652,50		657,10	657,10	45	1000	1000	605		605	0,1116		1264,91	0,1274	0,2679
821		B2	0,1064	0,1064	0,0822	4,97	5,22	566,00		571,22	571,22	38	875	875	605		605	0,0822		1099,60	0,1088	0,2285
87																						
35																						
52																						

			kapc	sigsl	xsitest	zetatest	z_test	zeta	z_	muflex	Mu_flex	Vu_flex	betaflex	BB	sin2thp	thp	cotthp	vup	gamwp	cotthu	thu	nueu	betalb	VB	A2+A3
-	-	-	-	MPa	-	-	mm	-	mm	-	-	kN	-	-	-	[°]	-	-	-	-	[°]	-	-	-	-
504	Levi;	RC 30 A1	0,905	329	0,164	0,918	863	0,876	823	0,1971	3723,48	979,86	0,69		0,21	27,39	1,93	0,41	0,84	1,62	31,64	0,62	0,42		1
505	Marro	RC 30 A2	0,905	335	0,167	0,917	862	0,876	823	0,1971	3723,48	979,86	0,70		0,21	27,39	1,93	0,41	0,86	1,65	31,14	0,63	0,42		1
506	(1989/1993)	RC 60 A1	0,821	405	0,139	0,931	875	0,923	868	0,1168	4145,79	1091,00	0,91		0,11	19,60	2,81	0,32	0,84	2,35	23,09	0,59	0,34		1
507		RC 60 A2	0,821	382	0,131	0,935	879	0,923	868	0,1168	4145,79	1091,00	0,86		0,11	19,60	2,81	0,32	0,79	2,21	24,32	0,53	0,33		1
508		RC 60 B1	0,810	367	0,159	0,920	865	0,898	844	0,1485	5608,32	1475,87	0,80		0,16	23,48	2,30	0,37	0,82	1,89	27,93	0,58	0,34		1
509		RC 60 B2	0,810	387	0,168	0,916	861	0,898	844	0,1485	5608,32	1475,87	0,84		0,16	23,48	2,30	0,37	0,86	1,99	26,70	0,63	0,36		1
510		RC 70 B1	0,772	413	0,157	0,922	866	0,911	856	0,1255	5688,41	1496,95	0,89		0,13	21,33	2,56	0,34	0,83	2,12	25,24	0,58	0,35		1
511	Lyngberg	5A-0	0,902	522	0,158	0,921	497	0,907	490	0,1522	758,58	505,72	0,86		0,18	25,15	2,13	0,38	0,97	2,06	25,84	0,76	0,46		1
512	(1974)	5B-0	0,899	520	0,152	0,924	499	0,909	491	0,1490	768,48	512,32	0,85		0,17	24,16	2,23	0,37	0,96	2,14	25,00	0,75	0,44		1
525	Moayer; Regan (1974)	P20	0,845	553	0,080	0,960	268	0,954	266	0,0748	135,30	138,44	0,87		0,02	8,39	6,78	0,14	0,67	4,53	12,44	0,37	0,34		1
534	Özden (1967)	T6	0,873	418	0,159	0,920	274	0,916	273	0,1345	151,41	144,20	0,95		0,07	14,83	3,78	0,25	0,73	5,53	10,24	0,82	0,48		1
535		T7	0,884	435	0,179	0,911	271	0,908	271	0,1477	152,10	144,86	0,98		0,07	15,53	3,60	0,26	0,79	2,85	19,34	0,52	0,53		1
536		T9	0,884	444	0,183	0,909	271	0,909	271	0,1469	151,20	144,00	1,07	BB	0,07	15,53	3,60	0,26	0,86	3,11	17,81	0,61	0,58		1
564	Petersson (1972)	V1	0,856	677	0,235	0,883	284	0,831	267	0,2407	157,63	137,07	0,74		0,02	8,93	6,36	0,15	0,46	2,92	18,93	0,18	0,37		1
565		VL1	0,837	700	0,246	0,877	244	0,826	230	0,2405	228,53	253,93	0,75		0,09	17,84	3,11	0,29	0,68	2,13	25,19	0,41	0,40		1
605	Regan (1971)	T6	0,895	502	0,221	0,889	226	0,861	219	0,2139	221,65	249,22	0,82		0,11	18,95	2,91	0,31	0,92	2,69	20,40	0,69	0,46		1
608		T9	0,919	374	0,209	0,896	227	0,824	209	0,2662	212,16	238,55	0,65		0,07	15,29	3,66	0,25	1,08	3,96	14,19	0,93	0,43		1
613		T17	0,866	630	0,225	0,887	225	0,887	225	0,1731	228,40	126,62	1,06	BB	0,04	11,82	4,78	0,20	0,73	3,47	16,06	0,44	0,25		1
615		T20	0,880	505	0,198	0,901	229	0,876	223	0,1906	225,58	167,52	0,82		0,05	12,49	4,51	0,21	0,78	3,52	15,84	0,50	0,28		1
620		T26	0,803	412	0,108	0,946	240	0,917	233	0,1218	236,12	265,48	0,67		0,03	9,73	5,83	0,17	0,75	4,35	12,94	0,46	0,24		1
625		T32	0,893	532	0,231	0,884	225	0,863	219	0,2114	222,10	249,72	0,86		0,10	18,81	2,94	0,31	0,97	2,85	19,34	0,76	0,48		1
630		T37	0,886	508	0,209	0,896	227	0,870	221	0,1998	224,06	251,92	0,83		0,10	18,18	3,04	0,30	0,89	2,72	20,17	0,66	0,44		1
631		T38	0,883	588	0,237	0,882	224	0,873	222	0,1957	224,73	252,68	0,94		0,10	17,97	3,08	0,29	1,02	3,15	17,62	0,83	0,49		1
637		W1	0,886	356	0,140	0,930	248	0,876	234	0,1920	237,13	266,44	0,60		0,67	54,76	0,71	0,47	0,94	1,46	34,39	0,76	0,29		1
640		W5	0,870	425	0,148	0,926	247	0,890	238	0,1700	240,86	270,63	0,70		0,87	69,05	0,38	0,33	1,38	1,01	44,70	0,74	0,31		1
647	Reineck (1991)	Stb III	0,770	394	0,111	0,945	557	0,928	547	0,1033	1734,31	688,22	0,78		0,13	21,44	2,55	0,34	0,80	2,03	26,21	0,55	0,10		1
648		Stb I	0,770	443	0,144	0,928	547	0,917	540	0,1176	1965,35	781,45	0,87		0,20	26,57	2,00	0,40	0,88	1,76	29,63	0,65	0,12		1
658	Roller; Russel (1990)	7	0,725	378	0,142	0,929	708	0,911	694	0,1174	2144,36	966,22	0,82		0,01	6,50	8,78	0,11	0,39	3,45	16,15	0,13	0,71		1
660		9	0,524	282	0,106	0,947	722	0,911	694	0,0848	2680,57	1207,83	0,62		0,01	4,94	11,58	0,09	0,28	3,22	17,24	0,07	0,44		1
661		10	0,524	371	0,172	0,914	696	0,900	686	0,0942	2978,08	1341,88	0,87		0,01	6,00	9,51	0,10	0,37	3,54	15,79	0,12	0,63		1
671	Sarsam;	BS4-H	0,696	502	0,268	0,866	202	0,866	202	0,1612	119,90	205,84	1,01	BB	0,03	9,11	6,24	0,16	0,60	3,73	15,01	0,30	0,63		1
674	Al-Musawi (1992)	CS3-H	0,718	511	0,354	0,823	192	0,812	189	0,2192	151,02	259,26	0,95		0,02	8,19	6,95	0,14	0,90	6,25	9,08	0,65	0,64		1
675		CS4-H	0,712	442	0,303	0,849	198	0,814	190	0,2155	151,44	259,98	0,85		0,03	9,37	6,06	0,16	0,67	4,06	13,83	0,37	0,57		1
697	Shin; Lee; Moon; Gosh (1999)	HB 2.5-50	0,723	410	0,309	0,846	182	0,846	182	0,1886	75,56	143,58	1,04	BB	0,04	11,37	4,97	0,19	0,61	3,05	18,17	0,32	0,77		1
702	Soerensen (1974)	T-23	0,870	420	0,156	0,922	275	0,922	275	0,1254	144,93	138,03	1,02	BB	0,05	12,38	4,55	0,21	0,85	3,89	14,41	0,59	0,49		1
711	Stroband (1997)	2	0,668	409	0,081	0,959	621	0,940	608	0,0751	1304,59	788,60	0,69		0,06	14,32	3,92	0,24	0,62	2,41	22,50	0,33	0,25		1
712		3	0,664	467	0,135	0,932	603	0,919	595	0,0991	1740,67	1052,20	0,85		0,14	22,22	2,45	0,35	0,70	1,71	30,32	0,45	0,36		1
803	Yoon; Cook;	N2-N	0,863	253	0,245	0,878	575	0,806	528	0,2696	1483,37	702,19	0,69		0,02	7,77	7,33	0,13	0,77	0,00	0,00	0,00	0,30		1
806	Mitchell (1996)	M2-N	0,745	354	0,213	0,893	585	0,879	576	0,1580	1617,96	765,90	0,90		0,01	6,77	8,43	0,12	0,53	4,44	12,68	0,23	0,29		1
808		H2-S	0,669	298	0,154	0,923	605	0,897	587	0,1241	1649,54	780,85	0,77		0,01	5,48	10,43	0,09	0,42	4,38	12,86	0,15	0,22		1
809		H2-N	0,669	366	0,189	0,906	593	0,897	587	0,1241	1649,54	780,85	0,92		0,03	9,77	5,81	0,17	0,55	3,19	17,40	0,26	0,27		1
820	Kautsch (2010)	A2	0,883	468	0,123	0,939	675	0,939	675	0,1017	1009,81	524,58	1,253	BB	0,11	19,52	2,82	0,31	0,85	2,40	22,62	0,60	0,02		1
821		B2	0,881	468	0,121	0,940	676	0,940	676	0,1000	1010,83	525,11	1,09	BB	0,08	16,66	3,34	0,27	0,83	2,78	19,79	0,57	0,02		1
87																									87
35																									100,0
52																									

			kon34	KONA4	KONA5	A4+A5	Differenz	f1ck	fcm_cyl	fck	fc_prime	Vct_ACI	Vs_ACI	Vcal_sACI	Vmax_ACI	rat_smax	VACI_cal	gmod_ACI
-	-	-	-	-	-	-	-	MPa	MPa	MPa	MPa	kN	kN	kN	kN		kN	
504	Levi;	RC 30 A1	1	1	0	1	0	20,0	25,0	21,0	22,6	89,07	453,60	542,66	445,33	1,219	445,33	1,518
505	Marro	RC 30 A2	1	1	0	1	0	20,0	25,0	21,0	22,6	89,07	453,60	542,66	445,33	1,219	445,33	1,545
506	(1989/1993)	RC 60 A1	1	1	0	1	0	40,9	47,0	43,0	44,6	125,12	453,60	578,71	625,59	0,925	578,71	1,711
507		RC 60 A2	1	1	0	1	0	40,9	47,0	43,0	44,6	125,12	453,60	578,71	625,59	0,925	578,71	1,621
508		RC 60 B1	1	1	0	1	0	43,7	50,0	46,0	47,6	129,26	680,39	809,65	646,29	1,253	646,29	1,827
509		RC 60 B2	1	1	0	1	0	43,7	50,0	46,0	47,6	129,26	680,39	809,65	646,29	1,253	646,29	1,917
510		RC 70 B1	1	1	0	1	0	53,2	60,0	56,0	57,6	142,19	680,39	822,58	710,95	1,157	710,95	1,871
511	Lyngberg	5A-0	1	0	1	1	0	20,6	25,7	21,7	23,3	51,95	228,68	280,63	259,76	1,080	259,76	1,675
512	(1974)	5B-0	1	0	1	1	0	21,5	26,6	22,6	24,2	52,95	219,52	272,47	264,73	1,029	264,73	1,643
525	Moayer; Regan (1974)	P20	1	1	0	1	0	34,9	40,7	36,7	38,3	43,03	27,59	70,62	215,16	0,328	70,62	1,701
534	Özden (1967)	T6	1	1	0	1	0	27,9	33,4	29,4	31,0	30,29	54,47	84,76	151,45	0,560	84,76	1,620
535		T7	1	1	0	1	0	25,2	30,5	26,5	28,1	28,87	54,47	83,33	144,33	0,577	83,33	1,695
536		T9	1	1	0	1	0	25,2	30,5	26,5	28,1	28,87	54,47	83,33	144,33	0,577	83,33	1,848
564	Petersson (1972)	V1	1	1	0	1	0	32,3	38,0	34,0	35,6	55,84	39,21	95,05	279,21	0,340	95,05	1,062
565		VL1	1	1	0	1	0	37,1	43,0	39,0	40,6	35,30	102,29	137,59	176,52	0,779	137,59	1,386
605	Regan (1971)	T6	1	1	0	1	0	22,5	27,7	23,7	25,3	32,26	85,73	117,99	161,30	0,731	117,99	1,737
608		T9	1	1	0	1	0	16,5	21,3	17,3	18,9	27,89	43,47	71,36	139,45	0,512	71,36	2,158
613		T17	1	1	0	1	0	29,7	35,3	31,3	32,9	36,77	43,47	80,25	183,87	0,436	80,25	1,670
615		T20	1	1	0	1	0	26,3	31,7	27,7	29,3	34,69	43,47	78,16	173,43	0,451	78,16	1,766
620		T26	1	1	0	1	0	45,5	51,9	47,9	49,5	45,10	43,47	88,57	225,51	0,393	88,57	2,021
625		T32	1	1	0	1	0	22,9	28,1	24,1	25,7	32,51	85,73	118,24	162,56	0,727	118,24	1,827
630		T37	1	1	0	1	0	24,7	30,0	26,0	27,6	33,69	85,73	119,42	168,44	0,709	119,42	1,750
631		T38	1	1	0	1	0	25,4	30,7	26,7	28,3	34,12	85,73	119,85	170,60	0,703	119,85	1,986
637		W1	0	0	0	0	1	24,6	29,9	25,9	27,5	14,88	259,00	273,88	74,42	3,680	74,42	2,150
640		W5	0	0	0	0	1	28,8	34,3	30,3	31,9	16,03	388,50	404,53	80,15	5,047	80,15	2,371
647	Reineck (1991)	Stb III	0	0	0	0	1	53,8	60,6	56,6	58,2	57,56	279,65	337,22	287,81	1,172	287,81	1,865
648		Stb I	0	0	0	0	1	53,8	60,6	56,6	58,2	57,46	417,80	475,27	287,32	1,654	287,32	2,373
658	Roller; Russel (1990)	7	1	1	0	1	0	65,0	72,4	68,4	70,0	484,17	245,58	729,75	2420,86	0,301	729,75	1,080
660		9	1	1	0	1	0	115,2	125,3	121,3	122,9	641,51	245,58	887,09	3207,54	0,277	887,09	0,845
661		10	1	1	0	1	0	115,2	125,3	121,3	122,9	641,51	362,53	1004,03	3207,54	0,313	1004,03	1,167
671	Sarsam;	BS4-H	1	0	1	1	0	72,3	80,1	76,1	77,7	61,40	64,02	125,43	307,01	0,409	125,43	1,650
674	Al-Musawi (1992)	CS3-H	1	0	1	1	0	66,7	74,2	70,2	71,8	59,03	48,02	107,04	295,13	0,363	107,04	2,309
675		CS4-H	1	0	1	1	0	68,1	75,7	71,7	73,3	59,64	64,02	123,66	298,19	0,415	123,66	1,785
697	Shin; Lee; Moon; Gosh (1999)	HB 2.5-50	1	0	1	1	0	65,6	73,0	69,0	70,6	37,51	57,94	95,45	187,53	0,509	95,45	1,564
702	Soerensen (1974)	T-23	1	1	0	1	0	28,7	34,2	30,2	31,8	30,72	39,23	69,95	153,60	0,455	69,95	2,012
711	Stroband (1997)	2	0	0	0	0	1	79,2	87,4	83,4	85,0	89,17	236,56	325,73	445,83	0,731	325,73	1,682
712		3	0	0	0	0	1	80,1	88,3	84,3	85,9	89,66	558,88	648,55	448,31	1,447	448,31	1,987
803	Yoon; Cook;	N2-N	1	1	0	1	0	30,4	36,0	32,0	33,6	236,48	122,86	359,33	1182,38	0,304	359,33	1,344
806	Mitchell (1996)	M2-N	1	1	0	1	0	59,9	67,0	63,0	64,6	327,90	173,60	501,49	1639,48	0,306	501,49	1,374
808		H2-S	1	1	0	1	0	78,9	87,0	83,0	84,6	375,24	147,88	523,12	1876,18	0,279	523,12	1,143
809		H2-N	1	1	0	1	0	78,9	87,0	83,0	84,6	200,13	249,55	449,68	1000,63	0,449	449,68	1,603
820	Kautsch (2010)	A2	1	0	1	1	0	25,5	30,8	26,8	28,4	98,76	291,68	390,44	493,81	0,791	390,44	1,683
821		B2	1	0	1	1	0	26,0	31,4	27,4	29,0	99,73	218,76	318,49	498,65	0,639	318,49	1,794
87			80	56	24	80	7										m	1,556
35			92,0	64,4	27,6	92,0	8,0										s	0,361
52																	s²	0,131
																	v=s/m	0,232
															gmod-ACI <	3	0,05	0,962
															%	1,9	0,95	2,150